Counterfactual-Guided Diversified Path Integration for Transferable Neuron Attribution Attacks

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Abstract

Transferable adversarial attacks pose significant security threats to neural networks due to their strong black-box effectiveness against real-world models. While neuron attribution-based attacks have shown promise in enhancing transferability, existing methods are often constrained by integration paths that accumulate noise or lack structural diversity. Moreover, they typically rely on single-layer neuron attribution, failing to exploit multi-layer information. To address these limitations, we propose **CO**unterfactual-Guided Diversified **P**ath **I**ntegration Neuron Attribution **A**ttack, termed **COPIA**. Specifically, we incorporate counterfactual-guided paths into the neuron attribution framework, to effectively mitigate noise accumulation in integration trajectories and enable model-adaptive construction. Building on this, we diversify path update mechanisms and counterfactual samples to enhance path diversity and improve adversarial transferability. Furthermore, to capture the complementary semantics across different layers, we design a Cross-layer Attribution (CA) mechanism that progressively aggregates attribution information from deeper layers into the target attack layer via attribution flow. Extensive experiments on the ImageNet dataset across normally trained models, adversarially trained CNNs, ViTs, and various defense models demonstrate that COPIA consistently outperforms state-of-the-art methods in black-box attack settings. In addition, the proposed CA mechanism is highly generalizable and effectively improves the black-box transferability of existing single-layer neuron attribution attacks. Code and supplementary materials are available at https://github.com/AdvML-Group/COPIA.

CCS Concepts

• **Computing methodologies** → **Computer vision**.

Keywords

Adversarial Attack; Neuron Attribution; Path Integration; Attack Transferability; Counterfactual Guidance

ACM Reference Format:
Di Ming, Renqing Yang, Yuanchengzhi Liu, Peng Ren, Pengzhan Zhou, Yunlong Wang, and Xin Feng. 2026. Counterfactual-Guided Diversified Path Integration for Transferable Neuron Attribution Attacks. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26), November 10–14, 2026, Rio de Janeiro, Brazil.* ACM, New York, NY, USA, 37 pages. https://doi.org/10.1145/3767308.3835442

1 Introduction

Deep learning has achieved remarkable success in recent years. However, its inherent vulnerability to adversarial examples raises serious security concerns [2, 10, 24]. By introducing imperceptible perturbations, attackers can easily mislead well-trained models. In practical black-box scenarios, adversarial transferability—*i.e.*, the ability of perturbations crafted on a surrogate model to deceive unknown target models—has become a key evaluation measure for

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MM '26, Rio de Janeiro, Brazil.
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2213-4/2026/11
https://doi.org/10.1145/3767308.3835442

assessing real-world threat and effectiveness of adversarial attacks. To overcome the bottleneck of transferability in gradient-based attacks, feature importance-based methods [8, 21, 27, 40, 44, 47] introducing aggregate gradients to suppress critical object features, whereas neuron attribution-based [14, 17, 46] compute attribution scores along a path to perturb decision-critical neurons.

Despite these advances, existing feature-level attacks remain constrained by limited input perturbations, *e.g.*, FIA [39] restricts feature evaluation to a narrow input space, leading to surrogate model overfitting. Similarly, existing neuron attribution attacks expand the search scope via baseline integration, but remain constrained by their integration paths, In NAA [46], a black-image baseline leads to incomplete attributions [7, 29], and its linear integration path escapes saturated regions slowly [25], accumulating substantial noise during integration [49] and reducing attribution accuracy. NEAA [17] mitigates this issue by using a noisy version of the original image as the baseline and smoothing the integration through multiple passes, but still lacks model adaptivity in its linear integration path. While DANAA [14] increases model adaptivity by updating the integration path toward the maximum gradient direction, but its singular path limits attribution diversity. Furthermore, existing neuron attributions are restricted to single-layer analysis, failing to exploit diverse latent representations across layers.

Motivated by the limitations of integration paths in existing neuron attribution attacks, this paper introduces counterfactual paths within the neuron attribution framework. As illustrated at the top of Fig. 1, a counterfactual sample is defined as an image from the same dataset but belonging to a different category. Accordingly, a counterfactual path is constructed by iteratively optimizing the discrepancy between a benign sample and a counterfactual reference in the model's intermediate feature space. To address the limitation of singular integration path, we further propose two diversification strategies: (i) reference point diversification, which selects different counterfactual samples; and (ii) path generation diversification, which constructs multiple paths by maximizing latent feature differences between benign and counterfactual samples. Consequently, these counterfactual-guided diversification strategies yield more robust neuron attributions and improved attack transferability.

To alleviate the limitation of single-layer attributions in existing neuron attribution attacks, we further design a cross-layer attribution (CA) mechanism to fully exploit complementary attribution information across layers in the latent space. As illustrated at the bottom of Fig. 1, CA utilizes attribution flow to propagate attribution information from deeper to shallower layers, thereby aggregating diverse attributions at the target attack layer. The CA mechanism is highly generalizable: it not only significantly enhances the proposed counterfactual-guided attack but can also be applied to existing single-layer attribution methods, consistently improving transferability of adversarial examples across various black-box models. Overall, the proposed neuron attribution-based attack approach, termed COPIA[1], effectively integrates counterfactual-guided path, diversified attribution, and cross-layer aggregation into a unified framework. Our main contributions are summarized as follows:

[1] ***COPIA**: derived from the Latin word for abundance and also referring to the Roman personification of prosperity, the name reflects the rich and diversified attribution paths employed in the proposed adversarial attack method.*

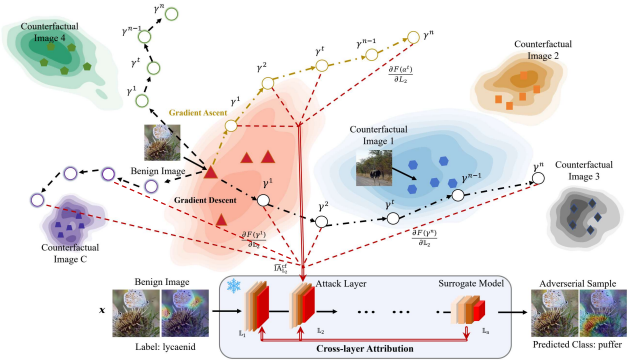


Figure 1: Overall framework of the proposed COPIA, incorporating diversified counterfactual-guided integration paths and cross-layer attribution aggregation.

- We introduce the counterfactual-guided path into the neuron attribution attack framework, enabling semantically guided path integration that mitigates noise accumulation and improves transferability in black-box settings.
- We propose a diversified path integration strategy that jointly explore different path update directions and an ensemble of counterfactual reference points, leading to more robust and transferable neuron attributions.
- We develop a cross-layer attribution aggregation mechanism that exploits complementary semantic information across multiple layers. Moreover, this general mechanism can be seamlessly integrated into existing single-layer attribution attacks to consistently enhance transferability.
- Extensive experiments on the ImageNet dataset demonstrate that COPIA achieves state-of-the-art performance in terms of black-box transferability across diverse model architectures, including CNNs, ViTs, and defense models.

2 Related work

Feature-level Attacks. These methods fool models by perturbing internal feature representations, disrupting high-level semantic activations to enhance transferability. FDA [8] perturbs feature maps at multiple layers to corrupt deep representations. FIA [39] estimates feature importance via gradient averaging over masked inputs and selectively perturbs salient features. MFAA [47] extends this idea by aggregating multi-layer features to compute more robust gradients. SMP [44] introduces semantics-guided, multi-granularity patchout with a two-stage optimization strategy. P2FA [21] reformulates the perturbation space by mapping pixel-level perturbations to latent feature distortions. SVD [40] and LRS [27] further enhance adversarial transferability through low-rank feature decomposition.

Neuron Attribution Attacks. Building upon Integrated Gradients (IG) [30], this line of work computes neuron-level attributions to guide the generation of adversarial perturbations. As an early representative, NAA [46] employs a fixed black-map baseline and a linear integration path, which lead to noise accumulation and hinder path adaptivity. DANAA [14] introduces model-adaptive, nonlinear paths to improve attribution accuracy but remains limited by its single-path formulation. NEAA [17] further adopts a

path-ensemble strategy to alleviate overfitting, though its baseline and paths are still non–model-adaptive.

Explainability Methods. Deep model predictions are primarily attributed to critical input regions or internal activations. CAM [48] uses intermediate activations for localization. Integrated Gradients (IG) [30] accumulates gradients along straight-line paths from a baseline input but suffers from noise and poor alignment with the data manifold [45]. To improve attribution quality, GuidedIG [15] adopts adaptive paths, MIG [45] aligns paths with the data manifold, and IG^2 [49] introduces counterfactual paths that better reside on the data manifold. IIA [1] refines attributions via iterative integration of inputs and internal states, while RISE [26] estimates saliency by averaging predictions over randomly masked inputs.

Existing neuron attribution attacks are limited by *non-informative baselines* and *single-layer formulations*, highlighting the need for *"more robust and diversified attribution trajectories"*. Motivated by recent advances [18, 37, 38, 49] in interpretability, we incorporate a counterfactual-based explaining strategy into the latent-space neuron attribution framework, as detailed in the following section.

3 Methodology

3.1 Preliminary

Notations. Let $\boldsymbol{x} \in [0,255]^{W\times H\times C}$ denotes a clean input image, where W, H, and C represent its width, height, and number of channels, respectively. The corresponding ground-truth label is denoted by $z \in \{1,\dots,K\}$, where K is the total number of classes. Given a surrogate model F, the output logit for class h are represented by $F(\boldsymbol{x})_h$, satisfying $\arg\max_h F(\boldsymbol{x})_h = z$. Let $\boldsymbol{y}(\boldsymbol{x})$ denote the output of an intermediate feature layer $\boldsymbol{y}$ in model F. We define three types of gradients: $\partial F(\boldsymbol{x})/\partial \boldsymbol{x}$ (the gradient of the output *w.r.t.* the input), $\partial F(\boldsymbol{x})/\partial \boldsymbol{y}(\boldsymbol{x})$ (the gradient of the output *w.r.t.* the feature layer), $\partial \boldsymbol{y}(\boldsymbol{x})/\partial \boldsymbol{x}$ (the gradient of the feature layer *w.r.t.* the input).

Adversarial Attack. To ensure both effectiveness and imperceptibility of an adversarial example $\boldsymbol{x}_{adv} = \boldsymbol{x} + \boldsymbol{\delta}$, adversarial attack [33] are typically formulated as the following constrained optimization problem induces misclassification such that $\arg\max_h F(\boldsymbol{x}_{adv})_h \neq z$:

$$\arg\max_{\boldsymbol{x}_{adv}} \ \mathcal{L}(F(\boldsymbol{x}_{adv}), z), \quad \text{s.t. } \|\boldsymbol{x}_{adv} - \boldsymbol{x}\|_p \leq \epsilon, \tag{1}$$

where $\mathcal{L}$ denotes the loss function (*e.g.*, cross-entropy), and ϵ bounds the perturbation $\boldsymbol{\delta}$ under the ℓ_p-norm. In this paper, we follow prior studies and adopt the ℓ_∞-norm setting (*i.e.*, $p = \infty$).

Neuron Attribution Attack. Given an input image $\boldsymbol{x}$ with $P = H \times W$ pixels in total, integrated gradient [30] defines the input-space attribution *w.r.t.* a baseline image $\boldsymbol{x}'$ as follows:

$$A = \sum_{i=1}^{P} (x_i - x_i') \int_0^1 \frac{\partial F(\boldsymbol{x}^t)}{\partial x_i} dt, \tag{2}$$

where $\boldsymbol{x}^t = \boldsymbol{x}' + t \cdot (\boldsymbol{x} - \boldsymbol{x}')$ lies on the straight-line integration path from $\boldsymbol{x}'$ to $\boldsymbol{x}$. Building on this, neuron attribution attack (NAA) [46] shifts the focus from output-level misclassification to latent neuron attribution. Extending Eq. 2 to an intermediate layer $\boldsymbol{y}$, the attribution of j-th neuron y_j is given by:

$$A_{y_j} = \sum_{i=1}^{P} (x_i - x_i') \int_0^1 \frac{\partial F}{\partial y_j}(y(\boldsymbol{x}^t)) \frac{\partial y_j}{\partial x_i}(\boldsymbol{x}^t) dt. \tag{3}$$

Accordingly, the layer-wise latent-space attribution is then obtained by aggregation, *i.e.*, $A = \sum_{y_j \in \boldsymbol{y}} A_{y_j}$. Minimizing A encourages $\boldsymbol{x}_{adv}$ to suppress informative neurons while amplifying misleading ones.

3.2 Counterfactual-Guided Neuron Attribution

Limitations of Existing Attribution Paths.

Existing attribution methods [4, 30, 46] typically construct paths between a baseline and the input via linear interpolation and estimate attributions along these paths. NEAA [17] further improve diversity by introducing multiple noisy baselines and averaging the resulting attributions. While these strategies can enhance transferability, they often suffer from attribution noise due to saturation effects and may yield incomplete explanations when the baseline lacks semantic relevance. Alternatively, DANAA [14] employs gradient-based nonlinear path that adaptively evolve with model gradients and capture richer intermediate representations. However, such methods remain limited by single-path formulations and rely heavily on surrogate gradients, restricting their robustness and generalization. A detailed comparison of different attribution paths is provided in Appendix A.1.

Path Integration under Counterfactual Guidance. As discussed above, existing methods contrastively estimate attributions by identifying neurons that drive the model prediction for the explicand [49], (*e.g.*, the original input) rather than a baseline prediction (*e.g.*, a black or noisy input) as reference. However, such baselines lack semantic relevance and often lead to incomplete or misleading attributions. Motivated by the role of counterfactual reasoning in human cognition [20, 36] for explaining unexpected outcomes, counterfactual samples [18, 37, 38, 49] can offer more discriminative and structured guidance for attribution. Based on this insight, we propose a counterfactual-guided path integration framework, which leverages both explicand and counterfactual gradients to construct semantically informed integration path, thereby improving attribution quality and attack transferability.

Consider a counterfactual sample $\boldsymbol{x}^{cf}$ that belongs to a different class from the explicand (*i.e.*, the original input) $\boldsymbol{x}$, *i.e.*, $z^{cf} \neq z$. Such a counterfactual provides a semantically meaningful reference for contrasting model behavior. Accordingly, a counterfactual-guided path $\boldsymbol{\gamma}$ with respect to $\boldsymbol{x}^{cf}$ is constructed to better explain $\boldsymbol{x}$, ranging from $\boldsymbol{\gamma}^0 = \boldsymbol{x}$ to $\boldsymbol{\gamma}^n \approx \boldsymbol{x}^{cf}$ (*i.e.*, $\boldsymbol{\gamma}^n$ is close to $\boldsymbol{x}^{cf}$), where n denotes the total number of intermediate steps and $\boldsymbol{\gamma}^t$ denotes the state at step t. This path is further integrated with latent-space attribution to enable more informative and transferable neuron attribution. Specifically, we introduce the counterfactual-guided neuron attribution in the latent representation space as follows:

$$\begin{aligned} A_{y_j}^{cf} &= \sum_{i=1}^{P} \int_0^1 \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\partial \boldsymbol{\gamma}_i^t}{\partial t} dt \\ &= \sum_{i=1}^{P} \int_0^1 \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \cdot \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot \frac{\partial \gamma_i^t}{\partial t} dt, \end{aligned} \tag{4}$$

where $y_j \in \boldsymbol{y}$ denotes the j-th neuron in the intermediate feature layer $\boldsymbol{y}$ of the surrogate model F. As can be seen, the resulting attribution is more semantically grounded and better aligned with transferable adversarial optimization through label flips along the counterfactual-guided integration path.

Constructing Counterfactual Attribution Path $\boldsymbol{\gamma}^0 \Rightarrow \boldsymbol{\gamma}^n$. To construct the path starting from $\boldsymbol{x}$ under the guidance of $\boldsymbol{x}^{cf}$, an ℓ_2-based similarity objective is first defined to progressively align the representation of the current state with that of the counterfactual sample, *i.e.*, $\arg\min_{\boldsymbol{\gamma}} \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf}) = \left\|G(\boldsymbol{\gamma}) - G(\boldsymbol{x}^{cf})\right\|_2^2$. Since the goal is to induce a label transition from the original class to the counterfactual class, the similarity is measured in the deep feature space $G(\cdot)$ (*e.g.*, the logit layer), where semantic and class-discriminative information is more explicitly encoded. The path is then generated efficiently through gradient descent, which iteratively moves the current state toward the counterfactual representation while maintaining a smooth transition in the input space. Accordingly, the update rule is defined as follows:

$$\boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^t - \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}^t} \cdot \frac{\eta}{W_t}, \tag{5}$$

where η is the step size , and W_t is the normalization coefficient.

To maintain practical attack efficiency under a limited attack budget (*e.g.*, the widely used 10 iterations), the computation of Eq. 4 is accelerated through the following approximations. First, following prior studies [14, 17, 46], the layers preceding and following $\boldsymbol{y}$ are assumed to be linearly independent, allowing the first two terms of Eq. 4 to be simplified into the product of two summation terms. Second, a finite-difference approximation is applied between the t-th and $(t+1)$-th integration steps, *i.e.*, $\partial \gamma_i^t / \partial t = (\gamma_i^{t+1} - \gamma_i^t)/(1/n)$. As a result, the counterfactual-guided neuron attribution with respect to y_j can be reduced to the following discretized form:

$$A_{y_j}^{cf} = \left(\sum_{t=1}^{n} \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)}\right) \cdot \left(\sum_{t=1}^{n}\sum_{i=1}^{P} \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t)\right). \tag{6}$$

For notional simplicity, we define the first term as the *integrated counterfactual-guided attribution*, denoted by $IA_{y_j}^{cf}(\boldsymbol{\gamma})$, obtained through gradient updates along the counterfactual-guided path. By applying the fundamental theorem of calculus for path integrals, the second term can be expressed as $\Delta_{y_j}^{cf}(\boldsymbol{\gamma}) \approx y_j(\boldsymbol{\gamma}^n) - y_j(\boldsymbol{\gamma}^0)$, representing the difference in latent-space representations at the endpoints of the counterfactual-guided path. Thus, both terms are influenced by counterfactual semantics. Detailed derivations from Eq. 4 to Eq. 6 are provided in Theorems 1-3 of Appendix A.2.

Consequently, after n gradient descent-based updates, the overall counterfactual-guided attribution of layer $\boldsymbol{y}$ is obtained by aggregating the contributions of all neurons:

$$A_{\boldsymbol{y}}^{cf}(\boldsymbol{\gamma}) = \sum_{y_j \in \boldsymbol{y}} A_{y_j}^{cf}(\boldsymbol{\gamma}) = \sum_{y_j \in \boldsymbol{y}} IA_{y_j}^{cf}(\boldsymbol{\gamma}) \cdot \Delta_{y_j}^{cf}(\boldsymbol{\gamma}). \tag{7}$$

Unlike prior studies [14, 17, 46], which rely on black-map or noisy inputs, this formulation fundamentally reformulates neuron attribution to align with the semantic information provided by counterfactual samples and their corresponding paths.

3.3 Diversified Path Integration

Building on the descent-based counterfactual path, the next step is to move beyond a single trajectory. While descent aligns the input with the counterfactual sample in representation space, relying on a single path limits attribution diversity. To capture richer neuron behaviors and further enhance transferability, the counterfactual-guided path is extended to a diversified path integration scheme.

Exploring Diversified Paths within a Single Counterfactual Sample. On the one hand, we diverge from the existing path (see Eq. 5) by instead maximizing the corresponding representation discrepancy, *i.e.*, $\arg\max_{\boldsymbol{\gamma}} \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf}) = \left\|G(\boldsymbol{\gamma}) - G(\boldsymbol{x}^{cf})\right\|_2^2$. Accordingly, starting from $\boldsymbol{\gamma}^0 = \boldsymbol{x}$, the counterfactual path is constructed via gradient ascent:

$$\boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^t + \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}^t} \cdot \frac{\eta}{W_t}. \tag{8}$$

Assuming that $\mathcal{L}_G$ is β-smooth and G is L-Lipschitz, when $0 < \eta < 2/\beta$, then the iterates $\{\boldsymbol{\gamma}^t\}$ are progressively driven away from $\boldsymbol{\gamma}^0 = \boldsymbol{x}$, *i.e.*, $\mathcal{L}_G(\boldsymbol{\gamma}^{t+1}, \boldsymbol{x}) \geq \mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x})$ (for $t = 1, \cdots, n-1$), while remaining guided by $\boldsymbol{x}^{cf}$. A rigorous proof of this nondecreasing distance property is provided in Theorem 4 of Appendix A.3.

Empirically, the effectiveness of the ascent-based path is validated in Fig. 2a: both gradient-descent and gradient-ascent paths move away from $\boldsymbol{\gamma}^0 = \boldsymbol{x}$ under the guidance of $\boldsymbol{x}^{cf}$, leading to label flips across decision boundaries. Beyond suppressing target-class confidence, these two paths exhibits complementary behaviors (see Fig. 2b), jointly contributing to more robust and accurate attribution. By combining gradient-descent and gradient-ascent paths $\mathbb{P} = \{\boldsymbol{\gamma}_{desc}, \boldsymbol{\gamma}_{asc}\}$ together, the counterfactual-guided latent-space attribution in Eq. 7 is extended as follows:

$$\widetilde{A}_{\boldsymbol{y}}^{cf} = \sum_{y_j \in \boldsymbol{y}} \left(\sum_{\boldsymbol{\gamma} \in \mathbb{P}} IA_{y_j}^{cf}(\boldsymbol{\gamma})\right) \cdot \left(\sum_{\boldsymbol{\gamma} \in \mathbb{P}} \Delta_{y_j}^{cf}(\boldsymbol{\gamma})\right), \tag{9}$$

where $IA_{y_j}^{cf}$ and $\Delta_{y_j}^{cf}$ are respectively averaged over two paths in $\mathbb{P}$ to mitigate potential attribution noise induced by a single path.

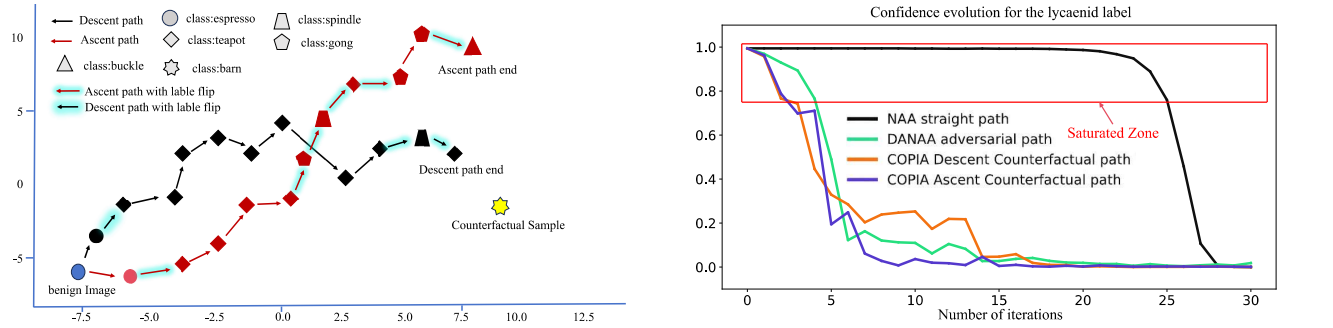
Exploring Diversified Paths across Multiple Counterfactual Samples. On the other hand, we adopt an ensemble strategy to diversify the counterfactual-guided path integration by incorporating multiple counterfactual samples. In this way, the semantic information from different counterfactual samples jointly contributes to the attribution of the explicand $\boldsymbol{x}$. Given C counterfactual samples, two gradient descent/ascent-based paths are constructed for each c-th sample, *i.e.*, $\mathbb{P}_c = \{\boldsymbol{\gamma}_{desc}^{(c)}, \boldsymbol{\gamma}_{asc}^{(c)}\}$. Accordingly, based on Eq. 9, the diversified counterfactual-guided attribution *w.r.t.* the ensemble integration paths $\{\mathbb{P}_1, \cdots, \mathbb{P}_C\}$ is formulated as follows:

$$\begin{aligned} \widetilde{A}_{\boldsymbol{y}}^{cf} &= \sum_{y_j \in \boldsymbol{y}} \left(\sum_{c=1}^{C} \sum_{\boldsymbol{\gamma} \in \mathbb{P}_c} IA_{y_j}^{cf}(\boldsymbol{\gamma})\right) \cdot \left(\sum_{c=1}^{C} \sum_{\boldsymbol{\gamma} \in \mathbb{P}_c} \Delta_{y_j}^{cf}(\boldsymbol{\gamma})\right) \\ &= \sum_{y_j \in \boldsymbol{y}} \widetilde{IA}_{y_j}^{cf}(\boldsymbol{\gamma}) \cdot \widetilde{\Delta}_{y_j}^{cf}(\boldsymbol{\gamma}). \end{aligned} \tag{10}$$

where $\widetilde{IA}_{y_j}^{cf}$ and $\widetilde{\Delta}_{y_j}^{cf}$ are respectively averaged over all $2C$ paths.

To capture the heterogeneous effects of neurons, we distinguish between positively and negatively contributing neurons according to Eq. 10, forming two sets: $\mathcal{N}_+^{cf} = \{y_j | \widetilde{A}_{y_j}^{cf} \geq 0, y_j \in \boldsymbol{y}\}$ and $\mathcal{N}_-^{cf} = \{y_j | \widetilde{A}_{y_j}^{cf} < 0, y_j \in \boldsymbol{y}\}$. Based on this partition, we formulate a weighted counterfactual-guided attribution $\widetilde{WA}^{cf}$ for layer $\boldsymbol{y}$ as:

$$\widetilde{WA}_{\boldsymbol{y}}^{cf} = \sum_{y_j \in \mathcal{N}_+^{cf}} f_p\left(\widetilde{A}_{y_j}^{cf}\right) - \lambda \cdot \sum_{y_j \in \mathcal{N}_-^{cf}} f_n\left(-\widetilde{A}_{y_j}^{cf}\right), \tag{11}$$



(a) Gradient-descent vs. gradient-ascent paths.

(b) Different paths for escaping saturation regions.

Figure 2: t-SNE visualization of intermediate counterfactual path steps illustrating decision boundary crossings, along with original class confidence curves to compare the efficiency of different paths in escaping saturation regions.

where λ is the hyperparameter to balance the relative importance between positive neuron attribution $f_p(\widetilde{A}^{cf}_{y_j})$ and negative neuron attribution $-f_n(-\widetilde{A}^{cf}_{y_j})$, and $f_p(\cdot)$, $f_n(\cdot)$ denote linear/nonlinear transformation functions.

3.4 Cross-Layer Attribution Aggregation

In preliminary experiments (see Fig. 5), the following observations can be drawn: (i) Deep-layer formulations yield accurate attributions but tend to overfit the surrogate model, resulting in limited transferability. (ii) Shallow or intermediate-layer formulations alleviate overfitting, yet still show limited performance in black-box settings due to insufficient attribution accuracy. These results suggest that single-layer attribution inherently trades off between attribution quality and transferability. To overcome this limitation, a cross-layer formulation is considered to exploit their complementary properties (see Fig. 7 of Appendix A.4).

From Single-Layer to Cross-Layer Neuron Attribution. Motivated by the above insights, we extend the proposed weighted counterfactual-guided neuron attribution to a cross-layer setting. Let $\mathbb{L}$ denote the set of L' selected layers from the total L layers of F. To capture cross-layer complementarity, we further define the *cross-layer integrated attribution* $\widetilde{mIA}^{cf}_{\boldsymbol{y}}$, which progressively propagates $\widetilde{IA}^{cf}_{\boldsymbol{y}}$ from deep to shallow layers. Specifically, the cross-layer attribution is formulated as follows: (i) For the last layer $\mathbb{L}[L']$, $\widetilde{mIA}^{cf}_{\mathbb{L}[L']} = \widetilde{IA}^{cf}_{\mathbb{L}[L']}$. (ii) For any intermediate layer $\mathbb{L}[l]$, where $l \in (1, L')$, the normalized gradient of the weighted attribution from layer $\mathbb{L}[l+1]$ is propagated and aggregated into layer $\mathbb{L}[l]$:

$$\widetilde{mIA}^{cf}_{\mathbb{L}[l]} = \widetilde{IA}^{cf}_{\mathbb{L}[l]} + \mu \cdot \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}^{cf}_{\mathbb{L}[l+1]} \Big/ \left\| \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}^{cf}_{\mathbb{L}[l+1]} \right\|_1, \quad (12)$$

where μ is the aggregation weight, $F_{\mathbb{L}[l]}$ is the activation output of layer $\mathbb{L}[l]$, and $\widetilde{mWA}^{cf}_{\mathbb{L}[l+1]}$ is the *cross-layer weighted attribution*, computed via Eq. 11 using the cross-layer attribution $\widetilde{mA}^{cf}_{\mathbb{L}[l+1]} = \sum_{y_j \in \boldsymbol{y}} \widetilde{mIA}^{cf}_{\mathbb{L}[l+1]_j}(\boldsymbol{y}) \cdot \widetilde{\Delta}^{cf}_{\mathbb{L}[l+1]_j}(\boldsymbol{y})$ as input.

As shown in Fig. 3, cross-layer aggregation mechanism improves counterfactual-guided neuron attribution over the single-layer variant (*i.e.*, without cross-layer aggregation), yielding more accurate

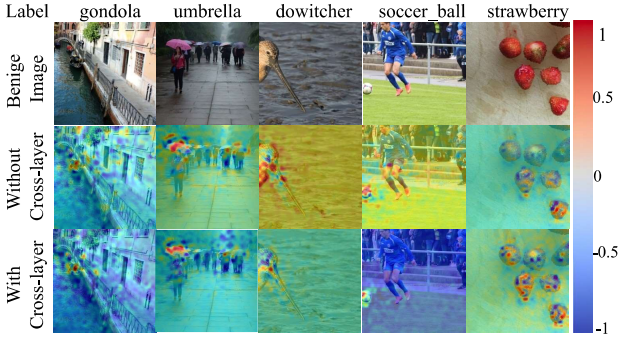


Figure 3: Comparison of neuron attribution with/without cross-layer aggregation. Red indicates positive attribution (stronger contribution with darker shades). Blue indicates negative attribution (stronger inhibition with deeper colors).

and object-focused results. Enabled by Eq. 12, it reduces layer-wise attribution noise while integrating complementary attribution information across layers. Quantitative results on 1000 images further confirm this: Deletion↓ [26] improves (9.2438 → 8.2693), Faithfulness↑ [16] improves (18.5447 → 19.5135), and Insertion↑ [26] remain comparable (27.7885 vs. 27.7828), validating its effectiveness.

The Overall Loss Function. Once the aggregation reaches first layer (denoted as $\boldsymbol{y} = \mathbb{L}[1]$), the cross-layer counterfactual-guided neuron attribution over the layer set $\mathbb{L}$ can be formulated as follows:

$$\widetilde{mWA}^{cf}_{\mathbb{L}} = \sum_{y_j \in \mathcal{N}^{cf}_{+}} f_p\left(\widetilde{mA}^{cf}_{y_j}\right) - \lambda \cdot \sum_{y_j \in \mathcal{N}^{cf}_{-}} f_n\left(\left(-\widetilde{mA}^{cf}_{y_j}\right)\right), \quad (13)$$

where $\mathcal{N}^{cf}_{+} = \{y_j | \widetilde{mA}^{cf}_{y_j} \geq 0, y_j \in \boldsymbol{y}\}$ and $\mathcal{N}^{cf}_{-} = \{y_j | \widetilde{mA}^{cf}_{y_j} < 0, y_j \in \boldsymbol{y}\}$. In summary, counterfactual-guided and cross-layer neuron attributions are jointly leveraged to produce adversarial examples with better transferability. In practice, we adopt MI-FGSM [5] to solve the following constrained optimization problem:

$$\arg\min_{\boldsymbol{x}_{adv}} \widetilde{mWA}^{cf}_{\mathbb{L}}(\boldsymbol{x}_{adv}), \quad \text{s.t. } \|\boldsymbol{x}_{adv} - \boldsymbol{x}\|_\infty \leq \epsilon. \quad (14)$$

The details of the corresponding optimization algorithm are provided in Algorithm 1 of Appendix A.5.

Table 1: Attack success rates (ASR, %) against normally trained and adversarially trained CNNs. Asterisks (*) denote white-box settings. ASR_{bb} denotes the average ASR over black-box models, excluding the white-box setting.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inc-v3$_{adv}$	IncRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR_{bb}
Inc-v3	FIA	98.5*	83.1	79.8	72.9	66.4	77.9	73.2	71.6	73.3	53.6	50.6	35.9	36.9	19.1	61.1
	NAA	96.6*	83.3	80.8	71.6	65.1	75.5	75.4	73.0	70.6	56.9	58.1	49.1	50.8	31.4	64.7
	DANAA	97.7*	86.7	84.1	75.2	70.7	79.5	78.6	74.8	72.5	64.4	66.1	54.0	53.7	32.0	68.6
	NEAA	99.1*	87.7	87.4	80.2	73.6	82.7	80.2	78.6	77.3	65.6	64.5	50.9	53.1	32.0	70.3
	SVD	98.6*	81.3	79.9	72.0	64.2	76.9	73.1	71.6	69.9	48.4	52.0	46.2	46.2	30.1	62.4
	MFAA	98.6*	86.5	85.3	79.3	74.7	83.2	80.2	76.8	78.0	60.4	62.9	45.1	44.6	24.7	67.8
	SMP	99.9*	92.1	91.1	80.0	74.5	84.7	82.4	81.4	80.4	58.8	63.4	43.2	41.6	21.7	68.9
	P2FA	**99.9***	**96.7**	**96.5**	**88.8**	83.2	**92.1**	**89.9**	**87.6**	**86.4**	57.0	68.7	33.9	34.9	16.8	71.7
	COPIA	98.8*	94.3	94.2	87.1	**84.8**	89.6	88.1	87.0	85.0	**71.9**	**76.3**	**64.1**	**63.7**	**41.9**	**79.1**
Inc-v4	FIA	84.2	95.7*	78.2	68.9	62.0	75.1	72.2	72.0	69.7	45.2	47.3	38.2	37.4	18.6	59.2
	NAA	81.7	96.0*	75.7	65.3	60.6	72.2	72.3	69.5	66.1	47.3	51.0	47.4	45.2	31.0	60.4
	DANAA	85.2	97.0*	81.8	70.8	65.1	76.9	76.6	73.2	71.2	52.9	59.4	53.2	51.1	31.9	65.3
	NEAA	89.3	98.6*	85.7	79.4	74.2	83.8	82.2	82.0	80.3	64.9	66.5	60.9	58.8	40.0	72.9
	SVD	83.5	96.1*	76.6	72.5	64.0	77.4	72.5	75.5	72.3	47.0	52.7	49.6	48.7	36.1	63.7
	MFAA	82.8	93.3*	78.9	75.0	69.4	79.1	76.5	75.7	73.5	58.4	60.4	54.0	52.5	31.9	66.8
	SMP	92.6	98.8*	88.9	76.2	68.5	84.1	82.2	81.0	79.7	48.8	57.1	46.3	42.8	24.1	67.1
	P2FA	**96.0**	**100.0***	92.3	81.9	72.8	89.0	87.7	82.5	81.9	45.5	54.9	29.9	32.1	14.4	66.2
	COPIA	95.3	99.0*	**93.7**	**87.5**	**83.6**	**89.7**	**89.6**	**88.6**	**87.0**	**71.1**	**76.3**	**70.0**	**66.6**	**51.5**	**80.8**
IncRes-v2	FIA	81.6	78.8	89.6*	71.5	66.5	75.5	71.8	71.8	71.5	60.3	62.9	48.7	44.3	31.7	64.4
	NAA	79.0	75.9	90.6*	68.2	66.0	71.9	70.3	68.7	65.7	56.6	63.6	55.0	52.6	44.4	64.5
	DANAA	82.3	79.6	91.4*	72.9	68.7	76.0	75.4	71.2	70.5	64.2	71.7	63.4	58.8	54.0	69.9
	NEAA	88.6	83.8	95.8*	79.9	76.6	83.0	79.7	79.1	77.9	72.0	75.0	65.8	60.0	52.7	74.9
	SVD	83.1	77.3	94.2*	70.4	65.6	73.9	72.4	70.8	68.0	49.6	58.9	52.0	49.4	41.6	64.1
	MFAA	80.5	79.0	93.9*	73.7	71.0	76.3	74.5	73.0	72.6	64.5	68.7	57.6	54.0	43.6	68.4
	SMP	89.4	87.3	95.8*	82.8	78.4	86.5	84.3	81.0	81.2	70.5	76.8	63.3	56.6	46.4	75.7
	P2FA	93.7	91.3	**99.9***	86.6	80.5	88.4	85.1	84.8	83.8	57.1	69.3	38.6	35.3	21.7	70.5
	COPIA	**96.3**	**94.8**	99.6*	**90.0**	**88.8**	**91.1**	**91.4**	**88.6**	**88.3**	**79.3**	**87.5**	**76.3**	**71.6**	**64.8**	**85.3**

4 Experiment

4.1 Experiment Setup

Benchmark Dataset. Following [46], we randomly sampled 1,000 images from distinct categories in the ILSVRC 2012 validation set.
Models. We used five classical CNNs as source models to generate adversarial examples: Inception-v3 (Inc-v3) [32], Inception-v4 (Inc-v4) [31], Inception-ResNet-v2 (IncRes-v2) [31], ResNet-v2-50 (Res-v2$_{50}$) [12], and ResNet-v2-152 (Res-v2$_{152}$) [12]. Adversarial transferability was evaluated on four categories of target models: *(i) Normally trained CNNs*: Inc-v3, Inc-v4, IncRes-v2, ResNet-v1-50 (Res-v1$_{50}$) [12], ResNet-v1-152 (Res-v1$_{152}$) [12], Res-v2$_{50}$, Res-v2$_{152}$, VGG16 [28], and VGG19 [28]. *(ii) Adversarially trained CNNs*: Inc-v3-adv, Inc-v3-ens3, Inc-v3-ens4, IncRes-v2-adv, and IncRes-v2-ens3. *(iii) Normally trained ViTs*: PiT [13], Swin [22], DeiT [34], CaiT [35], and ViT-Base [6]. *(iv) Defense models*: R&P [42], NIPS-r3, FD [23], JPEG [11], ComDefend [41], RS [3], and HGD [19].
Baseline. We compared the proposed COPIA with eight representative attacks, including FIA [39], NAA [46], DANAA [14], NEAA [17], SVD [40], MFAA [47], SMP [44], and P2FA [21].
Metric. Attack effectiveness is measured by the Attack Success Rate (ASR, %), defined as the percentage of cases where the model's prediction changes under adversarial perturbation.
Parameter Settings. Following NAA [46], we set the maximum perturbation to $\epsilon = 16$ and used 10 iterations with a step size of $\alpha = 1.6$. All baseline methods were optimized using momentum with a decay factor of 1.0. The positive-to-negative neuron ratio was set to $\lambda = 1$ with a linear transformation. For input transformations, DIM [43] was applied with a probability of 0.7, and PIM [9] used a scaling factor of 2.5 and kernel size 3. For COPIA, Gaussian noise (mean 0, variance 0.2) was added to the intermediate steps, the learning rate for counterfactual path construction was set to 0.025, and the number of counterfactual samples was fixed to 5.

4.2 Comparison with State-of-the-Art Methods

Attacking Normally Trained and Adversarially Trained CNNs. Table 1 compares ASR of different attacks on normally and adversarially trained models under three surrogate architectures. With Inc-v3 as the surrogate model, P2FA yields the highest ASR on normal models across multiple targets but drops significantly on adversarial models, indicating limited robustness of its generated adversarial examples. Although slightly lower on normal models, COPIA substantially outperforms P2FA on adversarial models, achieving the best performance and highest ASR_{bb}. With Inc-v4 and IncRes-v2, COPIA consistently attains the highest average ASR and best performance on almost all black-box target models.
Attacking Defended Models and Vision Transformers. As shown in Table 2, COPIA consistently achieves state-of-the-art (SOTA) performance in terms of DEF-ASR against various defense models across all three surrogate architectures. Furthermore, in transferability evaluations against vision transformers (ViTs), COPIA maintains its superiority across different surrogates and achieves the best ViT-ASR. Specifically, when IncRes-v2 is used as the surrogate, COPIA achieves an ViT-ASR of 67.3%, outperforming the current SOTA method SMP (55.0%) by a significant margin of 12.3%. Evaluations using Res-v2$_{50}$ and Res-v2$_{152}$ as surrogate models are provided in Appendix E. COPIA is extended to ViT-based architectures and evaluated against SOTA ViT attacks (Appendix B).

Results with PIDI data augmentation are reported in Appendix F. We further compare input transformation methods and cross-task attacks in Appendices D and G, respectively.

4.3 Generalizability of the Cross-Layer Attribution (CA) Mechanism

Existing neuron attribution attacks (*e.g.*, NAA, DANAA, NEAA) rely on attributions extracted from a single feature layer. To demonstrate

Table 2: Attack success rates (ASR, %) against defense models and vision transformers (ViTs). DEF-ASR and ViT-ASR denote the average ASR across defense models and ViT models, respectively.

Model	Method	Comdefend	FD	HGD	JPEG	NIPS-r3	R&P	RS	DEF-ASR	Pit	Swin	Deit	Cait	Vit_{base}	ViT-ASR
Inc-v3	FIA	71.3	69.4	11.0	69.0	28.0	17.8	32.6	42.7	43.3	29.9	27.4	36.0	34.9	34.3
	NAA	70.3	67.9	23.3	67.8	41.9	32.8	33.3	48.2	51.3	41.2	38.5	45.4	38.7	43.0
	DANAA	71.7	71.2	22.6	72	45.1	33.4	34.0	50.0	53.1	43.7	40.0	48.4	41.1	45.3
	NEAA	76.0	74.5	22.5	74.9	43.6	33.3	35.5	51.5	55.6	44.1	41.2	48.0	42.3	46.2
	SVD	66.7	64.5	**39.9**	64.5	36.8	31.5	32.0	48.0	50.3	42.9	37.5	45.3	36.7	42.5
	MFAA	76.4	74.3	12.0	75.4	37.7	25.8	34.2	48.0	55.3	40.9	39.4	47.6	44.0	45.4
	SMP	77.2	74.1	18.2	75.0	33.1	24.1	31.7	47.6	54.0	39.3	34.9	43.3	41.7	42.6
	P2FA	**84.1**	**81.9**	2.3	82.2	29.8	14.8	34.2	47.0	66.6	52.6	45.4	58.1	53.5	55.2
	COPIA	81.6	81.6	25.9	**82.3**	**55.9**	**44.6**	**38.0**	**58.6**	**70.4**	**58.2**	**56.7**	**65.5**	**53.8**	**60.9**
Inc-v4	FIA	64.4	64.8	16.6	64.2	27.9	27.9	31.8	42.5	48.6	41.6	30.1	38.4	33.6	38.5
	NAA	62.2	59.8	28.2	60.5	36.9	36.9	31.9	45.2	50.4	46.1	39.4	47.8	38.3	44.4
	DANAA	65.9	63.9	29.8	63.3	42.8	42.8	32.3	48.7	54.6	49.7	40.8	51.4	40.6	47.4
	NEAA	75.0	74.7	31.0	75.7	50.3	50.3	35.5	56.1	57.6	52.8	45.5	54.0	48.2	51.6
	SVD	66.7	65.3	**46.5**	64.5	41.0	37.9	32.0	50.6	51.6	49.3	39.3	47.7	41.4	45.9
	MFAA	71.3	69.5	19.7	71.1	41.8	41.8	34.9	50.0	56.5	47.9	41.2	51.6	46.2	48.7
	SMP	73.4	69.4	28.2	70.5	36.7	26.3	31.5	48.0	55.6	47.9	35.6	49.1	39.6	45.6
	P2FA	75.9	70.3	1.9	71.4	23.4	13.6	33.0	41.4	60.1	53.6	42.7	53.9	47.5	51.6
	COPIA	**82.9**	**82.9**	34.9	**82.0**	**61.3**	**51.9**	**37.4**	**61.9**	**73.6**	**65.9**	**59.0**	**69.3**	**58.2**	**65.2**
IncRes-v2	FIA	70.1	67.4	26.5	67.7	39.2	30.2	33.9	47.9	45.5	35.9	31.4	38.8	39.1	38.1
	NAA	66.5	62.6	39.3	62.9	48.1	41.9	33.8	50.7	49.9	42.6	40.9	47.9	41.4	44.5
	DANAA	68.3	66.6	47.3	66.3	56.2	51.3	34.2	55.7	57.6	49.2	44.5	52.5	45.8	49.9
	NEAA	78.9	77.4	39.9	78.6	57.4	49.0	**40.3**	60.2	57.5	47.8	45.5	53.2	52.9	51.4
	SVD	66.7	64.5	47.5	63.8	45.2	41.4	32.1	51.6	54.2	46.6	42.0	47.2	39.7	45.9
	MFAA	71.3	70.2	38.4	69.9	51.8	42.2	34.7	54.1	55.1	44.9	43.4	49.8	49.7	48.6
	SMP	80.3	76.7	44.3	77.5	55.2	46.5	34.5	59.3	63.6	50.9	48.1	57.2	55.4	55.0
	P2FA	80.8	76.4	3.6	79.2	32.8	19.8	35.3	46.8	57.7	46.9	41.3	52.1	52.1	50.0
	COPIA	**85.6**	**85.6**	**55.2**	**85.1**	**71.5**	**62.8**	38.8	**69.2**	**76.1**	**64.3**	**62.1**	**72.5**	**61.7**	**67.3**

the generality and effectiveness of the proposed CA mechanism, we integrate it into each baseline method. The resulting variants are denoted as NAA_{CA} (Cross-layer NAA), $DANAA_{CA}$ (Cross-layer DANAA), $NEAA_{CA}$ (Cross-layer NEAA), and S-COPIA (Single-layer COPIA). For fair comparison, we align the hyperparameters with those used in COPIA, including identical noise settings for NAA_{CA}, $DANAA_{CA}$, $NEAA_{CA}$, and the same learning rate for $DANAA_{CA}$.

Experimental results are summarized in Table 3, where CNN-ASR, ADV-ASR, ViT-ASR, and DEF-ASR denote the average ASR on normally trained models, adversarially trained models, vision transformers, and defended models, respectively. The results demonstrate the broad effectiveness of the proposed cross-layer attribution aggregation mechanism, with nearly all aggregated variants achieving improved performance. Notably, $DANAA_{CA}$ shows a slight decrease in ADV-ASR under certain surrogate settings, while all other methods consistently outperform their single-layer counterparts. These results indicate that CA effectively enhances the transferability of neuron attribution attacks. Detailed per-model results are provided in Appendix H. Besides, we compare the CA mechanism with direct multi-layer aggregation in Appendix I.

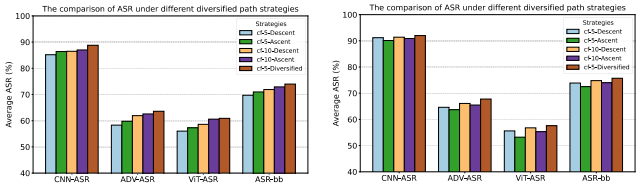
Table 3: Comparison of neuron attribution attacks with and without cross-layer attribution (CA) mechanism.

Model	Method	CNN-ASR	ADV-ASR	ViT-ASR	DEF-ASR
Inc-v3	NAA	74.4	49.3	43.0	48.2
	NAA_{CA}	78.2	52.7	49.8	50.1
	DANAA	77.8	54.0	45.3	50.0
	$DANAA_{CA}$	81.6	49.9	47.8	47.5
	NEAA	81.0	53.2	46.2	51.5
	$NEAA_{CA}$	85.5	57.9	54.0	55.2
	S-COPIA	83.7	58.5	52.0	54.0
	COPIA	**88.8**	**63.6**	**60.9**	**58.6**
IncRes-v2	NAA	70.7	54.4	44.5	50.7
	NAA_{CA}	75.6	57.8	49.5	53.8
	DANAA	74.6	62.4	49.9	55.7
	$DANAA_{CA}$	85.3	64.8	54.8	57.8
	NEAA	81.1	65.1	51.4	60.2
	$NEAA_{CA}$	85.0	68.4	58.4	63.7
	S-COPIA	85.9	71.9	59.4	64.9
	COPIA	**91.2**	**75.9**	**67.3**	**69.2**
$Res\text{-}v2_{50}$	NAA	83.5	55.3	46.7	59.5
	NAA_{CA}	87.4	61.6	50.3	63.6
	DANAA	84.5	60.9	47.6	60.5
	$DANAA_{CA}$	87.6	58.0	47.8	61.7
	NEAA	87.5	56.5	47.8	61.0
	$NEAA_{CA}$	90.5	61.1	52.3	65.4
	S-COPIA	89.6	61.2	51.6	64.9
	COPIA	**92.0**	**67.8**	**57.6**	**68.4**

4.4 The Effectiveness of the Diversified Path Integration Strategy

We conducted empirical validation to evaluate the effectiveness of the proposed diversified path integration strategy, as shown in Fig. 4. In the legend, "xr-5-descent" denotes using five counterfactual samples with standard gradient descent paths, while "xr-5-diversified" denotes using five counterfactual samples with both gradient ascent and descent paths. Other labels follow the same naming convention. As shown in Fig. 4, "xr-5-diversified" (10 paths in total) outperforms "xr-10-descent", which in turn outperforms "xr-5-descent". Similar trends are observed when using gradient ascent paths. These results demonstrate that increasing path diversity—rather than simply increasing the number of paths—more effectively improves the transferability of adversarial examples.

Please refer to Appendix J.1 for details on the selection strategy of counterfactual samples. The ablation study on the number of



(a) Inc-v3 (b) Res-v2$_{50}$

Figure 4: Effectiveness of diversified path integration: combining gradient descent and ascent paths improves transferability over single-direction paths.

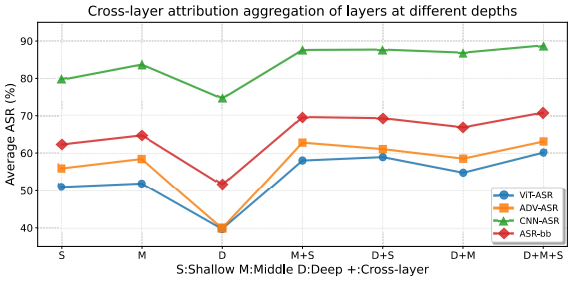


Figure 5: Comparison of average ASR across single-layer and cross-layer configurations at different depths. Cross-layer aggregation consistently outperforms single-layer settings.

Table 4: Quantitative evaluation of attribution quality using Insertion, Deletion, and Faithfulness metrics.

Model	Method	Insertion↑	Deletion↓	Faith ↑
Inc-v3	NAA	26.7671	**8.8119**	17.9552
	DANAA	26.8880	8.9443	17.9437
	S-COPIA	**27.7885**	9.2438	**18.5447**
IncRes-v2	NAA	48.4728	12.2342	36.2386
	DANAA	50.3073	12.0933	38.2140
	S-COPIA	**51.7154**	**11.9634**	**39.7520**
Res-v2$_{50}$	NAA	27.1209	10.6851	16.4358
	DANAA	27.0920	**10.3401**	16.7520
	S-COPIA	**27.7381**	10.8106	**16.9275**

counterfactual samples is provided in Appendix J.3. The selection of the feature layer for counterfactual path construction is discussed in Appendix J.2. An ablation study on the learning rate for path construction is provided in Appendix K.2. Additionally, a comparison of ASR between NEAA and S-COPIA under the same number of diversified paths is presented in Appendix J.4.

4.5 Effectiveness Analysis of COPIA

Single-layer neuron attribution is more accurate. To investigate whether counterfactual paths accumulate less noise during integration when computing intermediate-layer neuron attributions, we evaluated three representative single-layer attacks—NAA, DANAA, and S-COPIA—which correspond to straight-line, adversarial, and counterfactual paths, respectively, under a shared attack layer. For each method, the Integrated Attention (**IA**) was converted into a visual explanation using interpolation adapted from IIA [1]. We then assessed the quality of these explanations using three widely adopted interpretability metrics: Insertion [26], Deletion [26], and Faithfulness [16]. Higher values indicate better performance for Insertion and Faith, while lower values are preferred for Deletion. This evaluation framework enables us to measure the accuracy of **IA** and quantify the reliability of neuron attributions produced by different integration paths.

We computed these metrics for each image and report the averaged results in Table 4. Using IncRes-v2 as the surrogate model, S-COPIA (*i.e.*, the counterfactual path) achieves the best performance across all metrics. For both Inc-v3 and Res-v2$_{50}$ surrogate models, the counterfactual path consistently achieves the highest scores in Insertion and Faithfulness, further demonstrating its effectiveness in producing more reliable neuron attributions.

Investigation of Layer Selection for Cross-Layer Attribution Aggregation. The proposed Cross-Layer Attribution Aggregation mechanism leverages the diversity of neuron attributions across different layers in the latent space, as illustrated in Fig. 5. Using Inc-v3 as the surrogate model, we evaluated both single-layer configurations (shallow, intermediate, and deep) and cross-layer combinations (shallow+intermediate, shallow+deep, intermediate+deep, and shallow+intermediate+deep) across multiple black-box models. The results show that, for single-layer attacks, intermediate layers achieve better performance, while deep layers tend to cause overfitting and reduce transferability. In contrast, aggregating layers from different depths significantly improves performance by enhancing attribution diversity. These findings indicate that selecting layers from diverse depths is critical for capturing complementary information and improving adversarial transferability.

We provide ablation studies on the attack layer selection, the specific layer names for each surrogate model, the intensity of Gaussian noise added during the intermediate integration process, the positive-to-negative neuron ratio coefficients (λ), and the aggregation weight (μ) in Appendices K.4, K.5, K.1, K.6 and K.7, respectively. We also compare attack efficiency under the TensorFlow framework (Appendix L) and provide implementation details of COPIA under the PyTorch framework in Appendix M.

5 Conclusion

In this paper, we introduced counterfactual paths to mitigate noise accumulation during path integration and further enhanced path diversity through distinct optimization strategies and an ensemble of counterfactual examples. By leveraging these diversified paths, the transferability of neuron attribution attacks was significantly improved. To overcome the limitations of single-layer attribution, we further proposed a cross-layer attribution aggregation mechanism, which can be seamlessly integrated with both our counterfactual-guided framework and existing attribution-based attacks, consistently enhancing adversarial transferability. Extensive experiments demonstrated that the proposed COPIA achieves state-of-the-art performance across a wide range of black-box attack scenarios.

Acknowledgments

This work is partially supported by the National Natural Science Foundation of China (Grant No. 62441607, 62472059), the Natural Science Foundation of Chongqing, China (Grant No. CSTB2024NSCQ-MSX0341), the Science and Technology Research Program of Chongqing Municipal Education Commission (Grant No. KJQN202301142), and the Open Research Fund of Key Laboratory of Cyberspace Big Data Intelligent Security, Ministry of Education (Grant No. CBDIS202403).

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Counterfactual-Guided Diversified Path Integration for Transferable Neuron Attribution Attacks

Supplementary Material

Overview

In this supplementary material, we provide additional explanations, experiments, and analyses that further support the main findings of our paper. We also release the TensorFlow implementation, including demo scripts and pre-trained models, under various black-box attack settings.

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A Additional Explanation of the Proposed COPIA Approach

A.1 Comparison of Jump, Straight-line, Adversarial, and Counterfactual Integration Paths

Limitations of Existing Attribution Paths. *(i) Jump Integration Path.* To better distinguish feature-level attack strategies, FIA [39] can be reformulated as a random sampling-based integration process, where randomly replacing pixels with predefined colors can be viewed as generating noisy variants of the original input. Specifically, the jump integration path is approximated by sampling n points around the input, *i.e.*, $\boldsymbol{x}^t = \boldsymbol{x} + \Delta\boldsymbol{x}^t$, for $t = 1, \cdots, n$. In practice, this process often explores a relatively large and unstructured search region, *i.e.*,

$||\Delta \boldsymbol{x}^t||_2^2 \geq \epsilon$, rather than being strictly confined to a bounded neighborhood. As a result, it reflects a stochastic exploration of the input space that could induce label changes, there producing integrated gradient with adversarial semantics.

> Takeaway 1. *Such jump integration paths are highly stochastic, often causing rapid drops in label confidence without progressive refinement under semantic guidance. Consequently, they lack explicit structural direction, making it difficult to capture informative attributions.*

(ii) Straight-line Integration Path. Inspired by integrated gradient [4, 30], NAA [46] approximates the path integral via the Riemann sum and computes attributions along a linear interpolation path between a baseline $\boldsymbol{x}'$ and the input $\boldsymbol{x}$. Accordingly, a sequence of uniformly interpolated virtual samples is constructed, where the t-th virtual image is defined as: $\boldsymbol{x}^t = \boldsymbol{x}' + \sum_{k=1}^{t} \Delta \boldsymbol{x}^k$, $\Delta \boldsymbol{x}^k = (1/n) \cdot (\boldsymbol{x} - \boldsymbol{x}')$, for $t = 1, \cdots, n$. Building upon this formulation, NEAA [17] introduces an ensemble integration strategy to enhance path diversity by constructing multiple noisy baselines $\boldsymbol{x}'_{(e)} = \boldsymbol{x} + \mathcal{N}(0, \sigma)$ for $e = 1, \cdots, ens$, and averaging the attributions integrated along each baseline path, *i.e.*, $\boldsymbol{x}^t_{(e)} = \boldsymbol{x}'_{(e)} + \sum_{k=1}^{t} \Delta \boldsymbol{x}^k_{(e)}$, $\Delta \boldsymbol{x}^k_{(e)} = (1/n) \cdot (\boldsymbol{x} - \boldsymbol{x}'_{(e)})$, to capture neighborhood attribution around the original input $\boldsymbol{x}$.

> Takeaway 2. *Despite improving adversarial transferability, linear and ensemble-linear paths may introduces attribution noise due to saturation effects [25]. Moreover, the use of semantically weak baselines (e.g., black images) often leads to incomplete or misleading attributions [7, 29].*

(iii) Adversarial Integration Path. To overcome the limitations of linear paths, DANAA [14] introduces an adversarial nonlinear path that evolves with model gradients: $\boldsymbol{x}^t = \boldsymbol{x} + \sum_{k=1}^{t} \Delta \boldsymbol{x}^k$, $\Delta \boldsymbol{x}^k = \beta \cdot \text{sign}(\partial F(\boldsymbol{x}^k)/\partial \boldsymbol{x}^k) + \gamma$, where β is the step size and $\gamma \sim \mathcal{N}(0, \sigma)$ is the Gaussian noise. By leveraging such gradient ascent-based adaptive paths, DANAA captures richer model-specific intermediate representations and improves transferability.

> Takeaway 3. *Nonlinear paths still suffer from single-path constraints and exhibit excessive dependence on surrogate gradients, which limits their robustness and generalization.*

The Proposed Counterfactual-Guided Integration Attribution Path. Compared to the linear integration path, which is static and prone to saturation, and the adversarial path, which may be biased toward surrogate-specific artifacts, the proposed counterfactual-guided integration path provides more robust and accurate attributions from three key perspectives.

- First, it incorporates gradient information from both the original input and counterfactual samples to guide the construction of semantically meaningful integration trajectories, improving attribution reliability and transferability.
- Second, it enriches the integration process by introducing diversity in both update direction and counterfactual references, leading to more stable and transferable attribution patterns.
- Third, it aggregates attribution information across multiple layers to capture complementary semantic cues, enhancing the overall expressiveness of the attribution.

As a result, by iteratively aligning the integration trajectory with counterfactual feature distributions, the proposed approach suppresses irrelevant noise and highlights neurons with stronger transferability. Detailed visualizations of the aforementioned different integration paths are presented in Fig. 6.

A.2 Detailed Derivation of Counterfactual-Guided Neuron Attribution

(i) Simplification of Counterfactual-Guided Integration. In Sections 3.1–3.2, we examined the limitations of conventional neuron attribution methods that rely on standard path integration and introduced the counterfactual-guided attribution formulation. In the following, we provide a rigorous derivation that transitions from the classical path-integration formulation to our counterfactual-guided variant. First, we define the general attribution A_i for the i-th feature x_i as follows:

$$A_i = \int_0^1 \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\partial \boldsymbol{\gamma}_i^t}{\partial t} dt. \tag{15}$$

Under the counterfactual-guided path, we further establish the following inequality in the theorem.

> Theorem 1. *Assuming the integration path is determined by a counterfactual sample $\boldsymbol{x}^{cf}$, the general attribution A_i for the i-th feature x_i in Eq. 15 can be equivalently rewritten as:*
>
> $$A_i^{cf} = \sum_{t=0}^{n-1} \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial x_i} \cdot \left(\gamma_i^{t+1} - \gamma_i^t\right). \tag{16}$$

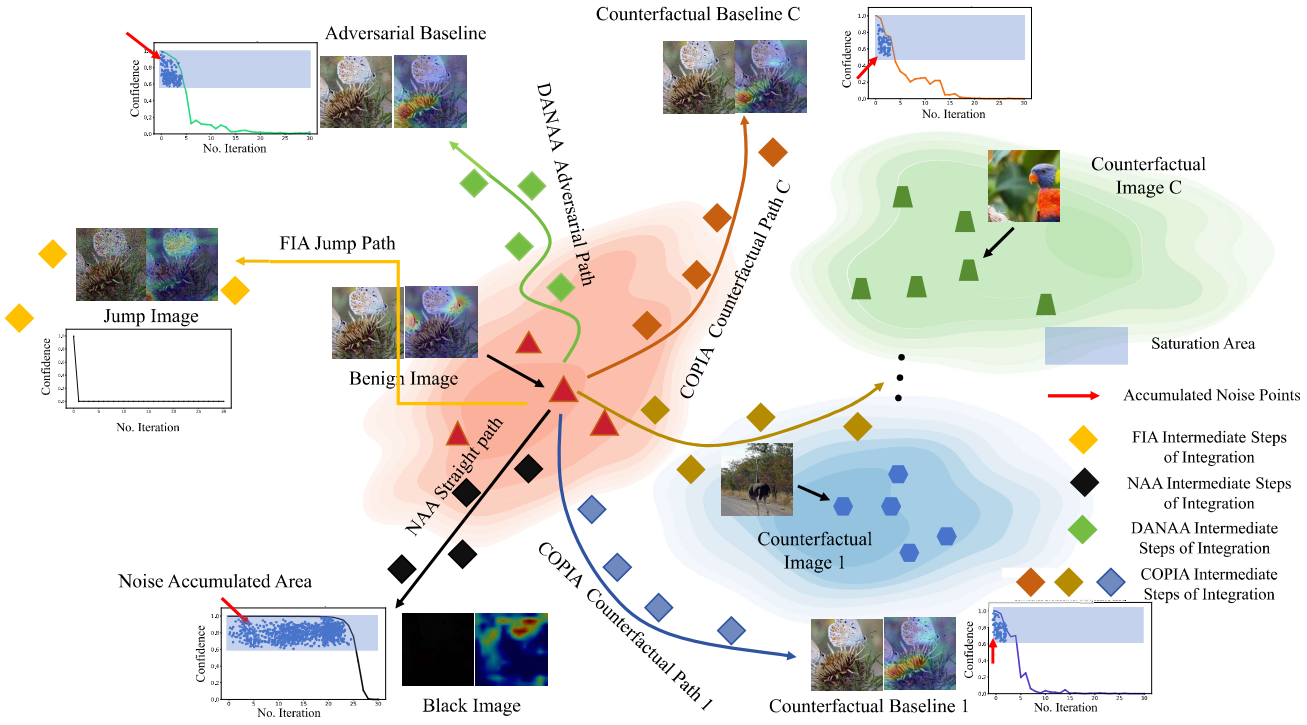


Figure 6: Visual comparison of different integration paths. (i) FIA introduces little accumulated noise, but its discontinuous jumps cause sharp confidence drops and unstable decision boundaries, hindering the accumulation of meaningful neuronal information. (ii) NAA's straight path escapes saturation slowly, leading to increased noise and reduced attribution accuracy. (iii) DANAA's adversarial path and COPIA's counterfactual path descend more efficiently, accumulate less noise, and better preserve label-critical neuronal information. (iv) Leveraging complementary update directions, COPIA achieves more robust and accurate attribution than DANAA.

Proof of Theorem 1. First, we construct a counterfactual-guided integration path based on the difference between the latent representation of the original clean sample $\boldsymbol{x}$ and its counterfactual sample $\boldsymbol{x}^{cf}$. Specifically, starting from $\boldsymbol{\gamma}^0 = \boldsymbol{x}$, each subsequent virtual point $\boldsymbol{\gamma}^{t+1}$, is obtained by performing gradient descent to minimize the discrepancy $\mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x}^{cf}) = \|G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})\|_2^2$. This iterative procedure yields an integration path consisting of n virtual images, guided by the semantics of the counterfactual sample, *i.e.*, $\{\boldsymbol{\gamma}^0, \boldsymbol{\gamma}^1, \cdots, \boldsymbol{\gamma}^n\}$.

Along the counterfactual path $\boldsymbol{\gamma}^0 \Rightarrow \boldsymbol{\gamma}^n$, the standard path integration formulation for computing the attribution of the i-th feature x_i [4, 30] can be numerically approximated using a Riemann sum discretization, yielding:

$$
\begin{aligned}
A_i &= \int_0^1 \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\partial \boldsymbol{\gamma}_i^t}{\partial t} dt \\
&= \sum_{t=1}^{n} \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}_i^t} \cdot \left(n \cdot \frac{\eta}{W_t}\right) \cdot \frac{(t+1-t)}{n} \\
&= \sum_{t=1}^{n} \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial x_i} \cdot \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}_i^t} \cdot \frac{\eta}{W_t} \\
&= \sum_{t=1}^{n} \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial x_i} \cdot \left(\boldsymbol{\gamma}_i^{t+1} - \boldsymbol{\gamma}_i^t\right) \\
&= A_i^{cf}.
\end{aligned}
\tag{17}
$$

In detail, the second equality follows from applying a finite difference approximation between t-th to $(t+1)$-th virtual points:

$$
\frac{\partial \boldsymbol{\gamma}_i^t}{\partial t} = \frac{\boldsymbol{\gamma}_i^{t+1} - \boldsymbol{\gamma}_i^t}{1/n} = \frac{\partial \mathcal{L}_G(\boldsymbol{\gamma}^t, \boldsymbol{x}^{cf})}{\partial \boldsymbol{\gamma}_i^t} \cdot \left(n \cdot \frac{\eta}{W_t}\right),
\tag{18}
$$

where η denotes the learning rate and W_t denotes the normalization factor. The third equality uses the fact that $\partial \boldsymbol{\gamma}_i^t = \partial(\boldsymbol{x} + \sum_{k=1}^t \Delta \boldsymbol{x}^k)_i = \partial x_i$, where $\Delta \boldsymbol{x}^k$ represents the gradient of the representation discrepancy $\mathcal{L}_G$ at step $\boldsymbol{\gamma}^k$. The fourth equality is obtained by substituting Eq. 5. Consequently, along the counterfactual path, the general attribution A_i can be equivalently expressed in its counterfactual-guided form A_i^{cf}, as defined in Eq. 16, thereby completing the proof. □

By applying Theorem 1, we obtain an effective simplification of the classical path integration formulation, leading directly to the counterfactual-guided neuron attribution. This reduced form substantially enhances attribution efficiency and provides the theoretical basis for the diversified path design and the cross-layer attribution aggregation strategy introduced later.

Unlike previous works [30, 46] that assign constant weight based on *"input feature differences" along a linear path*, our method uses counterfactual gradient to weight the Riemann summation, enabling the integration to focus on semantically critical features guided by *"latent representation differences" along the counterfactual path.*

(ii) From Input-Space to Latent-Space Counterfactual-Guided Neuron Attribution. From the above analysis, we observe that counterfactual-guided path integration captures attribution in the input space but primarily explains the model's decision for the explicand [49]. To better align with adversarial attack objectives, we further extend it into the latent space, enabling neuron-level attributions across feature layers to identify semantically meaningful neurons that improve transferability.

Building on the counterfactual-guided attribution developed in the previous section, we next reformulate the input-space formulation into its latent-space counterpart, which proves more advantageous for crafting adversarial examples with strong cross-model transferability.

> Theorem 2. *Under the assumption that the layers before $\boldsymbol{y}$ and the layers after $\boldsymbol{y}$ are linearly independent, the input-space attribution in Eq. 16 can be equivalently reformulated as the latent-space attribution defined in Eq. 6.*

Proof of Theorem 2. From Eq. 16, we extend A_i^{cf} to a latent-space formulation that accounts for all P pixels and the intermediate layer $\boldsymbol{y}$:

$$A^{cf} = \sum_{i=1}^{P} A_i^{cf} = \sum_{i=1}^{P} \sum_{t=1}^{n} \frac{\partial F\left(\boldsymbol{\gamma}^t\right)}{\partial x_i} \cdot \left(\gamma_i^{t+1} - \gamma_i^t\right) = \sum_{i=1}^{P} \sum_{t=1}^{n} \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \cdot \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t), \tag{19}$$

where the final equality is derived by applying the chain rule to the gradients back-propagated through y_j.

Since the layers preceding and following $\boldsymbol{y}$ are linearly independent, we can derive the following identity relationship. Let $\boldsymbol{a}$ and $\boldsymbol{b}$ denote the quantities associated with the layers before and after $\boldsymbol{y}$, respectively. If the two variables $\boldsymbol{a}$ and $\boldsymbol{b}$ have zero covariance, then $\sum_{i=1}^n (a_i - \hat{a})(b_i - \hat{b}) = 0$, where $\hat{a} = 1/n \cdot \sum_{i=1}^n a_i$ and $\hat{b} = 1/n \cdot \sum_{i=1}^n b_i$. This yields the identity: $\sum_{i=1}^n a_i b_i = 1/n \cdot (\sum_{i=1}^n a_i) \cdot (\sum_{i=1}^n b_i)$. Consequently, Eq. 19 can be further reformulated as follows:

$$A^{cf} = \sum_{t=1}^{n} \left[\left(\frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \right) \cdot \left(\sum_{i=1}^{P} \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t) \right) \right] = \left(\sum_{t=1}^{n} \frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)} \right) \cdot \left(\sum_{t=1}^{n} \sum_{i=1}^{P} \frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t} \cdot (\gamma_i^{t+1} - \gamma_i^t) \right) = A_{y_j}^{cf}, \tag{20}$$

As a result, we obtain $A^{cf} = \sum_{i=1}^P A_i^{cf} = A_{y_j}^{cf}$. which complete the proof. □

As demonstrated by Theorem 2, we extend the counterfactual-guided attribution from the input space to the latent space, consistent with prior findings [14, 17, 39, 46] showing that intermediate feature representations are more conducive to producing transferable adversarial examples. Moreover, the original formulation in Eq. 19 is computationally expensive to evaluate. By applying the linear-independence assumption [46][53, 56] , the latent-space neuron attribution under counterfactual guidance in Eq. 20 can be computed in a substantially more efficient manner.

> Takeaway 4. *By combining the results from Theorems 1-2, we establish the equivalence between Eq. 4 and Eq. 6, i.e., connecting the integral formulation to its accelerated discretized counterpart, thereby enabling efficient optimization under limited iteration budgets in practical attack scenarios.*

(iii) The Derivation of a Compact Form for Counterfactual-Guided Neuron Attribution. To further strengthen the practical computational efficiency of the proposed formulation, we derive a more compact representation of the counterfactual-guided neuron attribution, as detailed below.

> Theorem 3. *Given that the integration path $\boldsymbol{\gamma}^0 \Rightarrow \boldsymbol{\gamma}^n$ is continuous and differentiable, the summation of $A_{y_j}^{cf}$ in Eq. 6 over all neurons $y_j \in \boldsymbol{y}$ is mathematically equivalent to the formulation in Eq. 7.*

Proof of Theorem 3. Consider a general case where $\gamma(m) = x(m)$ for $m \in [0, 1]$. Given the continuity and differentiability of $x(m)$ *w.r.t.* m, the path $\gamma(m)$ is necessarily continuous and differentiable as well.

By the chain rule, we have $\frac{d}{dm}f(\gamma(m)) = \nabla f(\gamma(m))\frac{d\gamma(m)}{dm}$. Integrating both sides over the interval $[0,1]$ gives $\int_0^1 \frac{d}{dm}f(\gamma(m))dm = \int_0^1 \nabla f(\gamma(m))\frac{d\gamma(m)}{dm}dm$. Applying the Newton–Leibniz formula, the left-hand side simplifies to $\int_0^1 \frac{d}{dm}f(\gamma(m))dm = f(\gamma(1)) - f(\gamma(0))$. The right-hand side represents the integral along the path $\gamma(m)$, and can be equivalently rewritten as $\int_0^1 \nabla f(\gamma(m))\frac{d\gamma(m)}{dm}dm = \int_0^1 \nabla f(\gamma(m))d\gamma(m)$. Putting the above together, we conclude that:

$$\int_0^1 \nabla f(\gamma(m))d\gamma(m) = f(\gamma(1)) - f(\gamma(0)). \tag{21}$$

This establishes that, as long as the counterfactual-guided path remains continuous and differentiable, the path integral *w.r.t. m* reduces to the difference in the function values at the two endpoints, *i.e.*, $f(x(1)) - f(x(0))$.

According to the above derivations, the change in the j-th intermediate representation along the counterfactual path $\{\boldsymbol{\gamma}^0, \boldsymbol{\gamma}^1, \cdots, \boldsymbol{\gamma}^n\}$ can be defined as:

$$\Delta_{y_j}^{cf}(\boldsymbol{\gamma}) = \sum_{i=1}^{P}\left(\sum_{t=1}^{n}\frac{\partial y_j(\boldsymbol{\gamma}^t)}{\partial \gamma_i^t}(\gamma_i^{t+1} - \gamma_i^t)\right) \approx \sum_{i=1}^{P}\left(y_j(\boldsymbol{\gamma}_i^n) - y_j(\boldsymbol{\gamma}_i^0)\right) = y_j(\boldsymbol{\gamma}^n) - y_j(\boldsymbol{\gamma}^0). \tag{22}$$

Let $IA_{y_j}^{cf}(\boldsymbol{\gamma}) = \sum_{t=1}^{n}\frac{\partial F(\boldsymbol{\gamma}^t)}{\partial y_j(\boldsymbol{\gamma}^t)}$ denotes the integrated attribution *w.r.t.* y_j, accumulated through iterative gradient updates along the counterfactual-guided path $\{\boldsymbol{\gamma}^0, \boldsymbol{\gamma}^1, \cdots, \boldsymbol{\gamma}^n\}$. Consequently, the j-th counterfactual-guided attribution $A_{y_j}^{cf} = IA_{y_j}^{cf}(\boldsymbol{\gamma}) \cdot \Delta_{y_j}^{cf}(\boldsymbol{\gamma})$ can be extended to the compact formulation defined in Eq. 7, which aggregates the contributions of all neurons $y_j \in \boldsymbol{y}$, thereby completing the proof. □

A.3 Insights into Diversified Path Integration

(i) Gradient Ascent for Diversified Path Integration. Given a counterfactual sample $\boldsymbol{x}^{cf}$, conventional path construction typically employs *"gradient descent"* to move from the original input $\boldsymbol{x}$ toward $\boldsymbol{x}^{cf}$, thereby forming a counterfactual-guided integration path. While such a descent-based path effectively captures the transition toward the reference point, it is inherently limited to a single trajectory and may not sufficiently explore the surrounding feature space.

To further enhance black-box transferability, we aim to enrich the integration paths by introducing additional trajectories. Specifically, we continue to use $\boldsymbol{x}^{cf}$ as the reference point, but instead construct a complementary path via *"gradient ascent"*. By leveraging the geometric relationship between the current iterate $\boldsymbol{\gamma}^t$ and the counterfactual sample $\boldsymbol{x}^{cf}$, the ascent process progressively drives $\boldsymbol{\gamma}^t$ away from the initialization $\boldsymbol{\gamma}^0 = \boldsymbol{x}$, while still being guided by the reference representation. As a result, this ascent-based construction generates an *alternative counterfactual-guided integration path* that explores directions distinct from the descent trajectory. Such diversified paths enable a more comprehensive characterization of neuron attribution and provide stronger support for improving attack transferability.

To support the claim of path diversification, we next provide a rigorous analysis to explain why gradient ascent can progressively drive $\boldsymbol{\gamma}^t$ away from the initialization $\boldsymbol{x}$. We begin by defining the objective function $J(\boldsymbol{\gamma}) = \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf}) = \left\|G(\boldsymbol{\gamma}) - G(\boldsymbol{x}^{cf})\right\|_2^2$, where $G(\cdot)$ denotes the feature representation function. For notational simplicity, we use $J(\boldsymbol{\gamma})$ to represent $\mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf})$. Under the β-smoothness assumption, we first establish the following result.

> THEOREM 4. *Assume that $J(\cdot)$ is β-smooth, i.e., $\|\nabla J(\boldsymbol{u}) - \nabla J(\boldsymbol{v})\|_2 \le \beta\|\boldsymbol{u} - \boldsymbol{v}\|_2, \forall \boldsymbol{u}, \boldsymbol{v}$. Then, for any $\boldsymbol{x}, \boldsymbol{z}$, the following inequality holds: $J(\boldsymbol{z}) \ge J(\boldsymbol{x}) + \nabla J(\boldsymbol{x})^\top(\boldsymbol{z} - \boldsymbol{x}) - \frac{\beta}{2}\|\boldsymbol{z} - \boldsymbol{x}\|_2^2$.*

Proof of Theorem 4. Consider the linear interpolation path $\boldsymbol{\gamma}(t) = \boldsymbol{x} + t(\boldsymbol{z} - \boldsymbol{x}), t \in [0,1]$. By the fundamental theorem of calculus, we have

$$J(\boldsymbol{z}) - J(\boldsymbol{x}) = \int_0^1 \nabla J(\boldsymbol{\gamma}(t))^\top(\boldsymbol{z} - \boldsymbol{x})\, dt. \tag{23}$$

Adding and subtracting $\nabla J(\boldsymbol{x})$ yields

$$J(\boldsymbol{z}) - J(\boldsymbol{x}) = \nabla J(\boldsymbol{x})^\top(\boldsymbol{z} - \boldsymbol{x}) + \int_0^1 \left(\nabla J(\boldsymbol{\gamma}(t)) - \nabla J(\boldsymbol{x})\right)^\top(\boldsymbol{z} - \boldsymbol{x})\, dt. \tag{24}$$

By the inequality $a \ge -|a|$, together with the Cauchy-Schwarz inequality and the β-smoothness condition, we obtain

$$\begin{aligned}\left(\nabla J(\boldsymbol{\gamma}(t)) - \nabla J(\boldsymbol{x})\right)^\top(\boldsymbol{z} - \boldsymbol{x}) &\ge -\|\nabla J(\boldsymbol{\gamma}(t)) - \nabla J(\boldsymbol{x})\|_2\, \|\boldsymbol{z} - \boldsymbol{x}\|_2 &(25)\\ &\ge -\beta\|\boldsymbol{\gamma}(t) - \boldsymbol{x}\|_2\, \|\boldsymbol{z} - \boldsymbol{x}\|_2 &(26)\\ &= -\beta t\|\boldsymbol{z} - \boldsymbol{x}\|_2^2. &(27)\end{aligned}$$

Substituting this bound into the above expression in Eq. 24 gives

$$J(\boldsymbol{z}) - J(\boldsymbol{x}) \ge \nabla J(\boldsymbol{x})^\top(\boldsymbol{z} - \boldsymbol{x}) - \int_0^1 \beta t\|\boldsymbol{z} - \boldsymbol{x}\|_2^2\, dt \ge \nabla J(\boldsymbol{x})^\top(\boldsymbol{z} - \boldsymbol{x}) - \frac{\beta}{2}\|\boldsymbol{z} - \boldsymbol{x}\|_2^2, \tag{28}$$

which completes the proof. □

Theorem 4 establishes a quadratic lower bound on the objective improvement under the β-smoothness assumption. Building upon this result, we next formalize the optimization dynamics of gradient ascent for maximizing $J(\boldsymbol{\gamma}) = \mathcal{L}_G(\boldsymbol{\gamma}, \boldsymbol{x}^{cf}) = \left\|G(\boldsymbol{\gamma}) - G(\boldsymbol{x}^{cf})\right\|_2^2$, *i.e.*, $\boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^t + \eta \nabla J(\boldsymbol{\gamma}^t)$, and show that the distance between $\boldsymbol{\gamma}_t$ and $\boldsymbol{\gamma}_0$ (initialized as the original clean image $\boldsymbol{x}$) increases along the optimization trajectory, thereby providing a theoretical justification for the effectiveness of ascent-based diversified path construction.

LEMMA 5. *Assume that $J(\cdot)$ is β-smooth and $G(\cdot)$ is L-Lipschitz. If $0 < \eta < 2/\beta$, then the iterates $\{\boldsymbol{\gamma}^t\}$ are progressively driven away from $\boldsymbol{\gamma}^0$. More specifically, the distance $\|\boldsymbol{\gamma}^t - \boldsymbol{\gamma}^0\|_2$ admits a nondecreasing lower bound as the optimization proceeds.*

Proof of Lemma 5. According to Theorem 4, since $J(\cdot)$ is β-smooth, it satisfies $J(\boldsymbol{z}) \geq J(\boldsymbol{x}) + \nabla J(\boldsymbol{x})^\top (\boldsymbol{z} - \boldsymbol{x}) - \frac{\beta}{2}\|\boldsymbol{z} - \boldsymbol{x}\|_2^2, \forall \boldsymbol{x}, \boldsymbol{z}$. By letting $\boldsymbol{x} = \boldsymbol{\gamma}^t$ and $\boldsymbol{z} = \boldsymbol{\gamma}^{t+1} = \boldsymbol{\gamma}^t + \eta \nabla J(\boldsymbol{\gamma}^t)$, we obtain

$$J(\boldsymbol{\gamma}^{t+1}) \geq J(\boldsymbol{\gamma}^t) + \nabla J(\boldsymbol{\gamma}^t)^\top \left(\boldsymbol{\gamma}^{t+1} - \boldsymbol{\gamma}^t\right) - \frac{\beta}{2}\|\boldsymbol{\gamma}^{t+1} - \boldsymbol{\gamma}^t\|_2^2 \tag{29}$$

$$= J(\boldsymbol{\gamma}^t) + \eta \|\nabla J(\boldsymbol{\gamma}^t)\|_2^2 - \frac{\beta\eta^2}{2}\|\nabla J(\boldsymbol{\gamma}^t)\|_2^2 \tag{30}$$

$$= J(\boldsymbol{\gamma}^t) + \eta \left(1 - \frac{\beta\eta}{2}\right) \|\nabla J(\boldsymbol{\gamma}^t)\|_2^2. \tag{31}$$

This inequality characterizes the improvement of the objective under a single ascent step. Therefore, when $0 < \eta < 2/\beta$, it follows that $J(\boldsymbol{\gamma}^{t+1}) \geq J(\boldsymbol{\gamma}^t)$, which implies that the objective value $J(\boldsymbol{\gamma}^t)$ is nondecreasing along the optimization trajectory.

We next relate the growth of the objective to the distance in the input space. By the L-Lipschitz continuity of $G(\cdot)$, we have $\|G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})\|_2 \leq L\|\boldsymbol{\gamma}^t - \boldsymbol{x}^{cf}\|_2$. Rearranging this inequality yields:

$$\|\boldsymbol{\gamma}^t - \boldsymbol{x}^{cf}\|_2 \geq \frac{1}{L}\|G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})\|_2 = \frac{\sqrt{J(\boldsymbol{\gamma}^t)}}{L}. \tag{32}$$

To further connect this result with the initialization $\boldsymbol{\gamma}^0 = \boldsymbol{x}$, we apply the triangle inequality, $\|\boldsymbol{\gamma}^t - \boldsymbol{\gamma}^0\|_2 \geq \|\boldsymbol{\gamma}^t - \boldsymbol{x}^{cf}\|_2 - \|\boldsymbol{\gamma}^0 - \boldsymbol{x}^{cf}\|_2$, which leads to

$$\|\boldsymbol{\gamma}^t - \boldsymbol{\gamma}^0\|_2 \geq \frac{\sqrt{J(\boldsymbol{\gamma}^t)}}{L} - \|\boldsymbol{\gamma}^0 - \boldsymbol{x}^{cf}\|_2. \tag{33}$$

Finally, since $J(\boldsymbol{\gamma}^t)$ is nondecreasing, the right-hand side is also nondecreasing with respect to t. This implies that $\|\boldsymbol{\gamma}^t - \boldsymbol{\gamma}^0\|_2$ admits a nondecreasing lower bound, indicating that $\boldsymbol{\gamma}^t$ is progressively driven away from $\boldsymbol{\gamma}^0$ (*i.e.*, the original clean image $\boldsymbol{x}$) during optimization. This completes the proof. □

(ii) Update Direction Analysis under Gradient Ascent. To further understand the behavior of the ascent trajectory and its role in path diversification, we analyze the update direction induced by the objective function. By the chain rule, the gradient of $J(\boldsymbol{\gamma})$ is given by $\nabla J(\boldsymbol{\gamma}) = 2\, J_G(\boldsymbol{\gamma})^\top \left(G(\boldsymbol{\gamma}) - G(\boldsymbol{x}^{cf})\right)$, where $J_G(\boldsymbol{\gamma})$ denotes the Jacobian of G at $\boldsymbol{\gamma}$. Substituting this expression into the gradient ascent update yields $\boldsymbol{\gamma}^{t+1} - \boldsymbol{\gamma}^t = 2\eta\, J_G(\boldsymbol{\gamma}^t)^\top \left(G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})\right)$, which explicitly characterizes the update direction at iteration t. This formulation reveals that the optimization direction is explicitly governed by the counterfactual reference $\boldsymbol{x}^{cf}$ through the feature discrepancy $G(\boldsymbol{\gamma}^t) - G(\boldsymbol{x}^{cf})$, while the current iterate $\boldsymbol{\gamma}^t$ determines both this discrepancy and the local geometry via the Jacobian $J_G(\boldsymbol{\gamma}^t)$.

Although the initialization $\boldsymbol{\gamma}^0$ does not appear explicitly in the single-step update, it implicitly shapes the trajectory through the recursive optimization dynamics, which can be written as $\boldsymbol{\gamma}^t = \Phi_t(\boldsymbol{\gamma}^0; \boldsymbol{x}^{cf})$. In particular, at the first iteration, we have $\boldsymbol{\gamma}^1 - \boldsymbol{\gamma}^0 = 2\eta\, J_G(\boldsymbol{\gamma}^0)^\top \left(G(\boldsymbol{\gamma}^0) - G(\boldsymbol{x}^{cf})\right)$, showing that the initial ascent direction is jointly determined by the local structure at $\boldsymbol{\gamma}^0$ and the feature discrepancy induced by the reference $\boldsymbol{x}^{cf}$. Therefore, the update direction is explicitly influenced by the counterfactual reference $\boldsymbol{x}^{cf}$ and implicitly shaped by the initialization $\boldsymbol{\gamma}^0 = \boldsymbol{x}$ through the optimization trajectory. This interplay induces diverse optimization paths under different initializations and reference points, thereby providing additional insight into the effectiveness of ascent-based diversified path integration. The comparison of the intermediate integration processes for counterfactual paths, the NAA straight-line path, and the DANAA adversarial path through saturated regions is shown in Fig. 14. Benefiting from the complementary nature of gradient ascent and descent paths, diversified counterfactual-guided integration captures more informative attributions.

A.4 Mechanisms of Cross-Layer Attribution Aggregation

Neuron attributions computed at different layers inherently capture diverse and complementary information, as each layer emphasizes distinct characteristics of the input. To substantiate this observation, we visualize neuron attributions extracted from multiple layers for two randomly selected input images, as shown in Figure 7. From left to right, the visualizations exhibit a clear hierarchical progression: *"shallow layers" primarily highlight low-level textures and local structures, whereas "deeper layers" increasingly focus on semantically meaningful and task-relevant regions.*

To systematically exploit these complementary representations across the network hierarchy, we introduce a cross-layer attribution aggregation mechanism for neuron attributions. This mechanism integrates multi-scale information in a principled manner by combining

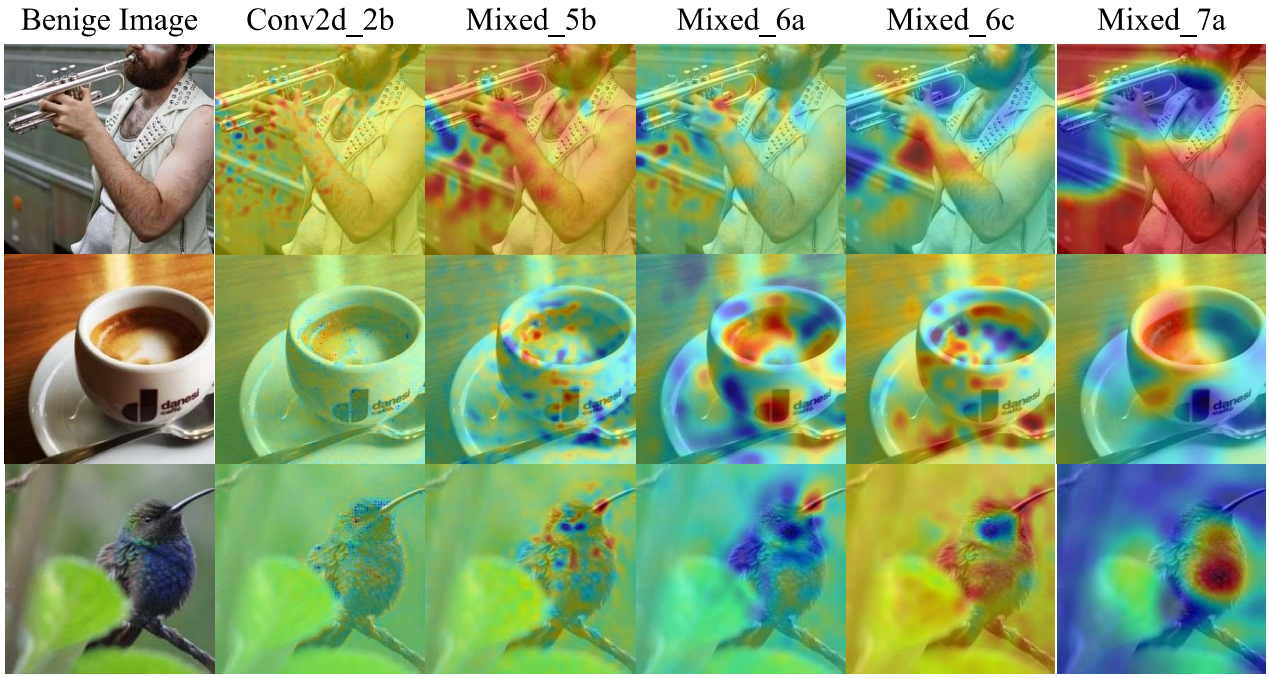


Figure 7: From shallow to deep layers, neuron attribution becomes increasingly interpretable, with progressively shifting regions of focus, thereby making attributions from shallow and deep layers complementary.

the fine-grained details captured by shallow layers with the high-level semantic cues extracted by deep layers, thereby producing a more comprehensive and robust attribution representation (see Fig. 8). In this supplementary section, we provide a detailed explanation of the proposed layer-aggregation mechanism, using the configuration employed in our practical implementation as an illustrative example. Let $\mathbb{L} = \{l_1, l_2, l_3, l_4, l_5\}$ denote the set of layers selected for aggregation, where l_1 corresponds to the shallowest layer and l_5 to the deepest layer closest to the classification head. The specific layer selections for different surrogate models are summarized in Appendix K.5.

Under counterfactual guidance, the cross-layer attribution aggregation procedure is outlined as follows:

(i) Initialization Step: We begin by computing the counterfactual-guided integrated attribution (denoted $\widetilde{IA}$) for each selected layer, yielding $\widetilde{IA}^{cf}_{l_1}$, $\widetilde{IA}^{cf}_{l_2}$, $\widetilde{IA}^{cf}_{l_3}$, $\widetilde{IA}^{cf}_{l_4}$, and $\widetilde{IA}^{cf}_{l_5}$ after diversified counterfactual path integration, followed by normalization. Next, the aggregation process propagates attribution information from deeper to shallower layers, *i.e.*, from l_5 to l_1. We initialize the cross-layer integrated attribution at the deepest layer l_5 as $\widetilde{mIA}^{cf}_{l_5} = IA^{cf}_{l_5}$. Accordingly, the cross-layer attribution for layer l_5 is computed as $\widetilde{mA}^{cf}_{l_5} = \sum_{(l_5)_j \in l_5} \widetilde{mIA}^{cf}_{(l_5)_j} \cdot \Delta^{cf}_{(l_5)_j}$. Following Eq. 11, we further obtain the cross-layer weighted attribution $\widetilde{mWA}^{cf}_{l_5}$.

(ii) Propagation Step: To propagate attribution information to the preceding layer l_4, we compute the gradient of $\widetilde{mWA}^{cf}_{l_5}$ *w.r.t.* the latent feature representation F_{l_4}: $\nabla_{F_{l_4}} \widetilde{mWA}^{cf}_{l_5} = \partial \widetilde{mWA}^{cf}_{l_5} / \partial l_4$. For numerical stability, this gradient is further normalized as $\nabla_{F_{l_4}} \widetilde{mWA}^{cf}_{l_5} / \| \nabla_{F_{l_4}} \widetilde{mWA}^{cf}_{l_5} \|_1$. The cross-layer integrated attribution for layer l_4 is then updated via $\widetilde{mIA}^{cf}_{l_4} = IA^{cf}_{l_4} + \mu \cdot \nabla_{F_{l_4}} \widetilde{mWA}^{cf}_{l_5} / \| \nabla_{F_{l_4}} \widetilde{mWA}^{cf}_{l_5} \|_1$. The resulting $\widetilde{mA}^{cf}_{l_4}$ and $\widetilde{mWA}^{cf}_{l_4}$ are computed analogously to the case of l_5.

(iii) Finalization Step: The above recursive aggregation continues layer-by-layer until reaching the shallowest layer l_1, at which point we obtain $\widetilde{mA}^{cf}_{l_1}$ and its weighted form $\widetilde{mWA}^{cf}_{l_1}$. Finally, the proposed COPIA approach is performed by minimizing the weighted attribution objective $\widetilde{mWA}^{cf}_{l_1}$.

A.5 The Summary of Optimization Algorithm

The overall optimization algorithm of the proposed COPIA approach is summarized in Algorithm 1. To diversify the counterfactual-guided integration paths, we first select a set of C counterfactual samples, denoted by $\mathbb{S}^{cf} = \{\boldsymbol{x}^{cf_1}, \boldsymbol{x}^{cf_2}, \cdots, \boldsymbol{x}^{cf_C}\}$. For each counterfactual sample $\boldsymbol{x} \in \mathbb{S}^{cf}$, we construct discrete counterfactual path sequences through gradient-based descent following Eqs. 5 & 8. This yields a gradient-descent path $\boldsymbol{\gamma}^1_{desc}, \boldsymbol{\gamma}^2_{desc}, \cdots, \boldsymbol{\gamma}^n_{desc}$, and a gradient-ascent path $\boldsymbol{\gamma}^1_{asc}, \boldsymbol{\gamma}^2_{asc}, \cdots, \boldsymbol{\gamma}^n_{asc}$, which together provide diverse nonlinear trajectories for counterfactual path integration.

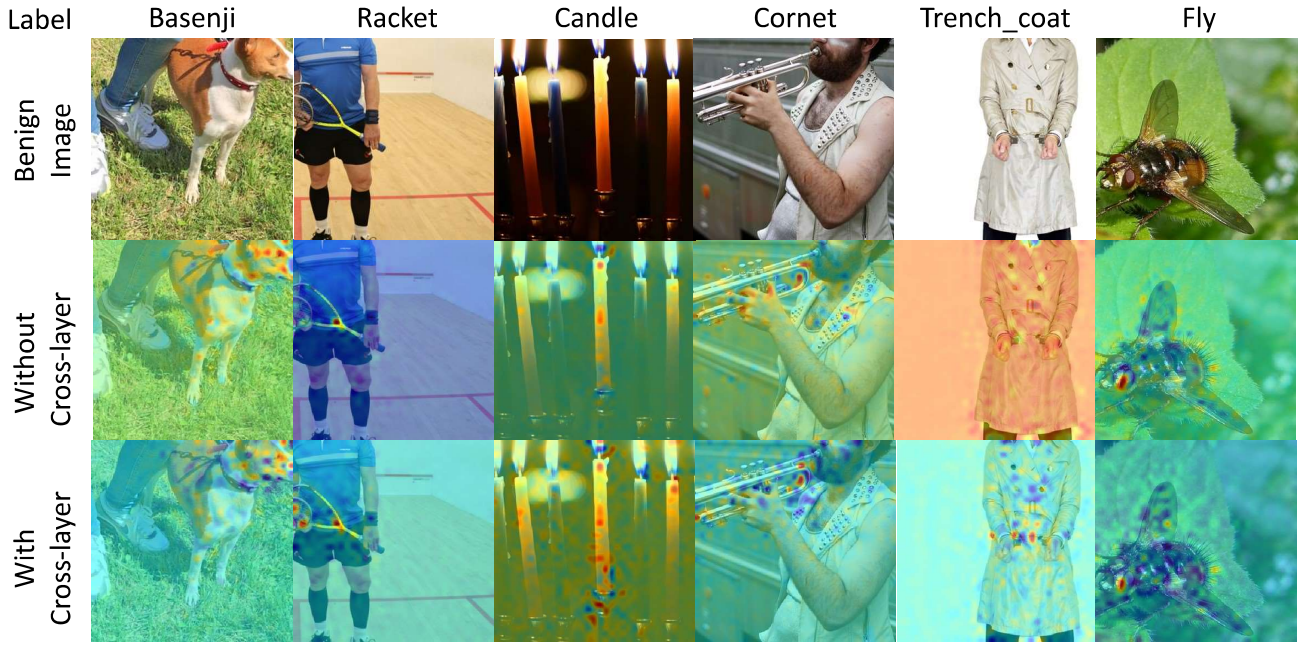


Figure 8: Through the cross-layer attribution aggregation mechanism, high-level semantic information from deeper layers is propagate to shallower layers via the attribution flow. This process suppresses noise in shallow attributions while integrating deep features, resulting in more precise neuron attribution.

Before generating the adversarial example, the first stage computes the layer-wise integrated counterfactual-guided attribution, as described in Lines 2-10. Specifically, for every selected layer, we accumulate the integrated attributions across all counterfactual examples and their associated descent/ascent-based counterfactual paths.

Following this initialization phase, the second stage employs momentum-based iterative gradient descent to generate the adversarial example within a total budget of T iterations. In particular, Lines 13-19 propagate the gradients of the weighted counterfactual-guided attribution *w.r.t.* the representations of the preceding layer, proceeding from the deepest layer to the shallowest. This enables the construction of the overall cross-layer counterfactual-guided neuron attribution loss. Finally, the adversarial example is updated using the momentum-accumulated gradient together with a clipping operator to enforce the perturbation bound.

B Generalization of COPIA to Vision Transformer (ViT) Architectures

Generalization to ViT Architectures. COPIA demonstrates state-of-the-art (SOTA) transferability not only when using CNNs as surrogate models but also when extended to ViT architectures. To evaluate this generalization, we adapt CNN-specific attribution attacks (including FIA, NAA, MFAA, and P2FA) to the ViT framework, and compare them with dedicated ViT-based SOTA methods such as ATT [57] and FPR [59]. For our ViT-adapted COPIA, we maintain hyperparameters consistent with our primary experiments: a learning rate of 0.025 for counterfactual path generation, a set of $C = 5$ counterfactual samples, and a joint strategy for path diversification. The primary modification in our ViT adaptation lies in the cross-layer attribution aggregation mechanism. In CNN architectures, COPIA utilizes an attribution flow that aggregates information from deep to shallow layers. This design leverages the *pyramidal hierarchy* of CNNs, where deep layers capture high-level semantic information, while shallow layers focus on fine-grained textures. By back-propagating attribution flows, we infuse shallow layers with multi-layer semantic guidance. However, unlike CNNs, ViT architectures maintain a *uniform resolution* and capture both local and global information within each block across layers. Empirically, we observe that layer-by-layer aggregation in ViTs can lead to the entanglement of local and global features, which hinders or even degrades transferability. Consequently, for ViT adaptation, COPIA directly aggregates multi-layer neuron attributions at the input space to facilitate information interaction while avoiding inter-layer interference. This design effectively preserves the integrity of multi-scale features and further enhances the transferability of the generated adversarial examples.

Evaluation on ViT-based Surrogate Models. To evaluate the generalization of COPIA across different Transformer-based architectures, we employ three representative models as surrogates: Vit_{base} [6], Deit [34], and Visformer [50]. For each surrogate model, we select representative functional layers for attribution: (1) for ViT_{base}, the layer is {Block2, Block3, Block4}; (2) for Deit, the layer is {Block2, Block3, Block4}; and (3) for Visformer, the layer is {Stage1.Block3, Stage1.Block4, Stage2.Block3}. To ensure a fair and rigorous comparison,we used the ATT

Algorithm 1 The Overall Optimization Algorithm of COPIA

Input: original clean image $\boldsymbol{x}$, class label z, surrogate model F, the set of L' selected layers $\mathbb{L}$, positive and negative transformation function $f_p()$, $f_n()$, hyperparameter λ, perturbation budget ϵ, iteration number T, integrated step n, counterfactual samples $\boldsymbol{x}^{cf_1}, \boldsymbol{x}^{cf_2}, \cdots, \boldsymbol{x}^{cf_C}$.
Output: adversarial example $\boldsymbol{x}_{adv}$

Initialize $\boldsymbol{x}_{adv}^0 = \boldsymbol{x}$, $\alpha = \frac{\epsilon}{T}$, $g_0 = 0$, $\mu = 1$, $\widetilde{IA}^{cf}_{\mathbb{L}[1]} = \cdots = \widetilde{IA}^{cf}_{\mathbb{L}[L']} = 0$.
for $\boldsymbol{y} = \mathbb{L}[L']$ to $\mathbb{L}[1]$ **do**
  for $i = 1$ to C **do**
    Fetch i-th counterfactual example $\boldsymbol{x}^{cf} = \mathbb{S}^{cf}[i]$.
    Obtain the gradient descent-based counterfactual path $\boldsymbol{\gamma}^1_{desc}, \cdots, \boldsymbol{\gamma}^n_{desc}$, via Eq. 5.
    Obtain the gradient ascent-based counterfactual path $\boldsymbol{\gamma}^1_{asc}, \cdots, \boldsymbol{\gamma}^n_{asc}$, via Eq. 8.
    Compute the integrated counterfactual-guided attributions $IA^{cf}_{y_j}(\boldsymbol{\gamma}_{desc})$ and $IA^{cf}_{y_j}(\boldsymbol{\gamma}_{asc})$ respectively.
    Update the layer-wise integrated counterfactual-guided attribution: $\widetilde{IA}^{cf}_{\boldsymbol{y}} = \widetilde{IA}^{cf}_{\boldsymbol{y}} + (IA^{cf}_{y_j}(\boldsymbol{\gamma}_{desc}) + IA^{cf}_{y_j}(\boldsymbol{\gamma}_{asc}))$.
  end for
end for
Initialize $\widetilde{mIA}^{cf}_{\mathbb{L}[1]} = \cdots = \widetilde{mIA}^{cf}_{\mathbb{L}[L']} = 0$
for $t = 1$ to T **do**
  for $\boldsymbol{y} = \mathbb{L}[L']$ to $\mathbb{L}[1]$ **do**
    Compute the representation difference $\widetilde{\Delta}^{cf}_{y_j}(\boldsymbol{\gamma})$ *w.r.t.* C counterfactual-guided paths.
    Compute the diversified path integration-based attribution $\widetilde{mA}^{cf}_{\boldsymbol{y}}$ via Eq. 10.
    Compute the weighted counterfactual-guided attribution $\widetilde{mWA}^{cf}_{\boldsymbol{y}}$ via Eq. 11
    Update the cross-layer integrated attribution $\widetilde{mIA}^{cf}_{\boldsymbol{y}}$ via Eq 12.
  end for
  Compute the overall counterfactual-guided neuron attribution loss $\widetilde{mWA}^{cf}_{\mathbb{L}[1]}$ via Eq 13.
  Update the momentum gradient $\boldsymbol{g}_t = \mu \cdot \boldsymbol{g}_{t-1} + \dfrac{\nabla_{\boldsymbol{x}^t_{adv}} \widetilde{mWA}^{cf}_{\mathbb{L}(1)}}{\|\nabla_{\boldsymbol{x}^t_{adv}} \widetilde{mWA}^{cf}_{\mathbb{L}(1)}\|_1}$
  Update the adversarial example $\boldsymbol{x}^t_{adv} = Clip^{\epsilon}_{x}\{\boldsymbol{x}^{t-1}_{adv} - \alpha \cdot \text{sign}(\boldsymbol{g}_t)\}$
end for
return $\boldsymbol{x}^T_{adv}$

experimental environment and test environment. Moreover all baseline methods strictly adhere to the hyperparameter configurations specified in their original publications. We conduct comprehensive transferability evaluations by launching attacks from these ViT-based surrogates against three types of target models: other ViT variants, normally trained CNNs, and adversarially trained CNNs.

Attack Transferability across ViT Architectures. To evaluate the transferability within Transformer-based architectures, we select seven representative ViT variants as black-box targets: Vit_{base} [6] , Pit [13], Cait [35], Visformer [50],Deit [34], LeViT_{256} [52], and $\text{ConViT}_{\text{base}}$ [51]. The experimental results, detailed in Table 5, demonstrate that COPIA consistently achieves the highest average Attack Success Rate (ASR) across all three ViT-based surrogate models. Specifically, when using Vit_{base} as the surrogate, COPIA outperforms all baseline methods on every black-box ViT target. Notably, for the Pit model, COPIA achieves an ASR of 75.7%, surpassing the state-of-the-art (SOTA) method ATT (62.9%) by a significant margin of 12.8%. Similarly, when using Deit and Visformer as surrogates, COPIA maintains its dominance by achieving the highest ASR for each black-box model. These results provide strong evidence that the adversarial examples generated by COPIA possess exceptional transferability and generalize effectively across a wide range of diverse ViT architectures.

Cross-Architecture Transferability to CNNs. For black-box evaluation on CNN architectures, we select naturally trained models (Inc-v3, Inc-v4, IncRes-v2, and Res-v2_{152}) and adversarially trained variants ($\text{Inc-v3}_{\text{ens3}}$, $\text{Inc-v3}_{\text{ens4}}$, and $\text{IncRes-v2}_{\text{adv}}$) as targets. As shown in Table 6, when using Vit_{base} as the surrogate, we observe that the neuron attribution-based method NAA achieves an average Attack Success Rate (ASR) of 49.3%, outperforming the ViT-specific ATT (47.2%). This highlights the superior *architecture-agnostic compatibility* of the neuron attribution paradigm. In this scenario, COPIA achieves an average ASR of 65.6%, surpassing ATT by a substantial 18.4% and achieving performance comparable to the state-of-the-art (SOTA) FPR (66.4%, with a marginal difference of 0.8%). However, when using Vit_{base} as the surrogate, FPR achieves an average Attack Success Rate (ASR) of only 69.9% on black-box ViT models—a significant gap compared to COPIA's 88.1%. This indicates that COPIA exhibits more balanced and robust performance than FPR. The superiority of COPIA is further emphasized when using Deit as the surrogate, where it achieves the highest ASR on every black-box CNN target. Specifically, COPIA achieves an average ASR of 76.2%, outperforming ATT (68.0%) and FPR (66.0%) by margins of 8.2% and 10.2%, respectively. Furthermore, when using Visformer as the surrogate, COPIA maintains its leading position by achieving the highest ASR on the majority of targets and achieving the best overall

Table 5: Attacking ViT models using ViT as the surrogate model. Among the baselines, FIA, NAA, MFAA, and P2FA are attack methods designed for CNN architectures, while ATT and FPR are tailored for ViT architectures. In the task of transfer-based attacks against ViT models, COPIA achieves state-of-the-art performance compared to these baseline methods.

Model	Method	Vit_{base}	Pit	Cait	Visformer	Deit	$Levit_{256}$	$Convit_{base}$	ASR_{bb}
Vit_{base}	FIA	93.1	50.8	73.3	54.5	72.1	55.3	74.0	63.3
	NAA	87.7	52.9	71.1	55.5	70.3	57.0	71.0	63.0
	MFAA	97.4	41.3	76.5	48.2	77.2	53.6	77.4	62.4
	P2FA	69.6	46.7	44.5	51.2	43.1	54.2	45.3	47.5
	ATT	99.9	62.9	91.4	68.1	91.8	70.6	92.7	79.6
	FPR	82.3	69.7	65.9	73.1	67.2	77.1	66.3	69.9
	COPIA	**100.0**	**75.7**	**97.3**	**80.1**	**97.2**	**81.3**	**97.0**	**88.1**
Deit	FIA	86.0	75.6	94.7	80.2	99.3	81.9	94.4	85.5
	NAA	84.8	79.0	92.3	80.6	97.7	81.5	91.8	85.0
	MFAA	84.2	65.6	94.6	77.5	99.8	83.3	94.1	83.2
	P2FA	54.5	59.8	60.2	62.1	70.6	65.1	59.5	60.2
	ATT	95.9	88.4	99.2	91.1	100.0	92.4	99.1	94.4
	FPR	82.3	69.7	65.9	73.1	67.2	77.1	66.3	72.4
	COPIA	**97.8**	**88.7**	**99.6**	**92.0**	**100.0**	**94.2**	**99.6**	**95.3**
Visformer	FIA	35.7	45.2	44.8	72.7	41.6	50.7	42.9	43.5
	NAA	40.8	50.7	49.8	78.9	48.4	54.6	47.9	48.7
	MFAA	33.3	57.1	48.4	100.0	43.9	59.6	44.0	47.7
	P2FA	44.5	59.4	52.0	94.8	48.7	63.9	46.8	52.6
	ATT	55.4	82.1	78.0	100.0	75.5	87.3	69.4	74.6
	FPR	47.0	61.6	52.9	91.7	52.2	62.2	52.4	54.7
	COPIA	**65.0**	**83.2**	**80.0**	**98.5**	**78.3**	**87.9**	**74.5**	**78.2**

Table 6: Attacking a normally trained and adversarially trained CNN models using ViT as the surrogate model. Among the baselines, FIA, NAA, MFAA, and P2FA are attack methods designed for CNN architectures, while ATT and FPR are tailored for ViT architectures. In the task of transfer-based attacks against CNN normally trained and adversarially trained models, COPIA achieves state-of-the-art performance compared to these baseline methods.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	$Res\text{-}v2_{152}$	$Incv3_{ens3}$	$Incv3_{ens4}$	$IncRes\text{-}v2_{adv}$	ASR_{bb}
Vit_{base}	FIA	50.9	49.1	44.5	48.8	39.2	37.1	34.6	43.5
	NAA	56.2	54.8	50.0	53.7	45.4	44.5	40.7	49.3
	MFAA	54.3	48.9	41.6	48.7	35.3	34.5	25.5	41.3
	P2FA	56.4	50.0	44.8	48.9	36.4	36.8	34.3	43.9
	ATT	57.7	53.8	49.8	54.5	40.2	39.2	35.0	47.2
	FPR	**74.8**	67.7	67.0	68.3	**63.2**	**62.4**	**61.5**	**66.4**
	COPIA	74.5	**74.3**	**69.8**	**71.5**	60.1	57.4	51.3	65.6
Deit	FIA	75.1	70.7	69.5	70.1	60.6	56.6	50.2	64.7
	NAA	78.5	74.0	75.1	75.7	67.5	65.5	60.7	71.0
	MFAA	77.9	72.3	68.9	70.6	56.7	52.5	44.4	63.3
	P2FA	65.6	59.7	56.7	60.6	45.0	43.5	44.2	53.6
	ATT	77.8	74.1	72.9	76.1	63.4	58.1	53.7	68.0
	FPR	74.4	69.3	67.9	68.1	63.0	59.9	59.5	66.0
	COPIA	**86.7**	**84.1**	**82.2**	**83.2**	**69.7**	**68.1**	**59.7**	**76.2**
Visformer	FIA	50.1	50.7	46.3	50.8	39.4	38.0	33.4	44.1
	NAA	56.4	55.5	50.8	55.2	43.7	42.5	37.1	48.7
	MFAA	55.3	52.3	43.7	50.5	30.3	25.9	20.1	39.7
	P2FA	64.3	63.4	54.2	57.0	33.8	34.1	37.0	49.1
	ATT	**87.7**	**87.7**	80.2	82.0	60.1	50.9	39.9	69.8
	FPR	63.4	62.3	54.4	60.8	49.9	46.7	47.5	55.0
	COPIA	85.6	86.5	**81.2**	**82.6**	**64.0**	**61.4**	**51.1**	**73.2**

average ASR. These results further underscore the exceptional cross-architecture transferability and robust generalization of our proposed counterfactual-guided attribution framework.

C Analysis of Cross-Architecture Transferability

To understand why adversarial examples generated from CNN surrogate models can successfully attack ViT models, and vice versa, we visualize the neuron attributions produced by CNNs and ViTs for the same benign images. The results are shown in Fig. 9. Although CNNs and ViTs differ significantly in architecture, they tend to focus on highly overlapping discriminative regions when making predictions. As highlighted by the red rectangles in Fig. 9, the neuron attribution heatmaps generated by the two architectures exhibit substantial similarity. This observation suggests that different model architectures may rely on a common set of discriminative features for classification. Therefore, improving adversarial transferability requires identifying these shared discriminative features across models and perturbing them effectively. Since such features are important to both the surrogate and target models, perturbations guided by them are more likely to transfer successfully, thereby improving the attack success rate on black-box models.

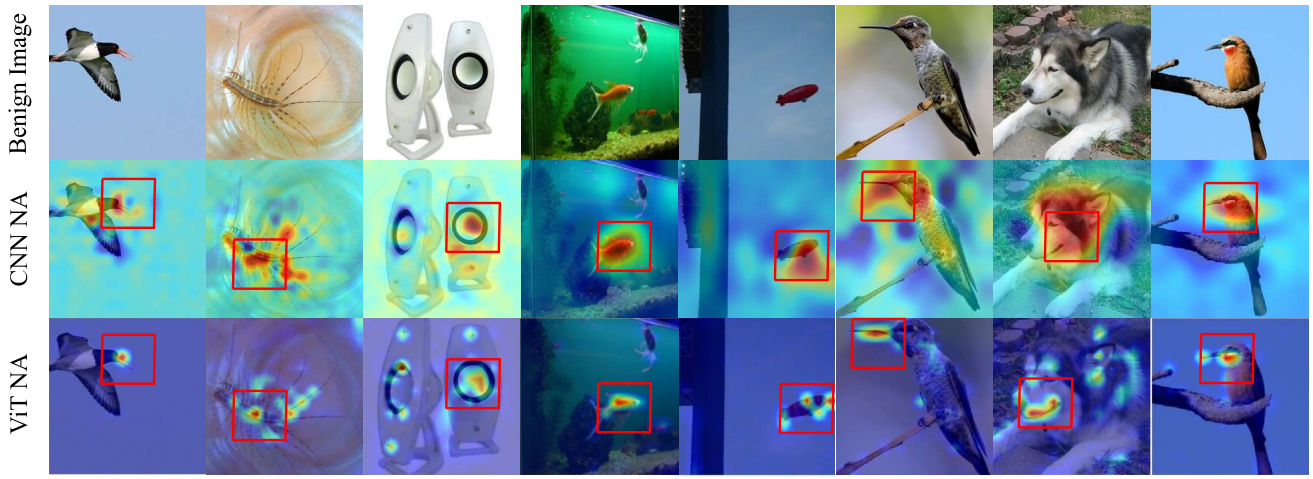


Figure 9: Neuron attributions (NA) generated by CNN and ViT models from the same benign sample exhibit similarity.

D Comparison with SOTA Input Transformation Attacks

Since our method incorporates the input diversity strategy (DIM), we further compare COPIA with two recent state-of-the-art(SOTA) input transformation attacks, SIA [61] and BSR [60], to provide a more comprehensive evaluation. Specifically, we use Inc-v3, Inc-v4, and IncRes-v2 as surrogate models to generate adversarial examples and evaluate their black-box transferability against normally trained models, adversarially trained models, and defense models. The implementations of SIA and BSR are reproduced from their official codebases. The experimental results are reported in Table 7. As shown in Table 7, COPIA consistently achieves superior transferability compared with existing input transformation attacks. Even when compared with recent SOTA methods, COPIA maintains the best overall attack performance, demonstrating the effectiveness of the proposed counterfactual-path framework.

Table 7: Comparison between COPIA and SOTA input transformation-based Attacks. Abbreviations for the target model: "C1: Inc-v3, C2: Inc-v4, C3: IncRes-v2, C4: Res-v1-50, C5:Res-v1-152, C6:Res-v2-50, C7:Res-v2-152, C8:Vgg16, C9: Vgg19. Ad1: Inc-v3-adv, Ad2: IncRes-v2-adv, Ad3: Inc-v3-ens3, Ad4: Inc-v3-ens4, Ad5: IncRes-v2-ens. V1: Pit, V2: Swin, V3: Deit, V4: Cait, V5: Vit-base."

Model	Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	Ad1	Ad2	Ad3	Ad4	Ad5	V1	V2	V3	V4	V5	$ASR_{bb}\uparrow$
Inc-v3	SIA	98.7	89.6	86.3	70.8	61.3	77.9	70.4	65.8	62.5	32.7	37.6	31.9	32.6	16.9	43.5	28.3	25.9	34.5	24.3	49.6
	BSR	**100**	93.5	92.4	85.7	77.9	86.9	85.9	85.1	83.7	53.6	58.6	54.0	53.7	32.1	66.0	45.6	47.2	58.7	44.1	66.9
	COPIA	98.8	**94.3**	**94.2**	**87.1**	**84.8**	**89.6**	**88.1**	**87.0**	**85.0**	**71.9**	**76.3**	**64.1**	**63.7**	**41.9**	**70.4**	**58.2**	**56.7**	**65.5**	**53.8**	**74.0**
Inc-v4	SIA	72.0	98.6	58.8	44.1	33.6	53.5	45.1	42.9	41.1	15.9	15.0	17.6	17.9	9.3	24.8	19.3	14.1	17.2	14.5	30.9
	BSR	94.0	**99.8**	91.4	83.7	74.9	87.3	86.2	87.1	84.1	52.3	60.6	57.1	55.5	36.9	68.1	54.2	49.3	61.2	50.1	68.6
	COPIA	**95.3**	99.0	**93.7**	**87.5**	**83.6**	**89.7**	**89.6**	**88.6**	**87.0**	**71.1**	**76.3**	**70.0**	**66.6**	**51.5**	**73.6**	**65.9**	**59.0**	**69.3**	**58.2**	**76.5**
IncRes-v2	SIA	71.0	64.1	99.6	50.6	40.6	55.5	47.0	50.8	47.4	22.9	26.0	20.3	21.9	11.7	26.8	16.3	14.6	18.7	18.1	34.7
	BSR	95.0	**95.2**	**99.7**	88.6	84.4	91.0	90.4	88.5	86.1	66.5	81.1	73.6	68.6	55.0	73.2	56.3	56.3	68.4	55.8	76.3
	COPIA	**96.3**	94.8	99.6	**90.0**	**88.8**	**91.1**	**91.4**	**88.6**	**88.3**	**79.3**	**87.5**	**76.3**	**71.6**	**64.8**	**76.1**	**64.3**	**62.1**	**72.5**	**61.7**	**80.3**

E Additional Black-box Attack Results on More Surrogate Models

Attack Effectiveness on ResNet-v2 Architectures. To further validate the effectiveness of COPIA, we conduct transferability attacks using Res-$v2_{50}$ and Res-$v2_{152}$ as surrogate models against normally trained, adversarially trained, defended, and Vision Transformer (ViT) models. SVD and P2FA are excluded from this comparison due to significant architectural and parameter discrepancies between their PyTorch-based ResNet implementations and the TensorFlow-based models, which preclude a fair comparison. For consistency, all hyperparameters for COPIA and the competing methods strictly follow those used in their original papers.

Attacking normally trained and adversarially trained models. The results are summarized in Table 8. When using Res-$v2_{50}$ as the surrogate, SMP achieves the highest Attack Success Rate (ASR) on nearly all normally trained models in black-box scenarios. However, the performance gap between COPIA and SMP is marginal (e.g., 91.5% for SMP vs. 91.4% for COPIA on Res-$v1_{152}$). Crucially, COPIA consistently outperforms SMP by a significant margin when attacking adversarially trained models. For instance, COPIA achieves an ASR of 50.8% on IncRes-$v2_{ens3}$, surpassing SMP (39.6%) by 11.2%. COPIA also achieves the highest average black-box ASR across all models. Similar trends are observed when using Res-$v2_{152}$ as the surrogate. While COPIA and SMP show comparable performance on normally trained models, COPIA

Table 8: Using Res-v2$_{50}$ and Res-v2$_{152}$ as surrogate models to attack both normally trained and adversarially trained models. COPIA consistently achieves the highest average Attack Success Rate (ASR) across all settings, demonstrating superior transferability to adversarially trained models.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inc-v3$_{adv}$	IncRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR$_{bb}$
Res-v2$_{50}$	FIA	85.3	82.6	79.5	85.9	80.0	99.4	94.3	80.5	80.3	56.3	52.9	44.4	40.8	25.2	68.3
	NAA	83.7	83.5	81.3	85.9	80.9	98.5	92.8	80.1	80.1	61.7	61.6	58.3	54.6	40.1	72.7
	DANAA	86.8	85.3	84.4	87.6	85.1	98.9	94.7	83.0	82.2	63.2	64.4	58.7	55.7	40.7	74.8
	NEAA	88.9	86.9	84.7	89.6	85.9	99.2	96.3	84.1	83.7	64.4	65.5	58.5	55.5	38.4	75.6
	MFAA	89.6	87.2	86.7	90.8	86.6	**99.8**	95.8	85.5	85.6	69.8	67.1	60.0	54.3	37.6	76.7
	SMP	**92.8**	**92.0**	**91.1**	**94.0**	**91.5**	99.7	97.3	**90.5**	**90.6**	69.3	71.6	64.0	56.2	39.6	80.0
	COPIA	92.4	91.8	90.6	93.2	91.4	98.4	**97.6**	90.2	88.6	**74.8**	**74.5**	**71.9**	**66.9**	**50.8**	**82.7**
Res-v2$_{152}$	FIA	83.5	81.9	78.5	83.7	79.3	94.2	99.2	78.4	78.7	58.8	58.1	51.5	46.0	28.1	69.3
	NAA	83.1	83.0	81.0	82.7	79.6	91.9	97.1	79.4	75.2	61.9	64.7	58.0	56.4	44.7	72.4
	DANAA	86.2	86.0	84.7	85.3	81.1	94.9	98.5	80.0	77.4	66.3	69.6	63.5	59.8	45.3	75.4
	NEAA	87.5	85.6	85.8	88.1	83.9	95.2	99.0	81.8	81.8	66.2	67.7	61.7	56.6	43.7	75.8
	MFAA	86.5	84.3	85.2	87.5	84.7	95.4	99.7	81.5	81.2	66.1	68.0	62.0	56.6	40.1	75.3
	SMP	**92.1**	**91.8**	90.3	**92.6**	88.9	97.4	99.6	87.7	**86.5**	68.0	73.7	63.9	58.9	43.7	79.7
	COPIA	91.8	91.7	**90.6**	91.8	**89.9**	**97.6**	98.9	**87.9**	86.3	**74.2**	**76.6**	**70.5**	**67.1**	**53.5**	**82.3**

Table 9: Using Res-v2$_{50}$ and Res-v2$_{152}$ as surrogate models to attack both defended and ViT models. COPIA consistently achieves the highest average Attack Success Rate (ASR) across all settings, demonstrating superior transferability to ViT models.

Model	Method	Comdefend	FD	HGD	JPEG	NIPS-r3	R&P	RS	DEF-ASR	Pit	Swin	Deit	Cait	Vit$_{base}$	ViT-ASR
Res-v2$_{50}$	FIA	81.7	78.7	28.7	79.7	34.3	26.7	32.4	51.7	45.1	31.5	29.9	35.6	37.3	35.9
	NAA	81.7	80.3	46.2	80.5	50.4	41.9	35.3	59.5	54.6	43.3	41.0	49.8	44.8	46.7
	DANAA	83.2	82.6	46.6	83.4	51.4	41.6	34.6	60.5	53.9	41.7	40.3	50.1	43.7	45.9
	NEAA	85.4	85.3	45.2	85.3	49.3	40.7	36.0	61.0	55.5	44.4	42.3	50.8	45.8	47.8
	MFAA	86.9	86.4	41.8	86.6	51.2	40.3	35.9	61.3	54.9	39.1	39.8	49.9	48.4	46.4
	SMP	**90.6**	89.2	54.6	**89.8**	52.8	44.2	34.0	65.0	60.9	46.8	44.2	54.9	50.9	51.5
	COPIA	89.2	**89.4**	**56.3**	89.7	**62.8**	**53.5**	**37.9**	**68.4**	**67.4**	**54.2**	**52.5**	**61.2**	**52.6**	**57.6**
Res-v2$_{152}$	FIA	78.5	77.6	33.7	78.9	38.4	29.6	34.1	53.0	44.7	34.5	31.6	39.1	37.7	37.5
	NAA	78.3	77.1	48.4	77.9	50.7	44.8	38.4	59.4	55.2	44.4	41.5	49.1	44.2	46.9
	DANAA	80.6	79.4	50.7	80.3	54.3	45.5	34.2	60.7	54.9	47.0	42.0	50.3	43.9	47.6
	NEAA	82.4	81.0	49.0	82.5	53.4	43.9	36.1	61.2	55.2	46.6	41.6	50.4	46.0	48.0
	MFAA	83.2	82.0	41.8	82.2	50.4	39.8	36.9	59.5	54.3	41.5	40.3	48.3	46.5	46.2
	SMP	**88.2**	**88.2**	55.9	**88.3**	56.7	46.2	34.0	65.4	60.6	51.5	47.9	57.9	49.6	53.5
	COPIA	87.1	87.2	**60.1**	87.0	**64.1**	**55.1**	**36.7**	**68.2**	**66.1**	**56.2**	**52.7**	**62.7**	**54.9**	**58.5**

achieves the state-of-the-art (SOTA) ASR on every adversarially trained model and maintains a superior average black-box ASR. These results underscore COPIA's robust transferability, particularly against advanced defense mechanisms.

Attacking normally trained ViT and defended models. The experimental results are summarized in Table 9. When employing Res-v2$_{50}$ and Res-v2$_{152}$ as surrogate models, COPIA and SMP exhibit competitive performance across various defended models, with each achieving superior Attack Success Rates (ASR) on different targets. Notably, however, COPIA attains a higher overall average ASR than SMP. A more significant advantage is observed when evaluating transferability to black-box ViT models. Under both Res-v2 surrogates, COPIA achieves the highest ASR across all ViT variants. Specifically, COPIA achieves average ASRs of 57.6% and 58.5% for Res-v2$_{50}$ and Res-v2$_{152}$, respectively, outperforming the state-of-the-art (SOTA) SMP (51.5% and 53.5%) by substantial margins of 6.1% and 5.0%. These results underscore the robustness and stability of COPIA in enhancing transferability across heterogeneous model architectures.

F Comparative Analysis of ASR under PIDI Data Augmentation

Integration with Data Augmentation. In the field of adversarial attacks, data augmentation (DA) techniques significantly enhance the transferability of adversarial examples and are generally compatible with various attack frameworks. In this subsection, we integrate the state-of-the-art DA method, PIDI, into COPIA and other baseline methods to evaluate their combined performance. Consistent with the experimental protocols in the main text, we utilize five surrogate models to assess black-box transferability against normally trained, adversarially trained, defended, and Vision Transformer (ViT) models. Due to significant architectural incompatibilities, SVD and P2FA are excluded from comparisons involving the ResNet-v2 series surrogates (Res-v2$_{50}$ and Res-v2$_{152}$).

Attacking normally trained and adversarially trained models with the PIDI data augmentation method. The results, summarized in Table 10, demonstrate that COPIA-PIDI consistently achieves the highest average black-box Attack Success Rate (ASR) across all five surrogate models. Notably, COPIA-PIDI outperforms all baseline methods on adversarially trained models, indicating that the adversarial examples generated by our method effectively withstand sophisticated defense mechanisms and exhibit superior robustness. Furthermore, when using Inc-v4 and IncRes-v2 as surrogates, COPIA-PIDI achieves the highest ASR on nearly all normally trained black-box models. For Inc-v3, Res-v2$_{50}$, and Res-v2$_{152}$ as surrogates, COPIA-PIDI delivers competitive performance on normally trained models, showing only a marginal gap compared to existing state-of-the-art (SOTA) methods.

Table 10: Integrating the PIDI data augmentation method into our attacks and baseline methods on both naturally and adversarially trained models. COPIA consistently achieves the highest average Attack Success Rate (ASR) across all five surrogate models.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inc-v3$_{adv}$	IncRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR$_{bb}$
Inc-v3	FIA-PIDI	98.5	87.8	85.5	82.7	78.2	84.1	80.3	82.9	83.6	60.2	57.4	38.9	38.5	20.7	67.8
	NAA-PIDI	97.0	86.9	85.2	79.3	76.8	82.0	80.9	81.9	80.7	63.0	64.6	53.0	53.4	34.3	70.9
	DANAA-PIDI	97.8	89.2	88.0	83.4	81.1	84.4	83.4	83.8	82.9	67.6	69.2	55.5	55.8	35.1	73.8
	NEAA-PIDI	98.9	92.3	91.3	87.8	84.4	89.4	87.8	87.3	87.8	72.1	71.8	58.9	56.7	37.1	77.3
	SVD-PIDI	99.1	86.4	84.1	79.8	72.6	81.5	77.3	79.4	78.4	64.2	64.5	57.5	59.3	44.4	71.5
	MFAA-PIDI	98.3	90.3	89.7	87.1	84.5	88.6	85.2	86.5	85.7	65.9	69.2	49.8	51.0	30.0	74.1
	SMP-PIDI	99.2	92.2	92.5	87.9	82.8	90.0	87.1	89.1	88.0	62.6	67.2	51.0	49.5	30.9	74.7
	P2FA-PIDI	**100.0**	**97.9**	**96.3**	90.3	85.9	**94.0**	**91.9**	90.1	89.1	64.9	71.5	47.7	48.4	27.5	76.6
	COPIA-PIDI	99.4	94.5	94.7	**91.6**	**90.3**	91.5	91.5	**91.2**	**90.0**	**78.3**	**81.9**	**70.9**	**72.2**	**52.3**	**83.9**
Inc-v4	FIA-PIDI	90.1	96.9	87.8	83.0	79.5	86.2	84.7	86.1	84.6	56.5	60.1	44.9	42.6	23.1	69.9
	NAA-PIDI	89.6	96.9	84.6	80.5	75.4	83.0	81.8	82.0	80.0	56.1	62.3	52.8	52.5	35.4	70.5
	DANAA-PIDI	87.9	96.3	85.8	81.9	77.3	84.5	83.7	83.6	82.0	60.8	67.8	57.6	56.2	37.3	72.8
	NEAA-PIDI	94.4	98.7	90.4	88.0	85.5	89.7	88.4	88.1	89.0	74.0	74.8	67.3	65.2	47.9	80.2
	SVD-PIDI	87.1	97.0	81.5	79.8	71.2	81.6	77.2	82.0	80.6	59.4	61.1	61.1	61.3	48.8	71.7
	MFAA-PIDI	87.7	94.4	84.8	81.9	79.4	84.9	82.4	83.4	81.6	62.6	66.1	55.9	54.8	35.0	72.3
	SMP-PIDI	94.4	98.6	92.3	86.8	83.5	90.1	89.1	90.4	90.1	59.2	66.2	53.0	52.1	35.1	75.6
	P2FA-PIDI	**96.4**	99.7	92.6	82.9	75.9	89.8	87.0	85.4	84.6	54.1	60.8	40.4	41.3	26.9	70.6
	COPIA-PIDI	96.3	**99.0**	**93.8**	**90.5**	**87.9**	**92.1**	**91.9**	**91.3**	**91.0**	**74.1**	**80.3**	**75.4**	**71.7**	**57.0**	**84.1**
IncRes-v2	FIA-PIDI	88.1	84.7	91.8	82.7	79.3	84.2	82.0	81.8	83.6	67.8	69.4	52.1	47.9	29.4	71.8
	NAA-PIDI	84.9	81.3	92.5	79.2	75.1	80.8	78.3	79.2	79.0	63.2	67.7	59.6	57.0	47.5	71.8
	DANAA-PIDI	84.0	82.5	90.2	78.9	77.4	80.6	79.6	80.2	78.9	67.8	72.6	64.2	62.2	52.7	74.0
	NEAA-PIDI	93.2	88.6	97.5	88.8	86.0	87.5	86.1	87.0	87.3	78.8	79.3	71.2	66.3	55.4	81.2
	SVD-PIDI	86.8	82.7	95.1	78.2	73.9	79.3	77.9	78.5	77.0	63.1	68.6	64.4	62.2	55.7	72.9
	MFAA-PIDI	88.0	86.0	93.9	84.6	82.9	85.8	83.4	83.6	83.4	72.0	74.6	63.8	59.0	44.2	76.3
	SMP-PIDI	89.6	86.1	93.7	85.9	83.8	86.5	85.6	85.1	86.8	70.2	74.7	63.6	58.8	48.3	77.3
	P2FA-PIDI	93.7	93.3	99.8	87.4	82.4	90.5	88.1	86.2	85.1	64.4	73.5	48.7	45.9	34.3	74.9
	COPIA-PIDI	**97.7**	**96.4**	99.5	**94.4**	**92.9**	**94.9**	**94.3**	**93.5**	**93.0**	**84.0**	**90.1**	**82.3**	**78.9**	**70.3**	**89.4**
Res-v2$_{50}$	FIA-PIDI	85.1	82.9	79.6	84.9	79.9	99.5	93.8	80.7	80.4	55.8	54.7	44.2	39.5	24.7	68.2
	NAA-PIDI	90.6	88.7	88.6	92.3	91.0	98.1	95.1	90.1	87.9	71.8	73.7	68.3	63.7	47.9	80.7
	DANAA-PIDI	93.2	91.0	90.9	94.0	92.2	98.6	96.4	91.1	90.8	74.0	76.3	69.4	66.0	46.8	82.5
	NEAA-PIDI	94.7	93.3	92.9	95.9	94.1	99.2	**97.9**	93.4	93.1	76.9	77.1	70.3	66.4	48.5	84.2
	MFAA-PIDI	94.4	92.7	92.5	95.6	93.8	99.6	96.9	93.0	93.2	77.7	78.5	68.0	63.2	42.7	83.2
	SMP-PIDI	**96.6**	**95.6**	**95.7**	**97.5**	95.3	**99.9**	**98.2**	**96.0**	**95.6**	79.4	81.6	74.4	69.0	53.0	86.8
	COPIA-PIDI	95.8	95.1	94.4	96.6	**95.8**	98.6	97.9	94.3	93.5	**83.1**	**83.7**	**81.6**	**75.7**	**60.2**	**88.3**
Res-v2$_{152}$	FIA-PIDI	89.7	87.9	88.1	92.6	89.9	96.3	98.9	89.3	88.6	69.5	70.8	57.7	56.5	36.4	77.9
	NAA-PIDI	89.4	88.5	87.7	90.1	88.4	94.9	97.9	87.9	86.1	71.0	75.4	69.0	65.1	53.0	80.5
	DANAA-PIDI	90.8	90.2	89.7	91.8	89.8	95.3	98.4	88.0	86.6	75.6	78.8	73.0	70.3	54.1	82.6
	NEAA-PIDI	93.1	91.6	93.0	94.3	92.6	96.8	99.3	92.1	90.9	77.5	79.6	71.0	68.4	54.0	84.2
	MFAA-PIDI	92.2	90.8	91.1	94.6	93.1	97.0	99.7	90.5	89.9	75.1	77.3	67.3	62.7	45.3	82.1
	SMP-PIDI	**95.6**	**95.2**	**95.1**	**96.3**	95.0	**97.7**	99.3	**95.0**	**95.0**	79.8	82.9	77.5	72.1	60.5	87.5
	COPIA-PIDI	94.7	93.9	95.0	95.4	**95.8**	97.4	98.9	92.9	91.7	**82.7**	**85.1**	**79.7**	**76.3**	**62.3**	**87.9**

Attacking defended and ViT models with the PIDI data augmentation method. The experimental results for attacking defended models and Vision Transformers (ViTs) using the PIDI data augmentation method are summarized in Table 11. Overall, COPIA-PIDI consistently achieves the highest average Attack Success Rate (ASR) against both defended models (DEF-ASR) and ViT models (ViT-ASR) across all five surrogate models. Specifically, when employing Inc-v3, Inc-v4, and IncRes-v2 as surrogates, COPIA-PIDI achieves the highest ASR on nearly all defended and ViT targets, with the sole exception of the HGD defense. Under these three surrogates, COPIA-PIDI achieves average ASRs of 71.9%, 73.6%, and 78.1% on ViT models, respectively. Compared to the current state-of-the-art (SOTA) NEAA-PIDI (60.4%, 66.4%, and 62.6%), our method achieves performance gains of 11.5%, 7.2%, and 15.5%, respectively. Furthermore, when using Res-v2$_{50}$ and Res-v2$_{152}$ as surrogates, although COPIA-PIDI and SMP-PIDI each achieve the best ASR on different individual target models, COPIA-PIDI demonstrates greater overall performance in terms of average ASR. These findings confirm that COPIA effectively synergizes with data augmentation techniques to further enhance transferability across diverse and robust architectures.

G Attacking Image Captioning Models

In recent years, image captioning models have been increasingly adopted in real-world applications. To provide a more comprehensive evaluation of attack effectiveness from a multimodal perspective, we further assess different attack methods on the image captioning model BLIP [54]. Specifically, adversarial examples generated from surrogate models are used to attack BLIP, and the similarity between captions generated from benign images and their corresponding adversarial images is measured using BLEU [58] and ROUGE [55]. Lower BLEU and ROUGE scores indicate lower similarity between the generated captions, implying stronger attack effectiveness.

In this experiment, we use Inc-v3, Inc-v4, and IncRes-v2 as surrogate models and evaluate a total of 1,000 benign–adversarial image pairs. The quantitative results are reported in Table 12, COPIA yields superior attack effectiveness against image-to-text models, which showcases the overall attack capability of its generated adversarial examples. When Inc-v3 is used as the surrogate model, P2FA achieves the best performance, while COPIA obtains the second-best results with only a small performance gap. When Inc-v4 and IncRes-v2 are used as surrogate models, COPIA achieves the lowest BLEU and ROUGE scores, indicating the strongest attack effectiveness. In addition to

Table 11: Integrating the PIDI data augmentation method into our attacks and baseline methods on both defended and ViT models. COPIA consistently achieves the highest average Attack Success Rate (ASR) across all five surrogate models.

Model	Method	Comdefend	FD	HGD	JPEG	NIPS-r3	R&P	RS	DEF-ASR	Pit	Swin	Deit	Cait	Vit_{base}	ViT-ASR
Inc-v3	FIA-PIDI	80.4	79.0	4.9	80.1	32.2	20.8	35	47.5	53.4	36.3	38.2	47.7	48.7	44.9
	NAA-PIDI	77.5	75.5	10.9	76.5	46.8	36.2	36	51.3	64.0	49.4	51.1	59.5	51.9	55.2
	DANAA-PIDI	79.3	79.1	9.8	79.1	48.9	36.4	37.6	52.9	65.5	52.7	50.9	62.1	54.8	57.2
	NEAA-PIDI	84.9	83.6	10.9	84	51.6	40.1	39.9	56.4	68.5	55.1	54.7	63.5	60.1	60.4
	SVD-PIDI	73.2	72.7	**57.3**	72.5	48.7	45.6	38.9	58.4	54.5	48.1	41.1	48.7	42.1	46.9
	MFAA-PIDI	84.1	84.4	8.8	84.5	46.4	31.3	37.9	53.9	65.8	50.1	51.6	62.0	58.5	57.6
	SMP-PIDI	85.4	84.0	11.8	84.7	45.8	33.8	35.3	54.4	66.2	49.8	49.6	58.1	59.4	56.6
	P2FA-PIDI	85.8	83.8	32.8	84.7	41.1	29.6	42.4	57.2	66.5	55.5	47.2	59.7	54.1	56.6
	COPIA-PIDI	**88.5**	**88.1**	26.4	**88.6**	**68.4**	**55.3**	**44.9**	**65.7**	**79.7**	**67.6**	**68.3**	**76.8**	**67.0**	**71.9**
Inc-v4	FIA-PIDI	79.1	79.3	7.6	80.0	36.6	23.8	33.7	48.6	65.5	55.2	47.2	61.3	54.6	56.8
	NAA-PIDI	76.3	75.8	16.5	75.3	48.4	38.0	34.1	52.1	69.1	59.1	55.9	65.5	55.1	60.9
	DANAA-PIDI	78.1	77.1	15.7	77.6	51.6	41.6	34.3	53.7	69.7	62.7	55.3	66.5	57.1	62.3
	NEAA-PIDI	79.5	80.2	14.9	79.9	59.9	38.4	37.5	55.8	73.7	64.0	59.2	69.7	65.3	66.4
	SVD-PIDI	72.7	74.0	**58.3**	73.4	50.5	49.9	38.4	59.6	58.5	52.8	47.1	53.0	45.1	51.3
	MFAA-PIDI	84.9	84.7	23.4	85.1	48.9	50.5	41.4	59.8	65.2	54.3	50.8	61.5	59.0	58.2
	SMP-PIDI	85.2	84.7	19.5	85.1	48.8	38.6	36.3	56.9	71.8	62.1	56.2	67.8	59.3	63.4
	P2FA-PIDI	78.3	74.3	31.6	77.0	33.4	26.3	42.1	51.9	60.4	54.5	42.5	52.2	46.4	51.2
	COPIA-PIDI	**87.1**	**87.0**	41.1	**87.2**	**69.9**	**61.6**	**43.4**	**68.2**	**81.1**	**71.4**	**68.8**	**78.1**	**68.7**	**73.6**
IncRes-v2	FIA-PIDI	80.5	79.4	13.3	79.0	45.0	32.5	37.9	52.5	58.9	43.5	41.9	52.5	53.9	50.1
	NAA-PIDI	74.5	73.8	25.1	74.9	56.7	50.0	37.1	56.0	64.6	52.8	53.2	61.9	56.6	57.8
	DANAA-PIDI	76.2	75.4	27.1	76.2	61.0	55.1	38.6	58.5	62.2	58.4	56.9	63.9	59.7	60.2
	NEAA-PIDI	81.9	81.6	23.1	82.1	59.3	48.8	39.2	59.4	69.5	53.8	57.2	65.4	67.3	62.6
	SVD-PIDI	73.8	72.2	**61.4**	71.8	56.8	55.3	39.6	61.6	58.9	52.8	46.1	54.1	45.0	51.4
	MFAA-PIDI	86.8	86.8	29.6	86.7	64.1	56.7	45.8	65.2	67.5	54.8	54.7	63.5	64.9	61.1
	SMP-PIDI	83.8	83.8	28.0	83.7	59.3	53.0	39.1	61.5	65.7	50.9	54.6	60.8	64.4	59.3
	P2FA-PIDI	82.1	78.5	31.9	80.3	43.3	33.0	40.5	55.7	61.1	50.2	43.2	51.8	51.9	51.6
	COPIA-PIDI	**89.5**	**90.2**	50.6	**90.5**	**79.1**	**73.3**	**48.2**	**74.5**	**84.2**	**74.6**	**74.2**	**81.9**	**75.8**	**78.1**
$Res\text{-}v2_{50}$	FIA-PIDI	82.6	79.4	28.4	80.3	35.1	24.9	32.7	51.9	44.4	29.6	29.5	37.1	37.4	35.6
	NAA-PIDI	89.9	90.2	33.9	90.4	61.1	51.8	39.9	65.3	70.5	56.7	58.8	66.7	62.3	63.0
	DANAA-PIDI	90.8	92.1	31.3	92.6	62.7	52.3	40.8	66.1	70.8	57.9	58.4	67.5	61.9	63.3
	NEAA-PIDI	93.1	92.7	32.2	92.9	61.7	51.6	42.9	66.7	72.7	57.7	59.5	68.2	65.0	64.6
	MFAA-PIDI	93.2	92.7	30.2	92.8	58.2	47.2	42.1	65.2	69.0	52.3	55.9	65.7	63.2	61.2
	SMP-PIDI	**95.8**	**95.8**	44.8	**95.9**	70.8	62.4	40.5	72.3	77.7	63.0	63.3	73.8	**72.6**	70.1
	COPIA-PIDI	94.2	94.1	**48.7**	94.5	**75.6**	**66.1**	**45.3**	**74.1**	**79.1**	**65.6**	**68.3**	**76.0**	72.2	**72.2**
$Res\text{-}v2_{152}$	FIA-PIDI	89.8	88.1	22.1	88.4	49.8	37.0	40.8	59.4	60.8	45.3	44.6	54.6	56.3	52.3
	NAA-PIDI	85.8	87.9	40.3	88.2	63.2	54.8	41.5	66.0	70.1	58.1	57.5	66.7	61.0	62.7
	DANAA-PIDI	89.8	88.1	22.1	88.4	49.8	37.0	40.8	59.4	69.6	62.2	59.3	67.7	61.8	64.1
	NEAA-PIDI	90.3	91.4	39.1	91.3	65.2	56.2	44.7	68.3	71.1	61.4	60.5	69.2	65.4	65.5
	MFAA-PIDI	91.3	91.5	32.1	91.8	60.3	48.6	42.1	65.4	66.7	54.3	53.0	64.9	64.0	60.6
	SMP-PIDI	**94.4**	**94.8**	50.3	**94.9**	73.8	66.8	42.4	73.9	76.6	**67.1**	63.4	74.2	**74.2**	71.1
	COPIA-PIDI	93.6	93.4	**53.0**	93.3	**77.4**	**67.3**	**46.7**	**75.0**	**78.6**	66.5	**69.2**	**77.4**	70.4	**72.4**

Table 12: Comparison of attack effectiveness on multimodal image-to-text models. The best results are highlighted in red, and the second-best results are highlighted in blue.

Model	Metrics	FIA	NAA	DANAA	NEAA	MFAA	SVD	P2FA	SMP	COPIA
Inc-v3	BLEU	0.3550	0.3712	0.3522	0.3509	0.3349	0.4408	**0.2941**	0.3368	0.3056
	ROUGE	0.5724	0.5902	0.5727	0.5714	0.5559	0.6417	**0.5241**	0.5591	0.5307
Inc-v4	BLEU	0.3508	0.3678	0.4358	0.3124	0.3201	0.4381	0.3067	0.3203	**0.2845**
	ROUGE	0.5680	0.5842	0.6937	0.5430	0.5442	0.6389	0.5343	0.5441	**0.5099**
IncRes-v2	BLEU	0.3497	0.3163	0.3540	0.3162	0.3391	0.4325	0.3075	0.2953	**0.2849**
	ROUGE	0.5712	0.5449	0.5737	0.5449	0.5597	0.6365	0.5357	0.5251	**0.5137**

the statistical comparison of BLEU and ROUGE scores over 1,000 images, we also provide a more intuitive comparison by presenting the texts generated by the BLIP image-to-text model before and after applying adversarial examples crafted by different attack methods, for the same input image. The experimental results are shown in Fig. 10. These results demonstrate that COPIA is effective not only against image classification models but also against image captioning models, suggesting that the proposed method generalizes well to multimodal vision-language tasks.

H Additional Analysis of Single-Layer vs. Cross-Layer Attribution Aggregation

We enhance single-layer neuron attribution attacks—NAA, DANAA, NEAA, and S-COPIA (the single-layer variant of COPIA)—using our proposed cross-layer attribution aggregation mechanism, producing NAA_{CA}, $DANAA_{CA}$, $NEAA_{CA}$, and the full COPIA. To ensure a fair comparison, Gaussian noise with zero mean and variance 0.2 is added during the integration process of NAA_{CA}, $DANAA_{CA}$, and $NEAA_{CA}$, consistent with COPIA, and the learning rate for $DANAA_{CA}$ is set to 0.025. Table 13 presents the results on normally and adversarially

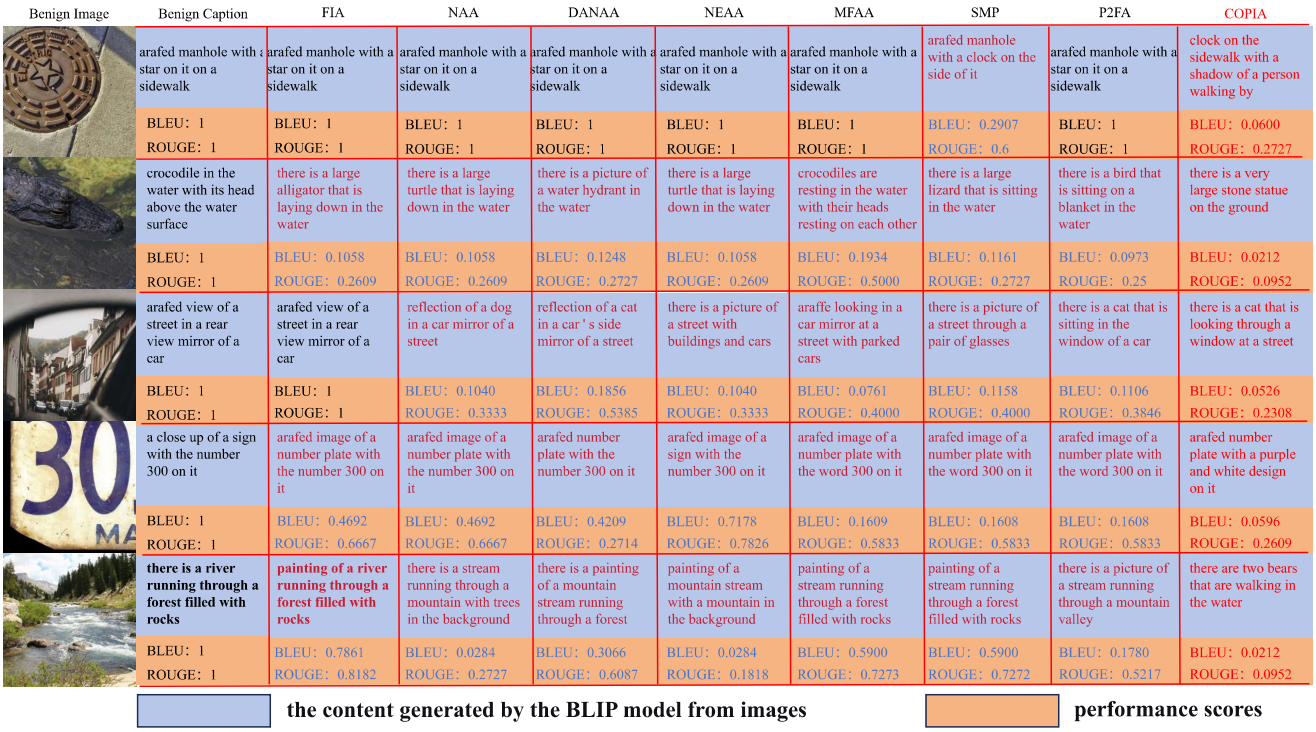


Figure 10: Example BLIP-generated texts from benign images and adversarial examples of different attacks, with BLEU/ROUGE metrics. COPIA yields the strongest attack, causing the most substantial changes in the output text.

Table 13: Comparison of attack success rates (ASR, %) before and after cross-layer attribution aggregation for various single-layer neuron attribution attacks on normally trained and adversarially trained models. Asterisks (*) indicate white-box settings. When equipped with joint cross-layer attribution aggregation, the transferability of each single-layer neuron attribution attack improves across different surrogate models, and COPIA consistently achieves the highest overall average ASR.

Model	Method	Inc-v3	Inc-v4	IncRes-v2	Res-v1$_{50}$	Res-v1$_{152}$	Res-v2$_{50}$	Res-v2$_{152}$	Vgg16	Vgg19	Inc-v3$_{adv}$	IncRes-v2$_{adv}$	Inc-v3$_{ens3}$	Inc-v3$_{ens4}$	IncRes-v2$_{ens3}$	ASR$_b$	ASR$_{bb}$
Inc-v3	NAA	96.6*	83.3	80.8	71.6	65.1	75.5	75.4	73.0	70.6	56.9	58.1	49.1	50.8	31.4	67.0	64.7
	NAA$_{CA}$	97.4*	84.8	84.5	76.0	71.4	78.8	77.9	76.3	75.9	59.8	63.5	52.9	52.9	34.6	70.5	68.4
	DANAA	97.7*	86.7	84.1	75.2	70.7	79.5	78.6	74.8	72.5	64.4	66.1	54.0	53.7	32.0	70.7	68.6
	DANAA$_{CA}$	98.5*	91.5	89.8	79.6	74.2	83.5	82.3	77.1	75.0	60.5	64.8	48.0	49.3	27.0	71.5	69.4
	NEAA	99.1*	87.7	87.4	80.2	73.6	82.7	80.2	78.6	77.3	65.6	64.5	50.9	53.1	32.0	72.4	70.3
	NEAA$_{CA}$	99.1*	91.9	92.1	83.7	80.9	85.9	85.4	82.5	81.4	69.1	70.2	57.0	58.3	35.1	76.6	74.9
	S-COPIA	**99.3***	90.5	88.7	81.8	77.9	84.8	82.9	82.7	80.0	69.9	70.8	58.0	57.4	36.3	75.8	74.0
	COPIA	98.8*	**94.3**	**94.2**	**87.1**	**84.8**	**89.6**	**88.1**	**87.0**	**85.0**	**71.9**	**76.3**	**64.1**	**63.7**	**41.9**	**80.5**	**79.1**
IncRes-v2	NAA	79.0	75.9	90.6*	68.2	66.0	71.9	70.3	68.7	65.7	56.6	63.6	55.0	52.6	44.4	66.3	64.5
	NAA$_{CA}$	83.6	80.2	95.8*	73.8	70.3	77.3	75.1	72.9	71.6	59.6	67.0	57.8	56.0	48.6	70.7	68.8
	DANAA	82.3	79.6	91.4*	72.9	68.7	76.0	75.4	71.2	70.5	64.2	71.7	63.4	58.8	54.0	71.4	69.9
	DANAA$_{CA}$	94.3	92.7	99.5*	83.4	81.0	87.4	86.2	79.6	77.9	66.4	80.6	65.6	60.3	51.3	79.0	77.4
	NEAA	88.6	83.8	95.8*	79.9	76.6	83.0	79.7	79.1	77.9	72.0	75.0	65.8	60.0	52.7	76.4	74.9
	NEAA$_{CA}$	92.3	89.3	99.2*	84.6	81.2	85.4	85.5	81.5	80.2	72.0	79.5	70.5	63.9	56.2	80.1	78.6
	S-COPIA	92.7	90.7	97.5*	85.0	81.7	86.4	86.3	82.1	81.9	76.3	82.1	72.9	67.0	61.4	81.7	80.5
	COPIA	**96.3**	**94.8**	**99.6***	**90.0**	**88.8**	**91.1**	**91.4**	**88.6**	**88.3**	**79.3**	**87.5**	**76.3**	**71.6**	**64.8**	**86.3**	**85.3**
Res-v2$_{50}$	NAA	83.7	83.5	81.3	85.9	80.9	98.5*	92.8	80.1	80.1	61.7	61.6	58.3	54.6	40.1	74.5	72.7
	NAA$_{CA}$	87.8	85.3	85.1	90.0	86.9	97.6*	94.0	86.2	83.9	68.2	67.5	64.8	60.6	46.9	78.9	77.5
	DANAA	86.2	86.0	84.7	85.3	81.1	98.5*	94.9	80.0	77.4	66.3	69.6	63.5	59.8	45.3	77.0	75.4
	DANAA$_{CA}$	89.4	86.9	85.9	90.0	86.4	97.7*	96.8	83.3	82.2	63.4	65.0	62.8	58.0	40.9	77.8	76.2
	NEAA	88.9	86.9	84.7	89.6	85.9	**99.2***	96.3	84.1	83.7	64.4	65.5	58.5	55.5	38.4	77.3	75.6
	NEAA$_{CA}$	**92.5**	89.5	89.8	91.7	89.2	98.6*	97.5	86.6	87.0	66.8	69.6	63.4	61.4	44.3	80.6	79.2
	S-COPIA	91.1	89.7	86.4	91.3	89.1	**99.2***	96.8	86.1	86.6	68.6	69.6	64.4	59.5	43.7	80.2	78.7
	COPIA	92.4	**91.8**	**90.6**	**93.2**	**91.4**	98.4*	**97.6**	**90.2**	**88.6**	**74.8**	**74.5**	**71.9**	**66.9**	**50.8**	**83.8**	**82.7**

Table 14: Attack success rates (ASR, %) for single-layer neuron attribution attacks before and after cross-layer attribution aggregation on defended models and Vision Transformer (ViT) models. By integrating the cross-layer attribution aggregation mechanism, consistent performance gains in transferability are observed across various single-layer methods. Notably, COPIA achieves the highest average ASR under all evaluated surrogate models.

Model	Method	Comdefend	FD	HGD	JPEG	NIPS-r3	R&P	RS	DEF-ASR	Pit	Swin	Deit	Cait	Vit_{base}	ViT-ASR
Inc-v3	NAA	70.3	67.9	23.3	67.8	41.9	32.8	33.3	48.2	51.3	41.2	38.5	45.4	38.7	43.0
	NAA_{CA}	73.0	70.1	20.4	71.1	45.1	36.1	34.8	50.1	59.0	47.4	45.3	53.3	44.0	49.8
	DANAA	71.7	71.2	22.6	72.0	45.1	33.4	34.0	50.0	53.1	43.7	40.0	48.4	41.1	45.3
	$DANAA_{CA}$	73.0	72.4	12.5	73.0	40.4	27.2	33.8	47.5	60.2	43.9	42.8	50.7	41.6	47.8
	NEAA	76.0	74.5	22.5	74.9	43.6	33.3	35.5	51.5	55.6	44.1	41.2	48.0	42.3	46.2
	$NEAA_{CA}$	80.0	78.9	26.5	78.5	49.3	36.9	36.4	55.2	63.2	51.4	48.9	58.0	48.4	54.0
	S-COPIA	78.3	76.7	24.8	78.6	48.4	35.9	35.2	54.0	61.1	50.9	45.9	55.8	46.1	52.0
	COPIA	**81.6**	**81.6**	**25.9**	**82.3**	**55.9**	**44.6**	**38.0**	**58.6**	**70.4**	**58.2**	**56.7**	**65.5**	**53.8**	**60.9**
IncRes-v2	NAA	66.5	62.6	39.3	62.9	48.1	41.9	33.8	50.7	49.9	42.6	40.9	47.9	41.4	44.5
	NAA_{CA}	70.3	67.5	38.6	68.9	51.0	45.6	34.7	53.8	56.8	44.7	45.5	53.0	47.4	49.5
	DANAA	68.3	66.6	47.3	66.3	56.2	51.3	34.2	55.7	57.6	49.2	44.5	52.5	45.8	49.9
	$DANAA_{CA}$	76.6	74.9	37.8	75.6	56.7	48.2	34.5	57.8	64.3	51.8	47.7	60.3	49.8	54.8
	NEAA	78.9	77.4	39.9	78.6	57.4	49.0	**40.3**	60.2	57.5	47.8	45.5	53.2	52.9	51.4
	$NEAA_{CA}$	81.2	79.8	51.8	80.3	60.9	53.0	39.0	63.7	66.2	54.9	53.4	62.7	55.0	58.4
	S-COPIA	79.5	79.7	**56.9**	79.4	64.4	58.2	36.2	64.9	68.3	58.3	53.6	63.6	53.1	59.4
	COPIA	**85.6**	**85.6**	55.2	**85.1**	**71.5**	**62.8**	38.8	**69.2**	**76.1**	**64.3**	**62.1**	**72.5**	**61.7**	**67.3**
$Res\text{-}v2_{50}$	NAA	81.7	80.3	46.2	80.5	50.4	41.9	35.3	59.5	54.6	43.3	41.0	49.8	44.8	46.7
	NAA_{CA}	85.0	84.4	48.3	85.4	55.9	47.9	38.0	63.6	58.0	45.7	45.9	53.3	48.7	50.3
	DANAA	83.2	82.6	46.6	83.4	51.4	41.6	34.6	60.5	54.9	47.0	42.0	50.3	43.9	47.6
	$DANAA_{CA}$	85.3	84.9	44.0	85.4	53.3	43.5	35.8	61.7	55.4	42.8	43.2	51.3	46.1	47.8
	NEAA	85.4	85.3	45.2	85.3	49.3	40.7	36.0	61.0	55.5	44.4	42.3	50.8	45.8	47.8
	$NEAA_{CA}$	88.1	87.4	52.1	87.9	57.0	47.2	38.0	65.4	61.5	46.4	47.6	57.3	48.6	52.3
	S-COPIA	86.5	86.6	51.5	87.0	56.6	45.9	36.1	64.3	60.3	48.0	45.5	56.8	47.6	51.6
	COPIA	**89.2**	**89.4**	**56.3**	**89.7**	**62.8**	**53.5**	**37.9**	**68.4**	**67.4**	**54.2**	**52.5**	**61.2**	**52.6**	**57.6**

trained models. When using Inc-v3 and IncRes-v2 as surrogates, all aggregated variants (NAA_{CA}, $DANAA_{CA}$, $NEAA_{CA}$) achieve higher average ASR than their single-layer counterparts, while COPIA retains the best attack performance across all black-box targets. Interestingly, when Inc-v3 and $Res\text{-}v2_{50}$ serve as surrogates, $DANAA_{CA}$ enhances transferability over DANAA on normally trained models but exhibits reduced ASR on adversarially trained models.

A possible explanation for the decrease in transferability of adversarial examples generated by DANAA combined with the cross-layer attribution mechanism (CA) against adversarially trained models and defense models is as follows. During path generation, the adversarial path of DANAA is updated along the direction of gradient ascent, which makes the objective unclear and introduces high uncertainty. The CA mechanism, which integrates neuron attributions from multiple layers, exacerbates this uncertainty. Since adversarially trained models and defense models are designed to defend against such uncertain perturbations, the transferability decreases. In contrast, the straight-line path of NAA, which goes from a black image to the original image, has well-defined starting and target points. Meanwhile, the counterfactual path is semantically guided by counterfactual samples, making the update direction more stable. As a result, NAA does not suffer from degraded attack performance against adversarially trained models and defense models when combined with the cross-layer mechanism. Therefore, the existence of the above phenomenon further highlights the necessity of finding better paths in neuron attribution-based attacks.

We further evaluate all methods on defended and Vision Transformer (ViT) models (Table 14). After cross-layer attribution aggregation, NAA, NEAA, and S-COPIA show consistent improvements across all surrogate settings. However, $DANAA_{CA}$ suffers performance degradation when attacking defended models with Inc-v3 as the surrogate. This behavior further indicates that neuron-attribution aggregation based solely on adversarial paths may compromise robustness under more challenging defense and architectural settings. Overall, the cross-layer attribution aggregation strategy generally strengthens transferability for most attribution-based attacks, and COPIA demonstrates the strongest and most stable performance across all target categories.

I Comparison Between Cross-Layer Attribution(CA) and Naive Multi-Layer Aggregation(MA)

To verify the difference between the proposed cross-layer attribution (CA) mechanism and naive multi-layer aggregation in neuron-attribution attacks (MA), we compare three variants of COPIA: (1) the full COPIA framework, which employs five counterfactual samples together with gradient-ascent and gradient-descent path updates; (2) a single-layer version without the cross-layer attribution (CA) mechanism, denoted as S-COPIA, which also uses five counterfactual samples; and (3) COPIA-MA, which directly aggregates neuron attributions from multiple layers through simple summation. To further strengthen the analysis, we conduct the same comparison on NAA. Specifically, we construct NAA-MA by directly summing neuron attributions from multiple layers and compare it with both the original single-layer NAA and the cross-layer version NAA-CA. Inception-v3 is used as the surrogate model in all experiments.

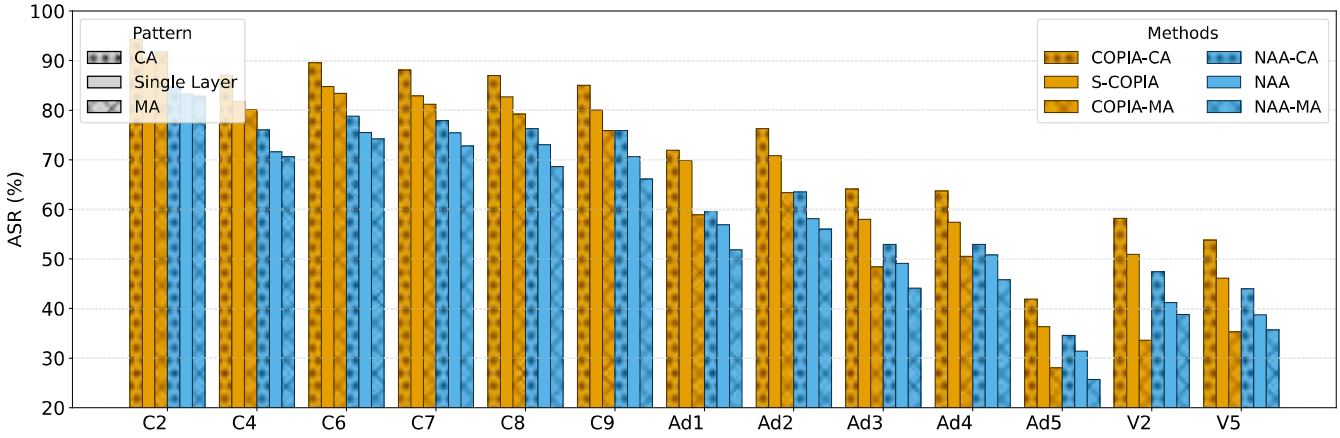


Figure 11: Comparison of ASR among cross-layer(CA) mechanism, single layer, and simple multi-layer aggregation(MA). Abbreviations for the target model: "C2: Inc-v4, C4: Res-v1-50, C6: Res-v2-50, C7: Res-v2-152, C8: Vgg16, C9: Vgg19. Ad1: Inc-v3-adv, Ad2: IncRes-v2-adv, Ad3: Inc-v3-ens3, Ad4: Inc-v3-ens4, Ad5: IncRes-v2-ens. V2: Swin, V5: Vit-base".

The results are shown in Fig. 11. Compared with single-layer attribution attacks, naive multi-layer aggregation fails to improve transferability and may even degrade attack performance. A possible explanation is that simple aggregation treats neuron attributions from shallow, middle, and deep layers equally when generating adversarial examples. However, neuron attributions from shallow and deep layers generally exhibit lower transferability than those from middle layers. Consequently, directly incorporating all layers introduces less transferable attribution information, reducing the transferability of the generated adversarial examples. In contrast, the proposed cross-layer attribution (CA) mechanism effectively exploits complementary information from multiple layers while mitigating the negative effects of less transferable attributions. As a result, CA consistently improves adversarial transferability. These findings demonstrate that the proposed CA mechanism is fundamentally different from naive multi-layer aggregation and provides a more effective way to leverage multi-layer attribution information.

J Further Investigation of Counterfactual Guidance

J.1 Strategy for Constructing the Counterfactual Sample Set

In our experiments, we employ the ILSVRC 2012 validation set, selecting one representative image from each of the 1,000 categories to construct a subset $\mathcal{D} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{1000}\}$ with corresponding labels $\mathcal{Z} = \{z_1, z_2, \ldots, z_{1000}\}$. By definition, a counterfactual sample should have a different ground-truth label from the given benign sample. Accordingly, for a benign sample $\boldsymbol{x}_t$ with label z_t, all samples in $\mathcal{D}$ whose labels differ from z_t can be naturally treated as valid counterfactual candidates. Based on this, we consider two practical strategies to construct a set of C counterfactual samples: (i) *Sequential Strategy*: C counterfactual samples are selected via sequential indexing, *i.e.*, $\{\boldsymbol{x}_{t+1}, \boldsymbol{x}_{t+2} \ldots, \boldsymbol{x}_{t+C}\}$ and $\{z_{t+1}, z_{t+2} \ldots, z_{t+C}\}$. (ii) *Random Strategy*: C counterfactual samples are randomly drawn from $\mathcal{D}$ such that $\{\boldsymbol{x}_{r_1}, \boldsymbol{x}_{r_2}, \ldots, \boldsymbol{x}_{r_C}\}$, and $\{z_{r_1}, z_{r_2}, \ldots, z_{r_C}\} \in \mathcal{Z} \setminus \{z_t\}$.

Empirically, using Inception-v3 (Inc-v3) as the surrogate model, we evaluate both strategies across normally trained, adversarially trained, and Vision Transformer (ViT) models. The sequential strategy achieves an average black-box Attack Success Rate (ASR) of 74.0%, while the random strategy yields 73.8%. Given the marginal performance difference and to ensure consistent reproducibility, we adopt the sequential selection strategy in all experiments, as it provides a deterministic and stable sampling process while avoiding additional randomness introduced by stochastic selection.

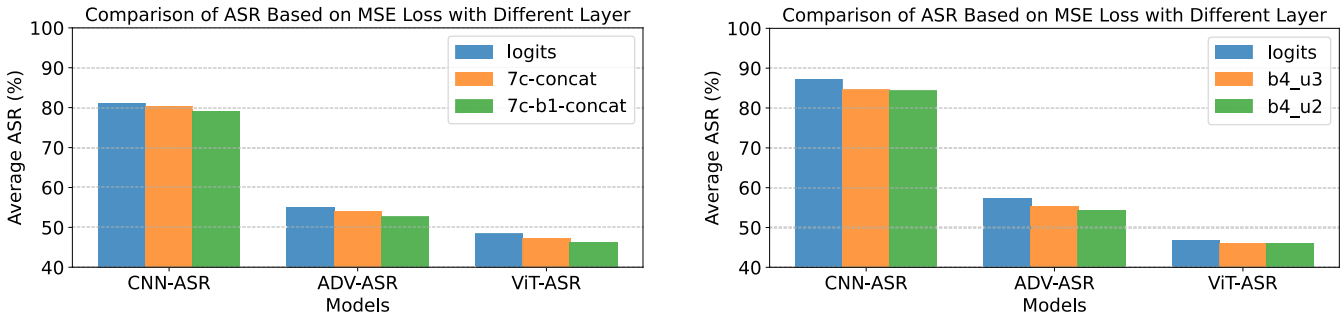
To further investigate the impact of counterfactual sample selection, we consider several selection strategies based on feature similarity and classification confidence. Specifically, we compute the feature similarity between counterfactual samples and the benign sample, and select samples with the highest and lowest similarity, denoted as *High-Similarity* and *Low-Similarity*, respectively. We also select samples with the highest and lowest classification confidence within the same class as the benign sample, denoted as *High-Confidence* and *Low-Confidence*. In addition, we consider *Sequential* and *Random* selection strategies.

The evaluation metric is the average attack success rate (ASR) over normally trained models, adversarially trained models, and ViT models. We evaluate two settings: using a single counterfactual sample and using five counterfactual samples. When only one counterfactual sample is used, the average ASRs are 65.3% (High-Confidence), 63.9% (Low-Confidence), 63.0% (High-Similarity), 64.2% (Low-Similarity), 63.6% (Sequential), and 63.4% (Random), with a variance of 0.5045. When five counterfactual samples are used, the average ASRs become 73.9% (High-Confidence), 73.3% (Low-Confidence), 74.2% (High-Similarity), 74.0% (Low-Similarity), 74.0% (Sequential), and 73.8% (Random), with a substantially smaller variance of 0.0628. These results indicate that COPIA is largely insensitive to the choice of counterfactual samples.

Moreover, both counterfactual sample diversification and counterfactual path diversification improve the stability of attack performance by exploiting complementary information from multiple counterfactual samples.

J.2 Additional Study on Feature Layers for Constructing Counterfactual Paths

Under the single-layer neuron attack setting (S-COPIA), we fix the number of counterfactual examples to 3, the learning rate to 0.025, and the Gaussian noise to zero mean with variance 0.25 (*i.e.*, scale = 0.25). We then perform an ablation study to examine how the choice of feature layers $G(\cdot)$ influences attack performance. Using Inc-v3 and Res-v2$_{50}$ as surrogate models, we evaluate each candidate layer across three metrics: CNN-ASR, ADV-ASR, and ViT-ASR. The results, presented in Fig. 12, show that constructing the MSE loss using the final output layer consistently yields the highest average ASR across all settings. Accordingly, the output layer is adopted in our final formulation.



(a) Inc-v3 **(b) Res-v2$_{50}$**

Figure 12: Effect of selecting different intermediate layers for constructing the MSE loss, evaluated under two surrogate models. The results demonstrate that guiding counterfactual path generation using the final output layer (logits) yields the optimal Attack Success Rate (ASR) across both surrogates.

J.3 Exploratory Experiment on the Number of Diversified Counterfactual Reference Points

Under the single-layer neuron attribution attack setup(S-COPIA)—with a fixed learning rate of 0.025, Gaussian noise of zero mean and variance 0.25, and gradient ascent for counterfactual path updates—we systematically vary the number of counterfactual examples from 1 to 8 using Inc-v3 and Res-v2$_{50}$ as surrogate models. To quantify the impact of varying the number of counterfactual reference points on attack performance, we evaluate the average Attack Success Rate (ASR) across a comprehensive suite of target models, including naturally trained CNNs, adversarially trained CNNs, and Vision Transformers (ViTs). This cross-category evaluation ensures that the observed trends in attack efficacy are consistent and robust across diverse model architectures and defense mechanisms. The results, shown in Fig. 13, indicate that attack performance improves substantially as the number of counterfactual examples increases initially. However, beyond a certain point, the average Attack Success Rate (ASR) exhibits diminishing returns, with only marginal improvements or slight degradation. Based on these observations, we adopt five counterfactual examples in our main experiments to strike a balance between attack effectiveness and computational cost.

J.4 Investigating the Effectiveness of Diversified Counterfactual Paths

To validate the superiority of multi-path aggregation in S-COPIA over the noisy straight-line path aggregation in NEAA, we conducted a controlled comparative experiment. For S-COPIA, we treated the gradient-ascent and gradient-descent updates generated from a single counterfactual example as two distinct paths. Both gradient-ascent and gradient-descent counterfactual paths accelerate escape from saturation regions, yielding diversified neuron attributions with inherent complementarity. For a visual comparison of the trajectories (including linear paths, adversarial paths, and our gradient-descent/ascent counterfactual paths) in navigating out of saturation regions, please refer to Fig. 14. Both gradient-ascent and gradient-descent counterfactual paths facilitate more rapid escape from saturation regions, effectively mitigating noise accumulation and yielding higher-fidelity neuron attributions.

To benchmark the attack performance of S-COPIA against NEAA, we conducted a comparison under an identical number of diversified paths. We progressively increased the number of counterfactual examples from 1 to 5, corresponding to 2, 4, 6, 8, and 10 aggregated paths in total. For a fair comparison, NEAA was configured to aggregate an identical number of paths (2, 4, 6, 8, 10). Attack performance was evaluated using three metrics: the average Attack Success Rate on normally trained models (CNN-ASR), adversarially trained models (ADV-ASR), and Vision Transformer models (ViT-ASR). As shown in Fig. 15, S-COPIA consistently outperforms NEAA across all path counts. These results provide strong empirical support for incorporating counterfactual paths into neuron attribution attacks and demonstrate that multi-path counterfactual aggregation is inherently more effective than straight-line aggregation.

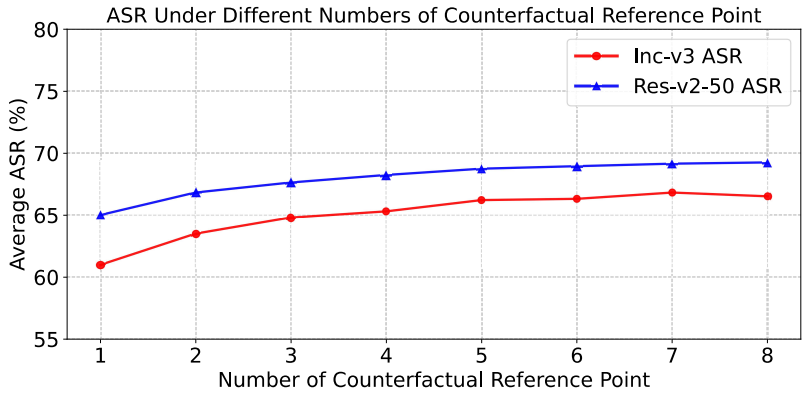


Figure 13: Effect of varying the number of counterfactual reference points on attack performance for different surrogate models. Increasing the number of counterfactual reference points initially enhances attack performance; however, the marginal gains gradually diminish, and excessive reference points may even lead to a decline in effectiveness.

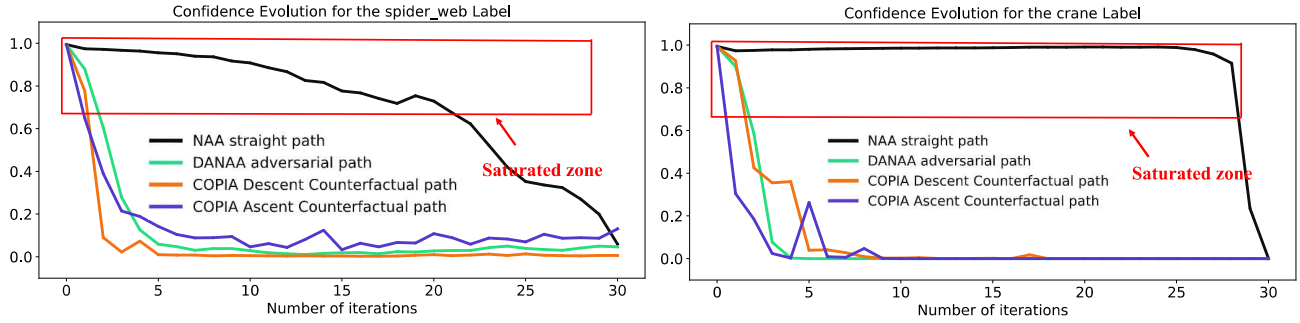


Figure 14: Comparison of the speed of escaping the saturation region via different paths. By leveraging both gradient-ascent and gradient-descent trajectories for counterfactual path generation, we obtain a more diverse set of neuron attributions. The resulting information from these dual paths exhibits strong synergy, providing complementary perspectives on feature importance.

K Additional Exploratory Study

K.1 Investigating the Effects of Gaussian Noise Strength in Intermediate Integration Phases

Following the COPIA configuration, we fixed the learning rate to 0.025, set the number of counterfactual examples to 5, and employed both gradient descent and gradient ascent updates prior to diversified integration. Gaussian noise with zero mean and variance scale was injected into the intermediate variables during path integration. To evaluate the influence of noise strength on the attack success rate (ASR), we varied scale from 0.1 to 1.0 with a step size of 0.1, using Inc-v3 as the surrogate model.

As shown in Figure 16, the attack performance exhibits a non-monotonic trend: ASR improves as scale increases in the low-noise regime, suggesting enhanced exploration and diversity along the integration path, but degrades when the noise becomes excessive due to the disruption of meaningful attribution signals. Based on this observation, we adopted scale = 0.2 for all subsequent experiments, as it provides a favorable balance between exploration and stability.

K.2 Exploration of Learning Rate for Counterfactual Path Construction

We investigated the effect of different learning rates on counterfactual path construction, evaluating three values: 0.25, 0.025, and 0.0025. All experiments followed the COPIA configuration, employed Gaussian noise (mean = 0, variance = 0.2, i.e., scale = 0.2), used 5 counterfactual examples, and updated counterfactual paths via both gradient descent and gradient ascent, followed by diversified integration.

As shown in Figure 17a, when Inc-v3 is used as the surrogate model, a learning rate of 0.025 achieves the highest average attack success rate across all target models. A consistent trend is observed when using Res-v2$_{50}$ as the surrogate (Figure 17b). Based on these findings, we adopted 0.025 as the learning rate in our final configuration.

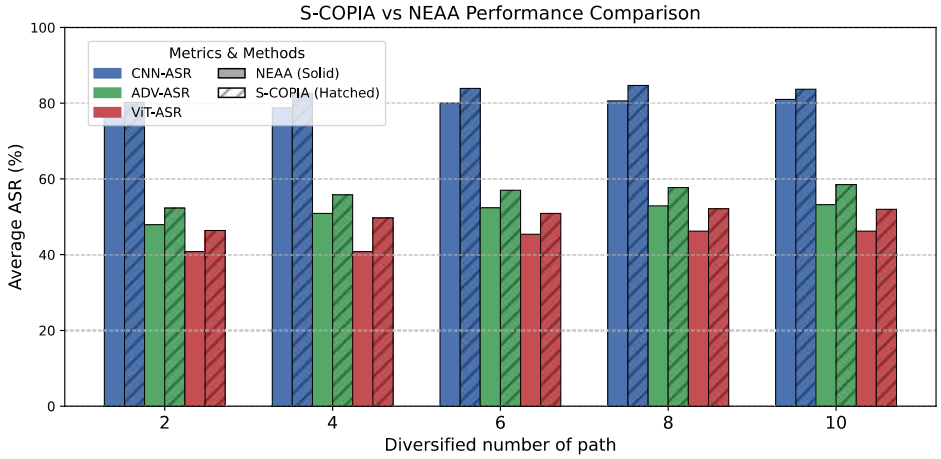


Figure 15: Comparison of attack performance between S-COPIA and NEAA under equal numbers of diversified paths, using Inc-v3 as the surrogate model. The results demonstrate that S-COPIA consistently outperforms NEAA in attack performance across all specified path counts.

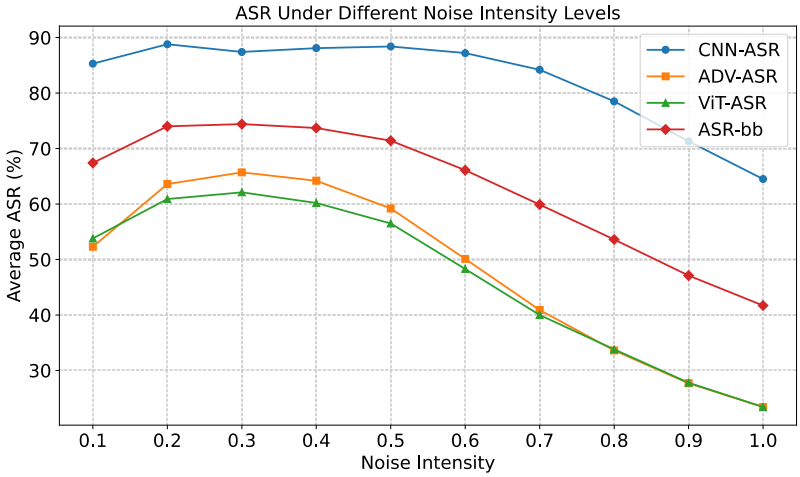
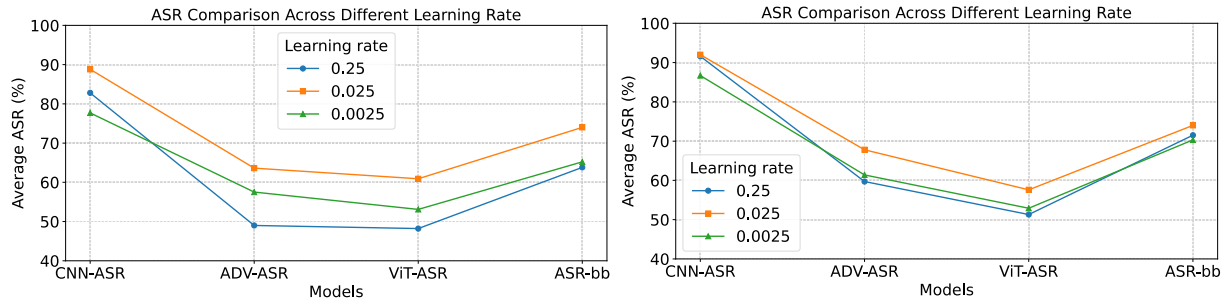


Figure 16: Effect of varying Gaussian noise variance on attack performance using Inception-v3 as the surrogate model. The results exhibit a non-monotonic trend, where moderate noise improves ASR while excessive noise degrades performance. A noise intensity of 0.2 achieves strong average ASR across all evaluated black-box models, indicating a favorable balance between exploration and stability.

K.3 An Analysis of Hyperparameter Effects on Path Integration and Attack Performance

To identify the optimal counterfactual path setup, we perform a grid search over different learning rates and path lengths. Specifically, we set the learning rate to 0.25, 0.025, and 0.0025, and vary the path length (number of iterations) among 10, 20, 30, 40, and 50. We evaluate all combinations using the resulting average attack success rate (ASR), as shown in Fig. 18.

When the learning rate is 0.25, it is relatively large, and increasing the path length does not consistently improve ASR; instead, the performance fluctuates with the number of iterations. In contrast, when the learning rate is 0.025 or 0.0025, ASR generally increases with longer paths but gradually saturates. Moreover, increasing the path length also introduces additional computational cost. Therefore, to balance efficiency and ensure a fair comparison with FIA, MFAA, NAA, DANAA, and NEAA, we set the path length to 30 in all experiments. In addition, we select 0.025 as the default learning rate, as it achieves the best overall ASR across settings.



(a) Inc-v3 **(b) Res-v2$_{50}$**

Figure 17: Effect of different learning rates on attack performance using Inc-v3 and Res-v2$_{50}$ as surrogate models. When the learning rate is set to 0.025, COPIA achieves the highest average Attack Success Rate (ASR) against all black-box models across different surrogates, demonstrating the strong generalizability of this parameter setting.

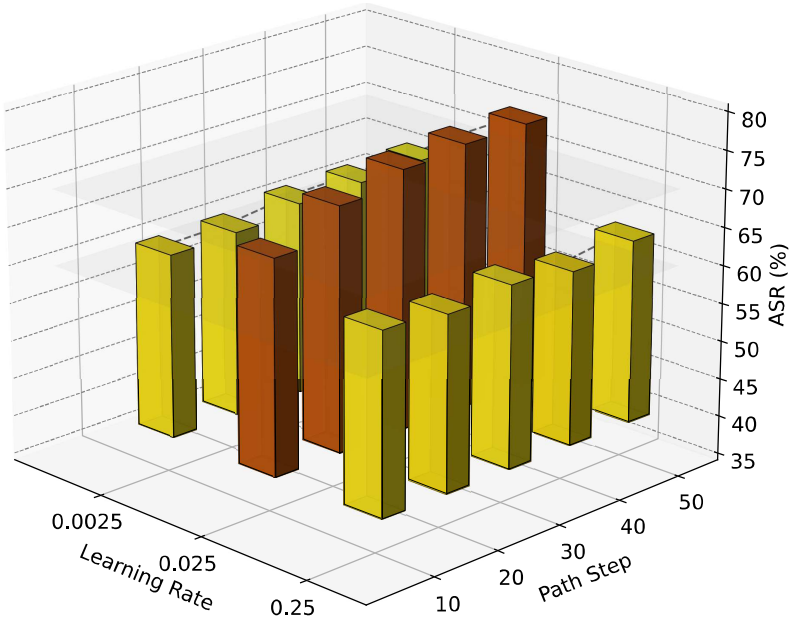


Figure 18: Effects of learning rate and path step on attack performance.

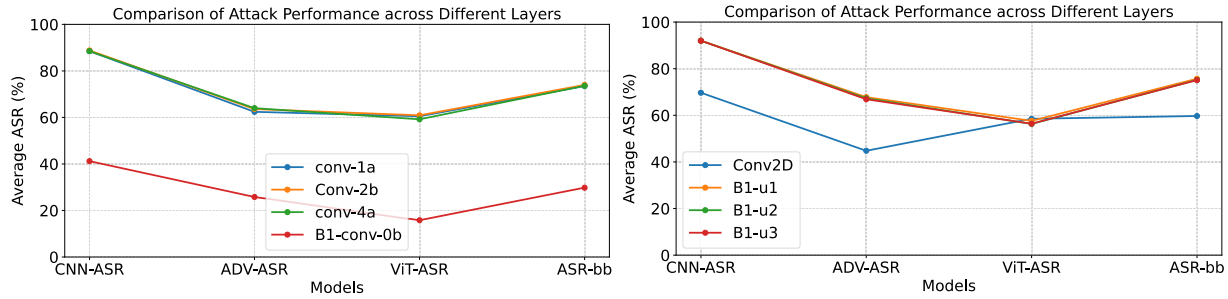
K.4 Additional Experiments on the Impact of Source-Model Attack Layers

Different attack layers on the surrogate model may lead to varying attack performance. To systematically investigate this effect, we conducted an ablation study on layer selection while following the COPIA configuration: a learning rate of 0.025, Gaussian noise with zero mean and variance 0.2 (i.e., scale = 0.2), five counterfactual examples, and hybrid counterfactual path updates using both gradient descent and gradient ascent.

Using Inc-v3 as the surrogate model, we evaluated attack performance across several representative layers while keeping all other parameters fixed. The layers examined were: InceptionV3/Conv2d-1a-3x3/Relu (conv-1a), InceptionV3/Conv2d-2b-3x3/Relu (conv-2b), InceptionV3/Conv2d-4a-3x3/Relu (conv-4a), and InceptionV3/Mixed-5b/Branch-1/Conv2d-0b-5x5/Relu (B1-conv-0b).

As shown in Figure 19a, the attack efficacy remains relatively stable for the shallower layers (conv-1a, conv-2b, conv-4a). However, once the attack layer exceeds a certain depth—most notably illustrated by the B1-conv-0b layer—performance drops sharply. Based on these observations, we selected *conv-2b* as the optimal attack layer in our final configuration.

For the Res-v2$_{50}$ surrogate model, we selected four representative layers for our attack-layer investigation: resnet-v2-50/block1/unit-1/bottleneck-v2/conv1/Conv2D (Conv2D), resnet-v2-50/block1/unit-1/bottleneck-v2/add (B1-u1), resnet-v2-50/block1/unit-2/bottleneck-v2/add (B1-u2), and resnet-v2-50/block1/unit-3/bottleneck-v2/add (B1-u3). As shown in Figure 19b, the results reveal a notable discrepancy: attacks initiated from the Conv2D layer yield the weakest performance on normally trained and defended CNNs, yet surprisingly achieve



(a) Inc-v3 **(b) Res-v2$_{50}$**

Figure 19: Comparison of attack performance across different attack layers for (a) Inc-v3 and (b) Res-v2$_{50}$, illustrating clear performance variation as the attack layer changes.

the highest success rate on Vision Transformer (ViT) models. This suggests that optimal transferability requires aligning the chosen attack layer with the architectural characteristics of the target model. Consistent with our findings on Inc-v3, the attack performance remains largely unchanged when shifting from B1-u1 to B1-u2 or B1-u3, indicating that these shallow layers offer similar feature representations for constructing transferable attacks.

K.5 Feature Layers Used by Different Source Models

In this work, we employ five surrogate models throughout the main text and appendix: Inception-v3 (Inc-v3), Inception-v4 (Inc-v4), Inception-ResNet-v2 (IncRes-v2), ResNet-v2-50 (Res-v2$_{50}$), and ResNet-v2-152 (Res-v2$_{152}$). For our proposed COPIA method, each surrogate model utilizes multiple feature layers. Following the layer-selection strategy in previous studies [47], we fix the number of selected layers to five per model, where the first layer corresponds to shallow features and the last layer is positioned near the output. The specific feature layers chosen for each model, listed from shallow to deep, are as follows:

For Inception-v3:

- `InceptionV3/Conv2d_2b_3x3/Relu`
- `InceptionV3/Mixed_5b/concat`
- `InceptionV3/Mixed_6a/concat`
- `InceptionV3/Mixed_6c/concat`
- `InceptionV3/Mixed_7a/concat`

For Inception-v4:

- `InceptionV4/Conv2d_2b_3x3/Relu`
- `InceptionV4/Mixed_4a/concat`
- `InceptionV4/Mixed_5a/concat`
- `InceptionV4/Mixed_6a/concat`
- `InceptionV4/Mixed_7a/concat`

For Inception-ResNet-v2:

- `InceptionResnetV2/Conv2d_2b_3x3/Relu`
- `InceptionResnetV2/Mixed_5b/concat`
- `InceptionResnetV2/Mixed_6a/concat`
- `InceptionResnetV2/Mixed_7a/concat`
- `InceptionResnetV2/Conv2d_7b_1x1/Relu`

For ResNet-v2-50:

- `resnet_v2_50/block1/unit_1/bottleneck_v2/add`
- `resnet_v2_50/block2/unit_3/bottleneck_v2/add`
- `resnet_v2_50/block3/unit_6/bottleneck_v2/add`
- `resnet_v2_50/block4/unit_1/bottleneck_v2/add`
- `resnet_v2_50/block4/unit_3/bottleneck_v2/add`

For ResNet-v2-152:

- `resnet_v2_152/block2/unit_1/bottleneck_v2/add`
- `resnet_v2_152/block2/unit_8/bottleneck_v2/add`
- `resnet_v2_152/block3/unit_8/bottleneck_v2/add`

- `resnet_v2_152/block4/unit_1/bottleneck_v2/add`
- `resnet_v2_152/block4/unit_3/bottleneck_v2/add`

K.6 Ablation Study on the Attribution Contribution Ratio of Positive and Negative Neurons

In this section, we analyze the roles of positive and negative neuron attributions in adversarial attacks. To ensure a controlled study, we fix the number of counterfactual samples to 5 and the learning rate to 0.025. Under the full COPIA setting with path diversification and cross-layer attribution, we vary the ratio between positive and negative neuron contributions to examine its impact on black-box transferability. Specifically, using Inc-v3 as the surrogate model, we set the attribution ratio λ to 0.01, 0.1, 1, 10, and 100, and evaluate performance across multiple black-box models. The results are shown in Fig. 20. As observed, the attack performance consistently peaks when the ratio λ is set to 1. This suggests that balancing positive and negative neuron attributions is crucial for accurately capturing decision boundary vulnerabilities and achieving optimal transferability.

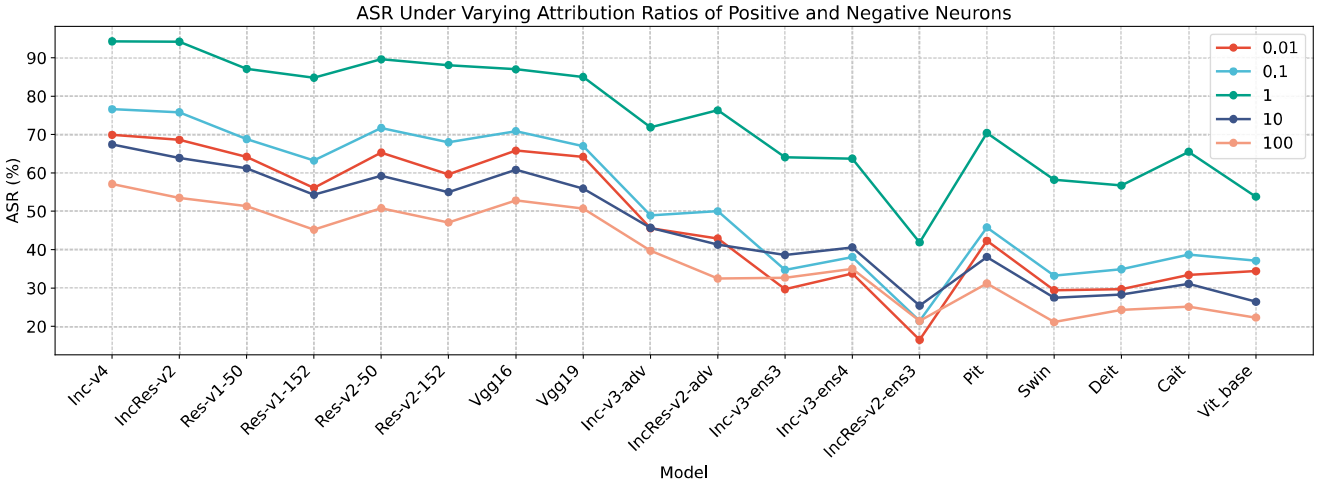


Figure 20: Analysis of the contributions of positive and negative neuron attributions to adversarial transferability. When the attribution ratio is set to 1, the attack success rate consistently peaks across diverse black-box models, indicating that balancing positive and negative influences enables more accurate identification of vulnerabilities in the model's decision boundaries.

K.7 Analysis of the Cross-Layer Attribution Coefficient

In the proposed cross-layer attribution (CA) mechanism, neuron attributions from deep, intermediate, and shallow layers are jointly exploited. The parameter μ controls the contribution of the attribution flow propagated from deeper layers to the attribution of the current layer. To investigate its effect, we conduct an ablation study using the full COPIA framework, including five counterfactual samples, gradient-ascent and gradient-descent counterfactual path generation, lr=0.025, and the CA mechanism.

Specifically, we set μ to 100, 10, 1, 0.1, and 0.01, and evaluate the transferability of the adversarial examples generated using Inception-v3 as the surrogate model against different black-box target models. The results are shown in Fig. 21. When μ=100, the propagated attribution from deeper layers dominates the optimization process, leading to severe overfitting to deep-layer representations and a substantial reduction in black-box transferability. This observation is consistent with our earlier finding that attribution attacks based solely on deep-layer features are generally less transferable than those based on intermediate-layer features. As μ decreases from 100 to 10 and then to 1, the transferability gradually improves. However, further reducing μ from 1 to 0.1 and 0.01 leads to a progressive decline in attack performance. In the extreme case of $\mu = 0$, no attribution information is propagated from deeper layers, and the method degenerates into a shallow-layer attribution attack. Such attacks exhibit lower transferability than intermediate-layer attribution attacks, which is consistent with our observations in the single-layer setting. Overall, the results suggest that neither excessive reliance on deep-layer attribution nor complete removal of cross-layer attribution flow is desirable. The best transferability is achieved when the propagated attribution from deeper layers contributes approximately equally to the attribution of the current layer, corresponding to $\mu = 1$. This finding validates the design principle of the proposed CA mechanism, which balances attribution information across different semantic levels to improve adversarial transferability.

L Further Evaluation of Practical Efficiency of Different Attack Methods on TensorFlow

In this section, we examine the per-image attack time and the corresponding average attack success rate (ASR) for black-box transfer attacks on normally trained, adversarially trained, and cross-architecture Vision Transformer (ViT) models across different methods. We

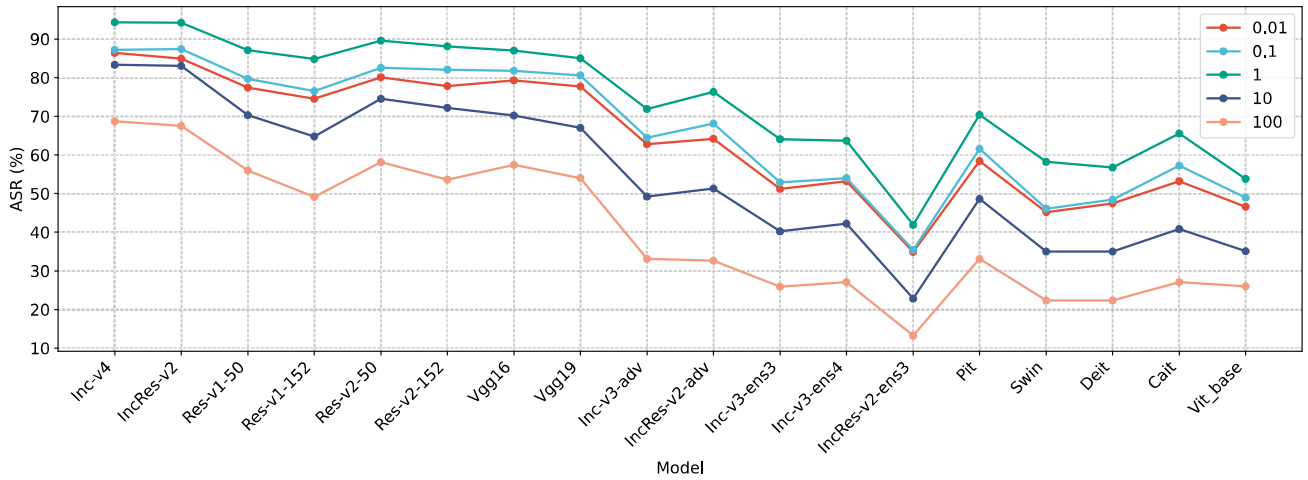


Figure 21: Comparison of ASR under different contribution ratios of neuron attribution across layers.

further compare both the computational cost and effectiveness of S-COPIA (the single-layer variant of COPIA) against baseline attacks. In addition, we analyze the runtime and ASR of existing neuron attribution attacks combined with a cross-layer attribution mechanism (*i.e.*, NAA_{CA}, DANAA_{CA}, NEAA_{CA}). The complete results are summarized in Table 15. In our notation, *S-COPIA-xr-5-Diversified* refers to the single-layer attribution attack using 5 counterfactual examples with diversified updates (combining gradient ascent and descent), whereas *S-COPIA-xr-1-Ascent* denotes the variant that performs path updates using only gradient ascent with 1 counterfactual example.

Specifically, the proposed COPIA strikes a favorable trade-off between efficiency and effectiveness, achieving strong performance while maintaining computational efficiency. For example, *S-COPIA-xr-1-Ascent* represents the most lightweight configuration with the highest efficiency, while *COPIA-xr-5-Diversified* corresponds to the most computationally intensive variant with the strongest effectiveness. Other configurations (*e.g.*, *S-COPIA-xr-1-Diversified*, *COPIA-xr-1-Ascent*, *etc.*) lie between these two extremes , offering balanced performance and efficiency.

Table 15: Attack time per image and corresponding average attack success rate (ASR) across different methods. ASR represents the average black-box transfer attack success rate evaluated over normally trained CNNs, adversarially trained CNNs, and Vision Transformer models.

Model	Method	Time(s)	ASR(%)	Model	Method	Time(s)	ASR(%)	Model	Method	Time(s)	ASR(%)
Inc-v3	FIA	4.8	53.7	IncRes-v2	FIA	6.8	57.1	Res-v2_{50}	FIA	**2.5**	59.3
	NAA	5.9	58.7		NAA	21.1	58.9		NAA	6.9	65.5
	DANAA	6.1	62.1		DANAA	15.3	64.4		DANAA	8.7	66.8
	NEAA	51.0	63.6		NEAA	203.3	68.4		NEAA	52.1	67.8
	MFAA	8.2	61.6		MFAA	20.4	62.9		MFAA	10.5	68.3
	SVD	**2.3**	56.9		SVD	**4.7**	59.0		SVD	––	––
	SMP	11.3	61.6		SMP	71.5	70.0		SMP	43.9	72.1
	P2FA	5.8	67.2		P2FA	9.8	64.8		P2FA	––	––
	NAA_{CA}	7.8	63.2		NAA_{CA}	20.7	63.4		NAA_{CA}	10.7	69.9
	DANAA_{CA}	13.1	63.4		DANAA_{CA}	39.2	71.1		DANAA_{CA}	17.3	68.3
	NEAA_{CA}	58.2	69.1		NEAA_{CA}	150.0	73.0		NEAA_{CA}	77.1	71.7
	S-COPIA-xr-1-Ascent	11.7	61.0		S-COPIA-xr-1-Ascent	41.4	67.5		S-COPIA-xr-1-Ascent	13.4	65.0
	S-COPIA-xr-1-Diversified	22.0	63.1		S-COPIA-xr-1-Diversified	78.0	69.4		S-COPIA-xr-1-Diversified	25.9	67.6
	S-COPIA-xr-5-Ascent	33.4	66.2		S-COPIA-xr-5-Ascent	116.3	73.0		S-COPIA-xr-5-Ascent	38.3	68.7
	S-COPIA-xr-5-Diversified	65.8	67.9		S-COPIA-xr-5-Diversified	232.7	74.6		S-COPIA-xr-5-Diversified	73.0	71.9
	COPIA-xr-1-Ascent	13.5	63.6		COPIA-xr-1-Ascent	40.8	71.1		COPIA-xr-1-Ascent	18.1	68.4
	COPIA-xr-1-Diversified	24.6	67.2		COPIA-xr-1-Diversified	64.9	73.5		COPIA-xr-1-Diversified	31.3	71.6
	COPIA-xr-5-Ascent	37.7	71.0		COPIA-xr-5-Ascent	101.2	78.7		COPIA-xr-5-Ascent	50.3	73.9
	COPIA-xr-5-Diversified	73.6	**74.0**		COPIA-xr-5-Diversified	184.2	**80.3**		COPIA-xr-5-Diversified	96.4	**75.7**

For example, with Inc-v3 as the surrogate, *COPIA-xr-1-Diversified* achieves a state-of-the-art ASR of 67.2% with a competitive runtime of 24.6s, significantly outperforming the most time-consuming baseline, *i.e.*, NEAA, which attains 63.6% ASR while requiring 51.0s. By further incorporating counterfactual sample diversification, *COPIA-xr-5-Ascent* increases the ASR to 71.0% at a runtime of 37.7s, surpassing other

approaches (*e.g.*, P2FA achieves the second-best ASR of 67.2%) by a large margin in black-box transferability. With additional path update diversification, *COPIA-xr-5-Diversified* further boosts ASR to 74.0%.

Additionally, when employing IncRes-v2 as the surrogate, *COPIA-xr-1-Ascent* yields an average ASR of 71.1% in 40.8s, already surpassing the SOTA SMP (70.0% ASR in 71.5s), demonstrating an excellent balance between efficiency and effectiveness. By further incorporating the cross-layer mechanism, *COPIA-xr-5-Diversified* elevates the average ASR to 80.3% with a runtime of 184.2s, achieving both higher efficiency and effectiveness compared to NEAA (203.3s; 68.4%). Notably, we observe an intriguing phenomenon where the per-adversarial-example generation time of *S-COPIA-xr-5-Diversified* exceeds that of *COPIA-xr-5-Diversified*, a trend also observed in other methods when combined with cross-layer attribution, such as NEAA vs. NEAA_{CA}. This is mainly due to the TensorFlow-based implementation of *S-COPIA-xr-5-Diversified*, which involves frequent CPU–GPU data transfers during neuron attribution computation, leading to inefficient GPU utilization. In contrast, although *COPIA-xr-5-Diversified* operates across multiple layers and consumes more GPU memory, it achieves better GPU efficiency, resulting in faster attack generation.

Finally, for the Res-v2-50 surrogate model, SVD and P2FA are excluded from the comparison due to substantial architectural and parametric divergences between their PyTorch-based ResNet implementations and the corresponding TensorFlow-based models. These discrepancies prevent a fair and meaningful comparison. *COPIA-xr-1-Diversified* achieves a competitive 71.6% ASR in 31.3s, comparable to the SOTA SMP (72.1%; 43.9s). By further incorporating counterfactual diversification, *COPIA-xr-5-Diversified* increases the average ASR to 75.7% with a runtime of 96.4s.

M Migration from TensorFlow to PyTorch

On the TensorFlow framework, the complete COPIA configuration achieves the highest transferability attack performance, while it is less computationally efficient compared with existing approaches. During implementation, we found that the TensorFlow version suffered from extremely low GPU utilization, with a large portion of computations being executed on the CPU, leading to significantly longer adversarial example generation time. To address this issue, we migrated the implementation from TensorFlow to PyTorch, which substantially improves GPU utilization and computational efficiency while preserving the original attack framework. Under the full COPIA setting, *i.e.*, using five counterfactual samples, gradient-ascent and gradient-descent counterfactual path generation, and the cross-layer attribution (CA) mechanism, the PyTorch implementation achieves significant acceleration across different surrogate models. Specifically, with Inception-v3 as the surrogate model, the generation time of a single adversarial example is reduced from 73.6 s in TensorFlow to 18.3 s in PyTorch. For IncRes-v2 and ResNet-v2-50, the generation time is reduced from 184.2 s to 35.5 s and from 96.4 s to 11.5 s, respectively. These results demonstrate that the PyTorch implementation substantially improves the computational efficiency of COPIA while maintaining its effectiveness and overall attack design.

M.1 Optimization Details of PyTorch

During counterfactual path generation, the model output layer is used to maximize (gradient ascent) and minimize (gradient descent) the discrepancy between the benign sample and the counterfactual sample. Five counterfactual samples are employed to construct diversified counterfactual paths. For cross-layer attribution optimization, the attribution flow is propagated from deeper layers to shallower layers. Specifically, the weighted neuron-importance loss from deeper layers is back-propagated to the current layer to compute the modified integrated attribution $\widetilde{mIA}^{cf}$ and the corresponding modified attribution loss ($\widetilde{mA}^{cf}$). Based on $\widetilde{mA}^{cf}$, the weighted modified attribution loss ($\widetilde{mWA}$) is further obtained.

Specifically, the cross-layer attribution is formulated as follows: (i) For the last layer $\mathbb{L}[L']$, $\widetilde{mIA}^{cf}_{\mathbb{L}[L']} = \widetilde{IA}^{cf}_{\mathbb{L}[L']}$, $\widetilde{mA}^{cf}_{\mathbb{L}[L']} = \widetilde{IA}^{cf}_{\mathbb{L}[L']} \cdot \Delta_{\mathbb{L}[L']}$, and we can get $\widetilde{mWA}^{cf}_{\mathbb{L}[L']}$. (ii) For any intermediate layer $\mathbb{L}[l]$, where $l \in (1, L')$, the normalized gradient of the weighted attribution from layer $\mathbb{L}[l+1]$ is propagated and aggregated into layer $\mathbb{L}[l]$:

$$\widetilde{mIA}^{cf}_{\mathbb{L}[l]} = \widetilde{IA}^{cf}_{\mathbb{L}[l]} + \mu \cdot \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}^{cf}_{\mathbb{L}[l+1]} \Big/ \left\| \nabla_{F_{\mathbb{L}[l]}} \widetilde{mWA}^{cf}_{\mathbb{L}[l+1]} \right\|_1, \tag{34}$$

where μ is the aggregation weight, $F_{\mathbb{L}[l]}$ is the activation output of layer $\mathbb{L}[l]$, and $\widetilde{mWA}^{cf}_{\mathbb{L}[l+1]}$ is the *cross-layer weighted attribution*, computed via Eq. 11 using the cross-layer attribution $\widetilde{mA}^{cf}_{\mathbb{L}[l+1]} = \sum_{y_j \in \boldsymbol{y}} \widetilde{mIA}^{cf}_{\mathbb{L}[l+1]_j}(\boldsymbol{\gamma}) \cdot \widetilde{\Delta}^{cf}_{\mathbb{L}[l+1]_j}(\boldsymbol{\gamma})$ as input.

Using the modified cross-layer aggregated attribution ($\widetilde{mIA}^{cf}_{\mathbb{L}[l]}$), we obtain the attribution loss of the current layer ($\widetilde{mA}^{cf}_{\mathbb{L}[l]}$), which is then weighted to derive the weighted attribution ($\widetilde{mWA}^{cf}_{\mathbb{L}[l]}$). Finally, we sum the weighted importance attributions from multiple layers layer by layer to obtain the final optimization objective :

$$\widetilde{mWA}^{cf}_{T} = \sum_{\boldsymbol{y}=\mathbb{L}[1]}^{\mathbb{L}[L]} \widetilde{mWA}^{cf}_{\boldsymbol{y}}. \tag{35}$$

By minimizing $\widetilde{mWA}_T^{cf}$, we can generate highly transferable adversarial examples. In practice, we adopt MI-FGSM [5] to iteratively solve the following constrained optimization problem:

$$\arg\min_{\boldsymbol{x}_{adv}} \widetilde{mWA}_T^{cf}(\boldsymbol{x}_{adv}), \quad \text{s.t. } \|\boldsymbol{x}_{adv} - \boldsymbol{x}\|_\infty \leq \epsilon. \tag{36}$$

M.2 Comparison of Attack Performance

We implement COPIA in the PyTorch framework using five surrogate models: Inception-v3, Inception-v4, Inception-ResNet-v2, ResNet50, and ResNet152. The input resolution is set to (299 × 299) for Inception-v3, Inception-v4, and Inception-ResNet-v2, and (224 × 224) for ResNet50 and ResNet152. For Inception-v3, Inception-v4, and Inception-ResNet-v2, adversarial examples generated by the TensorFlow implementation (COPIA-TF) and the PyTorch implementation (COPIA-PY) are evaluated under the same testing protocol. Both implementations achieve comparable white-box attack success rates (close to 100%), making their transferability performance directly comparable. We do not compare the TensorFlow and PyTorch implementations of the ResNet series for two reasons. First, PyTorch implementations do not explicitly distinguish between ResNet-v1 and ResNet-v2, resulting in substantial architectural and parameter differences. Second, the input resolution of PyTorch ResNet models is (224 × 224), whereas TensorFlow ResNet-v2-50 and ResNet-v2-152 use (299 × 299), making a fair comparison difficult. Therefore, we only report the black-box attack performance of the PyTorch ResNet models. The results are presented in Table 16.

As shown in Table 16, when Inception-v3 and Inception-v4 are used as surrogate models, COPIA-PY achieves higher average attack success rates than COPIA-TF, with improvements of 6.3% and 2.4%, respectively. When Inception-ResNet-v2 is used as the surrogate model, the two implementations achieve nearly identical performance, with average ASRs of 80.3% (COPIA-TF) and 80.1% (COPIA-PY). These results verify the correctness of the framework migration and demonstrate that the PyTorch implementation preserves the attack effectiveness of the original method while providing significantly higher computational efficiency.

To further validate the GPU resource utilization efficiency of COPIA implemented under the PyTorch framework, we compared the GPU memory usage and utilization of COPIA under both TensorFlow and PyTorch frameworks across different surrogate models, with experimental results presented in Table 17. The results demonstrate that COPIA under the PyTorch framework achieves higher GPU memory usage and utilization, thus improving the attack efficiency.

Table 16: Comparison of attack performance under TensorFlow and PyTorch frameworks. Abbreviations for the target model: "C1: Inc-v3, C2: Inc-v4, C3: IncRes-v2, C4: Res-v1-50, C5:Res-v1-152, C6:Res-v2-50, C7:Res-v2-152, C8:Vgg16, C9: Vgg19. Ad1: Inc-v3-adv, Ad2: IncRes-v2-adv, Ad3: Inc-v3-ens3, Ad4: Inc-v3-ens4, Ad5: IncRes-v2-ens. V1: Pit, V2: Swin, V3: Deit, V4: Cait, V5: Vit-base."

Model	Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	Ad1	Ad2	Ad3	Ad4	Ad5	V1	V2	V3	V4	V5	ASR_{bb}(%)↑	Time(S) ↓
Inc-v3	COPIA-TF	98.8	94.3	94.2	87.1	84.8	89.6	88.1	87.0	85.0	71.9	76.3	64.1	63.7	41.9	70.4	58.2	56.7	65.5	53.8	74.0	73.6
	COPIA-PY	99.5	**96.7**	**96.0**	**92.3**	**90.2**	**93.9**	**92.7**	**93.5**	**93.4**	**78.9**	**82.5**	**70.9**	**71.4**	**48.0**	**78.5**	**65.5**	**62.7**	**73.4**	**64.6**	**80.3**	**18.3**
Inc-v4	COPIA-TF	95.3	99.0	93.7	87.5	83.6	89.7	89.6	88.6	87.0	71.1	76.3	70.0	66.6	**51.5**	73.6	65.9	59.0	69.3	58.2	76.5	188.0
	COPIA-PY	**97.0**	99.7	**96.3**	**89.2**	**87.2**	**93.2**	**91.1**	**90.3**	**89.8**	**72.7**	**80.0**	**71.9**	**69.6**	46.5	**79.0**	**72.2**	**60.5**	**73.9**	**60.0**	**78.9**	**21.8**
IncRes-v2	COPIA-TF	96.3	94.8	99.6	90.0	**88.8**	91.1	91.4	88.6	88.3	**79.3**	**87.5**	**76.3**	**71.6**	**64.8**	**76.1**	64.3	62.1	72.5	61.7	**80.3**	184.2
	COPIA-PY	**97.0**	**96.5**	**99.9**	**90.7**	88.7	**92.5**	**93.0**	**89.6**	**90.0**	76.1	86.1	75.5	67.7	60.3	74.2	**65.4**	**63.2**	**72.9**	**63.0**	80.1	**34.5**
Res50	COPIA-PY	96.9	95.6	95.0	98.3	97.5	97.5	96.4	97.0	96.5	90.8	89.4	87.7	84.6	79.0	83.1	70.3	68.3	76.4	70.2	87.9	11.8
Res152	COPIA-PY	96.6	95.4	95.7	98.2	97.6	96.7	97.4	95.2	94.9	91.3	90.5	89.5	87.7	83.8	81.0	69.9	69.4	76.8	73.5	88.5	22.5

Table 17: Comparison of COPIA's GPU memory usage and utilization on TensorFlow vs. PyTorch. On TensorFlow, COPIA runs mostly on CPU, while on PyTorch it gains higher GPU utilization and efficiency.

Model	Method	GPU Memory(M)	GPU Utilization(%)	Model	Method	GPU Memory(M)	GPU Utilization(%)	Model	Method	GPU Memory(M)	GPU Utilization(%)
Inc-v3	COPIA-TF	0	0	Inc-v4	COPIA-TF	0	0	IncRes-v2	COPIA-TF	0	0
	COPIA-PY	2580	28		COPIA-PY	2748	45		COPIA-PY	2843	40

Table 18: Feature layers selected by COPIA (PyTorch implementation) across different surrogate models, ordered from shallow to deep layers.

Model	Layer	Model	Layer	Model	Layer	Model	Layer	Model	Layer
Inc-v3	Conv2d_2b_3x3	Inc-v4	features[2]	IncRes-v2	conv2d_2b	Resnet50	relu	Resnet152	relu
	Conv2d_4a_3x3		features[4]		conv2d_4a		layer1[-1]		layer1[-1]
	Mixed_5b		features[5]		mixed_5b		layer2[1]		layer2[2]
	Mixed_5d		features[9]		repeat[1]		layer2[-1]		layer2[-1]
	Mixed_6a		features[10]		repeat[9]		layer3[1]		layer3[4]

M.3 Implementation Details of Different Models in PyTorch

In the PyTorch implementation, Inception-v3, ResNet-50, and ResNet-152 are constructed using the `torchvision` library, while Inception-v4 and Inception-ResNet-v2 are implemented using the `timm` library. The input resolution is set to (299×299) for Inception-v3, Inception-v4, and Inception-ResNet-v2, and (224×224) for ResNet-50 and ResNet-152.

For each surrogate model, five representative layers are selected from shallow to deep layers for neuron attribution. The feature layers selected by each surrogate model are shown in Table 18. These layers are selected to cover hierarchical representations from shallow to deep stages, enabling the proposed cross-layer attribution mechanism to capture complementary attribution information across different semantic levels.

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